

# Jienluv2bake - change history

Every version of the backoffice app, from v54 to v179 - newest first. The green 'Engine v###' pill on the More screen tells you which build a phone is running.

*About this history: it honestly begins at version 54, when the two-phone shared-cloud era started (2 September 2026). The app grew through earlier versions before that, but they were never written down version by version, so they are left out rather than invented.*

What changed in each version of the backoffice app, newest first. Each version number is the "Engine" you can see on the app's **More** screen, so you can always tell which build a phone is running.

**23 Sep 2026 - engine v179, WORKING HOURS AND THE DELTA T (no database step).** Each person's line is now shaded across the hours they are actually working - drawn from their own jobs, from their first of the day to their last, with nothing for you to type. And the delta t is back across the whole chart: a press on batch 1 of a module that keeps its own time is a hold again, with its own way back. Your widest scale gives up the delta t before it gives up the batch number, on your own rule.

**1. What you asked, in your own words.** "can you shade their working hours in their line?" - and, asked which of two ways you meant it, you chose the one that needs nothing typed. And then, separately: "i ask for delta time, that function is not worker across the chart. And the return to original button is missing?" Both were real, and neither was your phone.

**2. The shade, drawn from the day itself.** Each person's row carries a wash from their own first job of the day to their own last, the gaps between jobs included - somebody who is back in an hour has not gone home. It is worked out from the day rather than typed, so it is drawn on every day you have ever built, including the days with no hours set on anybody. Where you HAVE typed the two hours on a person's card, those stay exactly as they were: the typed hours are the outer band and the working shape sits inside them.

**3. Measured live, and one thing to expect on a row.** On your own day at 1.6 pixels a minute, Wei's shade was drawn at 0 and 149 pixels wide, which is 93 minutes of a 4:00 am day - 4:00 am to 5:33 am - against her five jobs. A job shorter than four pixels is still drawn four wide so you can see it and tap it with a thumb, so on a row whose last job is one minute the last bar reaches a couple of pixels past the shade. Measured live, Wei's last job is the fold, one minute, drawn at 147 for 4 pixels and ending at 151, where the shade - which reads the minutes themselves - ends at 149. The shade is the true length of the work and the bar is the true size of your thumb, and the difference is only ever that.

**4. The delta t, and why it had gone.** A press on a batch used to be a hold measured from where the line puts that batch - a delta t. Since v154 that had been narrowed to batches below batch 1, and to batch 1 only of a module set to follow the one above. Every module of your own day is set to "Its own time", which is the one case the narrowing left out, so batch 1 of every module wrote an absolute start time instead: no hold was ever written, no delta t was drawn anywhere on your chart, and the card's own way back - which exists only while there is a hold to take off - disappeared with it.

**5. It is back, and the way back is called "Back onto the line".** That is its name from v151, and it can be renamed in one line if you meant the other wording. Measured live on your own day at 1.6 pixels a minute, pressing + 5 min on batch 1 of Dimple and top: the card read Batch 1 and delta t = +5 min with 4:05 am to 4:11 am, the bar's own tag read B1 delta t=+5, the note read "Dimple and top held 5 minutes later than the time you gave it - every batch of it moved with that.", and the three modules below it did not move a minute. Pressing Back onto the line put the batch back to 4:00 am and said so: "Every batch of Dimple and top is back at the time you gave it." Going there and back left your stored data byte for byte identical at 13,888 characters, because the hold is kept in the delta itself and none of your stored start times was rewritten.

**6. The badge now knows when to keep quiet.** Your own rule: "if the scale is too wide to show batch no. and

delta t then forgo delta t". So the batch number - the thing that tells one batch from another - is never given up, and the delta t is printed only above the widest stop. Measured live at the widest reading, 72 pixels an hour: the tag read exactly B1 while the bar kept its own tint, which takes no room and still answers which of the batches is held, and the tip on that tag still named the hold in full.

**7. Why the last module's batch 1 is different - the question you asked while this was being built.** It behaves differently on purpose, and it has since v152: tapping batch 1 at your LAST module does not open an ordinary batch card at all. That batch is the moment the whole day hangs from - it is where the day is worked backwards from - so its card is the day's own card, headed "the end of your first batch". Measured live, tapping batch 1 of Cutting and packing read "Cutting and packing - the end of your first batch", named the day's first batch ending at 4:12 am and the day finishing at 6:03 am, and listed all eight modules. On the FIRST module it is the other way round: batch 1 there has no module above it to be held back from, so a press is simply that module's own start time. Everything in between is the ordinary batch card with the delta t and its way back.

**8. And where "Put my start times back" lives, since you looked for it.** It is on that day card and it is a one-shot undo: it appears only once you have pressed "Pull them back to their latest start", because there is nothing to take back until something has been pressed. Measured live on your own day, every one of the eight modules was already ticked as late as the line allows, so the card said so - "Every module of your line is already as late as it can go - there is nothing to pull back." - and offered neither press, which is why you could not find it there.

**9. Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved day. Measured live, after working through the batch card, both presses and the widest scale, your stored settings were put back from a copy taken beforehand and compared: byte for byte the same 13,888 characters, and your three saved days still read 24, 12 and 24 pans.

**10. Every rule that stands on it was proved load-bearing, not assumed.** Thirteen faults were put back in all, and each was watched failing by a test that names it, then restored byte-identically. The shade: with it drawn from the whole day instead of from that person's own jobs it fails reading whose work it covers; with the bars painted before the shade it fails reading that the bars no longer cover the wash; with the shade given a stacking order it fails reading that the hours climb over the module titles, which is the fault this screen already had once. The delta t: with the hold narrowed back to the v154 rule it fails reading that a press on batch 1 wrote a start time instead of a hold; with the module's other batches left behind it fails reading that the hold did not take them with it; with the widest scale still printing the delta t it fails reading that the batch number was given up for it; and one fault in the tests' own stand-in screen was found and fixed on the way. The suite is 1294 passing with none failing.

**11. No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase. Both facts live inside the scenario, which already syncs whole, so they travel to your other phone with the rest of your plan.

**23 Sep 2026 - engine v178, REASSIGNMENT (no database step). "Share them out" now really hands the whole day back.** It used to clear every module and leave the stretches you had placed by hand exactly where they were - and a hand-placed stretch is read first, ahead of the module's own person, so the press said one thing and the chart drew another. Both full-day presses now take those stretches back, and they say how many they took back, because taking back a decision of yours is a change you are told about.

**What you asked, in your own words.** "how to rerun job assignment after we set the more details?" - and then, once that question had been measured and answered, "build v178, Name it as REASSIGNMENT". The answer to your question is the People box, and measuring it is what found the fault below.

**1. The fault: the press promised a day it did not draw.** The People box holds three answers, and two of them redraw the whole day's hands at once. "Share them out" cleared every module - but a stretch you had placed by hand is kept on the module as its own entry (that is what makes a hand-over move one stretch and not a whole row), and it is read BEFORE the module's own person. So the day kept your hand-placed stretch standing while

the press said it had handed everything back. Measured on your own day: after the press the toast said the day had been shared out, and the box's own closed words still read "Your own" - and your Dimple and top was still sitting on Wei, where you had put it.

**2. What it does now.** Both full-day presses - "Share them out" and "One a module" - take every hand-placed stretch back along with everything else. The stretch is not deleted quietly either: the press names how many it took back, and a day with none adds nothing to the sentence, because a count of zero is not news.

**3. Measured live on your own day,** starting at 4:00 am at 1.6 pixels a minute, with your one hand-placed stretch on Wei. Pressing "Share them out": the box went from "Your own" to "Sharing them out", your hand-placed entry was gone from the record, and the toast read "Sharing them out - every job reassigned to as few hands as can cover the day. 1 stretch you had placed by hand goes back on the day's own arrangement." Dimple and top moved off Wei's row onto the row the day picks for itself, and every other job on the day was untouched. Pressing "One a module" on the same day: six people, one to the six modules that hold hands, the box reading "One to a module", and the same sentence naming the one stretch it took back - and the stretch no longer overrides the person that press had just given that job.

**4. Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario was changed by building this, and nothing here blocks a sale or reads an order. The only writes are the ones your own press makes. Measured live: after both presses and a full pass of opens and closes, your stored data was put back from a copy taken beforehand and compared - byte for byte the same, 13,888 characters - and your three saved days still read "No fridge, 1 person" 24 pans, "My sister proposal 21/9/2026" 12 pans and "One baker day" 24 pans.

**5. Every rule that stands on it was proved load-bearing, not assumed.** Eight faults were put back in all, and each was watched failing by a test that names it, then restored byte identically. With the stretch left standing, the test fails reading that the old arrangement is still on the chart; with the count left out of the sentence, it fails reading what the press said; with the count worked out after the day had already been cleared, it fails the same way; with "One a module" left holding the stretch, it fails reading that one job is short of the person the press gave it; with the empty entry written back as an empty map instead of no entry at all, it fails reading that nobody has one spelling; and with the count written always in the plural it fails on the first hand-back. The suite is 1287 passing with none failing.

**6. No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase. Everything here is inside the Scenario planner, under More.

**23 Sep 2026 - engine v177 (no database step).** Each person now has two facts of their own, and both are yours to state once: the hours they are here, drawn as a band along their row and a real bound on who the day may hand work to, and which modules they are trained for. A job nobody is named for goes to somebody who can work it, inside their hours, and between two of those the day asks the most specialised person first. Nothing is refused and nothing is hidden: a job that lands outside somebody's hours or training is shown and said in words.

**What you asked, in your own words.** "back to the scenario planning. 1. I need each person to have the start work time and end work time, 2. which module they are dedicated for, and only taking over others job, as priority 1,2, or3." And then, while this was being built, the two sentences that settled the shape of it: "a person we should be able to specify their skill, by module, can be more than one, by module", and "module should select the one that specialised". Of the takeover order you said the thing that made it simple: "the person card superseded the module person assignment. maybe no priority rank is needed to be simple". So there is no 1, 2, 3 rank anywhere: the card says what each person can do, and the day picks by itself.

**1. Their hours, counted from your day's start and not from the clock.** You asked for a start time and an end time, and told me how to read them: "the working hours is relative to the chart, not exact hours, end also relative to the chart". So the two boxes are clock times you can read at a glance, and what is stored is the minutes from the start of your own day - which means moving your day's start slides every person's hours with it and nothing is retyped. Measured live on your own day, which starts at 4:00 am: typing 05:00 and 09:00 stored 60 and 300,

which is exactly five and nine o'clock less four.

**2. An empty box is a real answer, and it is "here all day".** Leave both empty and that person is here all day, which is what every person on every day you have ever built already is. The hint under the boxes says so in one line, and it also says which clock the times are read against, because two empty boxes with no explanation would read as two boxes you had not filled in. Typing the whole day by hand - 4:00 am to midnight - is stored as no entry at all, so "here all day" has one spelling rather than two that mean the same thing and could ever drift apart.

**3. And a pair that comes out back to front is refused and SAID.** An end earlier than the start is a mistake, and the app now puts both boxes back to the hours the day is actually using and says which way round they go. A box that took a number and quietly did something else with it is the kind of thing that reads as a fault for weeks, and this app does not do that.

**4. What the hours are for.** Your own aim, in your own words: "the person that have the most consistant work is the person handling oven, colling and packing, he only help do others when he has free slot, by one person not doing too many functions, it is easier to train that person as will. Probably can employ part-timers." So the hours and the training are what the day reads when it hands out a job nobody is named for - and they are a PREFERENCE, never a refusal. A job nobody qualified is free for still goes to a free pair of hands, and the row says why.

**5. Which modules each person can work, and the specialist is asked first.** The person card now carries a tick for every module that actually holds hands, so there is never a tick against a module switched off or against the proofer, which would be a tap that does nothing. Only six of your eight modules hold hands on your own day, and those six are exactly what the card offers. Tick nothing and that person can work anything, as they do today - the same absence rule this app already uses for automatic cycles and merged jobs. Tick one or more and the day only ever gives them those.

**6. And between two people who can both do a job, the more specialised one is asked.** Your own rule, in your own words: "say one person have all skill, person2 have have the skill, person3 have only 1 skill, module should select the one that specialised". Somebody trained for one module is asked before somebody trained for five, and somebody who can work anything is asked last of all - which is what leaves the specialist on their own job and the generalist on what is left over.

**7. Measured live, on your own day.** With everybody untrained and no hours set, your day packs to four people and 58 minutes of hand work, and the three stretch-and-fold slots all sit on Wei. Ticking Person 4 for "The rests and the stretch and folds" and nothing else moved all three folds off Wei and onto Person 4, even though Wei is the lower number and was free at that minute - the specialist was asked first, exactly as you asked for. Wei's row fell from five jobs to two and the other rows were untouched. Unticking put the day back to where it was, to the job.

**8. Their hours, drawn on their own row.** Each person's hours are painted as a wash along their own row, under everything the day draws on it, so you can see at a glance when somebody is here without reading a number. It carries no stacking order of its own on purpose: a band given one would climb over the pinned module names, which is the fault this screen already had once. Measured live at a phone's width on your own day: a shift of 05:00 to 09:00 was drawn at 96 pixels along the row and 384 wide, which is 60 minutes and 240 minutes at your own 1.6 pixels a minute. It takes no taps at all, so it can never swallow one meant for a bar over it.

**9. And when a job cannot fit somebody's hours, the card says so in words.** This is the part that matters most to the way you work, because it is the same rule this app has always kept: a plan that does not fit is shown, never quietly rearranged. Measured live: with Wei set to be here 4:00 am to 4:05 am while the dimple runs to 4:06 am, the card read "Wei is set to be here 4:00 am to 4:05 am, but Dimple and top runs to 4:06 am - a job that cannot fit inside those hours." The same sentence appears on their row's tip and on the Production line board, all three printed from one funnel so they can never disagree.

**10. And a job outside somebody's training is said in the same place.** Somebody given a job they are not

trained for is told so in the same words and the same place: the job is named, the times are named, and the note says it is said here rather than the job being quietly moved. The card also reads back what they are trained for - "Trained for 1 module: The oven swap and the bake" - so the answer you typed is the answer you can see.

**11. On the Production line, the worker's own hours.** The board a worker reads already says what they do next and how long they have; it now also says when they are here, and names any job that cannot fit inside those hours. Withholding that while printing their jobs would be perverse - when you are here is the first question of a working day. The board is still a READING and writes nothing: measured live, a tap on a person's row on the board left your stored settings character for character identical, and the board offers no hour boxes and no ticks.

**12. Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved day was changed by this version. Measured live: after a full pass of presses on the planner - the two hour boxes typed into and cleared, ticks set and taken off, cards opened and shut - your stored data was put back from a copy taken beforehand and compared, byte for byte the same 13,888 characters, with the two new keys absent again because writing them is something only a press you make can do. Merely opening the screen adds nothing. Your shelf itself is untouched: No fridge, 1 person still reads 24 pans, My sister proposal 21/9/2026 still reads 4, and One baker day still reads 24. And the two chart windows kept their pan at 300 across two of those presses, so a card write no longer throws either window back to the start of the day.

**13. Every rule that stands on it was proved load-bearing, not assumed.** Eighteen faults were put back in all, and each was watched failing by a test that names it, then restored byte-identically. Two of the eighteen were faults in earlier drafts that the fault sweep found, and both were unfalsifiable as first written - a test that could not fail - which is the same lesson this file has carried before about a stand-in being more forgiving than the real thing. Among the rest: with the band given a stacking order the test fails reading that the hours climb over the module titles; with the band drawn last it fails reading that the bars no longer cover the wash; with the boxes storing a clock reading instead of minutes from the day's start it fails reading where the band was drawn; with an untick leaving an empty list behind it fails reading that unticking is an answer and not a missing entry; with the copy of a day not carrying the two keys it fails reading that the hours of the day you left are still on the day you opened.

**14. No database step.** Not one stored field is added or changed on the server, so there is nothing to run in Supabase. Both new facts live inside the scenario, which already syncs whole, so they travel to your other phone with the rest of your plan and need no new table and no new SQL.

**23 Sep 2026 - engine v176 (no database step).** The Production line is now a board a worker reads, not a form you fill in: the day chart from your plan, a clock line showing where the real time falls in it, and a strip saying what each person does next and how long they have. The screen used to ask you for 21 numbers; it now asks for 12, and 8 of those are folded away.

**What you asked, in your own words.** "can we simplify the production portion, it ask for alot of key in" - and then the shape of what you wanted: "maybe one function for production line, 1. copy the chart into Prodcution line, but should not be editable, it become a dashboard for worker, give the worker good inform of what next for them, how long, and others, you can be creative". And of the boxes on the screen you said the thing that made this simple to build: "must of the question are answered in scenario plan". They are, and that is where the board now reads them from.

**1. The day chart is carried across, not drawn a second time.** The board and the Scenario planner are the same drawing, produced by the same code, so the two screens can never disagree about a day - not its modules, not its batches, not its times, and not who is on what. What the board drops is everything that changes the day: the Scale buttons, the People box, the backwards chip, the undo chip, the Start the day now button, New module, and Save changes. Measured on your own day: the planner wears 8 buttons above the chart, and the board wears none of them - with the two cards shut the whole screen carries two buttons, and those two are the fold heads at the foot. What the board keeps is everything that only reads: you can pan the day with a right press, the clock follows your finger along the ruler, and the tips on the module names still open. A board that could not be panned would be a worse board on a long day.

**2. A clock line, drawn with no press at all.** A board nobody has touched yet still has to say what time it is, so the line is placed the moment the screen opens and kept in place once a second. It is drawn exactly where the real clock falls in your day: at 96 pixels an hour on your own day, a press of nothing at all puts the line at 156 pixels for your 4:00 am start, and the clock it reads out matches the ruler under it. Measured live on your own day: at 5:00 am the line stands at 252 pixels and reads 5:00 am - which is 156 plus exactly one hour of 96. At 4:30 am it stands at 204 and reads 4:30 am. Before your day begins it parks at your day's own start and says "day starts 4:00 am"; after your day has ended it parks where the day ended and says "day ended 6:03 am". It never wraps around to the other end of the ruler, which is the tempting wrong answer and would have put the line in the middle of a day that is not the one on the screen.

**3. What is next, in clock times and not countdowns.** Above the chart the board says, for each person, the job they are on and the one after it, with the clock time each starts and how many minutes it runs: on your own day at 5:00 am that reads "Wei - the rests and the stretch and folds, 5:01 am, 1 min, then the rests and the stretch and folds at 5:32 am". Clock times and not a countdown on purpose: a countdown means rewriting a box somebody is reading every single second, and this app has a rule about that - a redraw must never move what you are looking at. On a day that is not shared out to people, the strip falls back to one list of what the line does next.

**4. And the strip answers AGAIN as the clock runs, without touching the chart.** This one was found while measuring, not reported: the strip was drawn once when the screen opened, so a worker opening the board at 4 am would still be reading "next: mix at 4:01" at eight in the morning. It now looks again whenever the clock has changed the answer, and only then - inside the working day it redraws at the exact minute a job starts or ends, and never in between. The chart itself is not redrawn, which is the planner's own rule and the reason a worker's scroll is never lost. Measured live across a job boundary: at 9:00 am the strip names Cutting and packing, at 9:10 am it names the oven swap, and the number of times the strip has been rebuilt does not change between them.

**5. Three things about a day the clock has already gone past.** The first was a real fault, found by opening the board in the evening. The strip had one fallback sentence for two different empty lists, so a day that finished at 6:03 am was being described as "Nothing on this day needs hands" - which is a different thing entirely, and the opposite of true. A day whose jobs have all been and gone now says so: "The day is finished - every job on it has been and gone." The second is the offer to make a sound. The board rings people one minute before each job, and it rings by working out from the clock which jobs are due - but on a day that is already over there is nothing left due, so switching calls on would ring for nobody. That offer is now withdrawn on a finished day rather than left promising a sound it cannot make. Measured live at 7:16 pm on your own day: no Start calling button, and the line under the strip reads "The day is finished". The third is the other half of that, and it matters more: calls you turned on earlier can ALWAYS still be turned off. Stop calling stays on a finished day, so a sound started before cannot be left ringing with no way to stop it.

**6. What left the screen, and where the numbers went.** Twenty-one boxes became twelve. Nine of them are now read off the day you built under More -> Scenario planner - how many pairs of hands, the pans a tub makes, oiling and weighing out, the dimple, the packing, the oven and the bake-and-swap - so those questions are asked once and answered on both screens. The twelve left are the ones genuinely yours: four behind the first fold (how long you will bake for, the pans you own, what your proofer holds, and the pace between batches you would like), and eight behind the second fold (the rests, the fold, the two proofs, the cooling, the clock you want the first batch at the oven, and the minutes of soft tolerance around a start). Both folds start shut and open in place, so opening one never redraws the chart above it. The line card still answers while it is shut, in one line: measured on your own day, "44 pans today · the proofer is the wall that holds it back", worked out from the day rather than typed anywhere.

**7. And the timetable card is folded, which is where it belongs.** The backwards timetable - the last moment each stage may start - was left on this screen on your word ("you decide, that is your baby") because it is worker information: it is the one thing on the screen that says when the dough has to be in the tub. But it is a page of advice, not a glance, and it was sitting out on the board in front of a worker who came to read what is

next. It now lives behind its own fold head. Measured at a phone's width with both cards as they come: the timetable card is 64 pixels tall, its head and nothing else, where it was 943 while the timetable sat outside it. That one change alone takes the whole screen from 2005 pixels to 1126.

**8. Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved day changed by this version, and nothing here blocks a sale or reads an order. The board is a READING of your day, and every number on it is computed rather than stored - the box you type a number into writes the one number you typed and nothing else. Measured live, going through a full pass of presses on the board: a bar tap, a module tap, a right press, and both cards opened and shut twice each, with your stored data compared before and after. Your data reads 13,888 characters, and it is byte for byte the same characters it read before the first press.

**9. Every rule that stands on it was proved load-bearing, not assumed.** Nine faults were put back in all, and each was watched failing by name. The strip: with the finished-day sentence taken off, the test fails reading that a finished day falls back to the sentence for a day with no hands on it; with the calling offer always drawn, it fails reading that a finished day offers a sound it cannot make; with the foot dropped whenever the day is over rather than only when calls are off, it fails reading that a sound already started cannot be turned off; with the strip never rebuilt, it fails reading that the strip still names a job the clock has passed; and with the live minute taken off the strip's own signature, it fails reading that the parked sentence no longer names the hour it is standing in. Two more put a null where the parked sentence goes, and each was caught by the word "null" appearing on the screen. The fold: with the timetable put back outside its fold, the test fails reading that a shut card is already showing the clock times its caret says are behind it; and with the test's own reader taken back to one that walks straight through a hidden box, the same test fails - the reader had been reading text the browser never paints, which is the same lesson this test file already carried about a stand-in being more forgiving than the real thing. Every one was restored byte-identically afterwards. The suite is 1266 passing with none failing.

**10. No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase. The day the board draws is the one already synced, and the four numbers behind the first fold were already synced with the rest of your production plan.

CHANGELOG and the guide both carry all of this, the app's own More screen reads Engine v176, and the Production line names itself as the board it now is.

**23 Sep 2026 - engine v175 (no database step). Two things you reported: the scenario windows no longer jump back to the start when you press something, and the right-button drag now works while a card is open on the chart. Both inside More -> Scenario planner.**

**What you reported, in your own words.** "Few problem of screen jump here and there. One obvious one is the scenario windows, when i click, window reset. And right button drag dont work". Both were real, and neither was your phone.

**1. Why a press threw the windows back to the start.** When you press something that changes the day - a batch card's + 5 min, the People box, the scale, opening a saved day - the screen is drawn again from scratch. That has always been how it works, and it is what keeps the whole panel agreeing with itself. But a box drawn again is a NEW box, and a new box starts its own scroll at nothing. So every one of those presses put both windows of the day chart back at the far left and the top, and would have put the list of saved days back at its first row with them. Measured live at a phone's width with the modules' window panned 300 pixels: one press of + 5 min left both windows reading 0.

**2. What it does now, and where you were is put back only after the measuring is done.** Where each of the three sliding boxes was - the two windows and the list of days - is read before the screen is drawn again, and written back afterwards. The order is the whole of it: the list of days is measured to work out its height, and a row's foot moves when the list under it slides, so measuring a list that had already been slid back down would come out short by exactly the amount it moved and would cut your fourth day in half. Nothing is slid until the measuring is finished, and the measuring all happens before anything is drawn on the screen, so you never see the day sitting at its start either. Measured live on your own day: panned 300, one press of + 5 min, and the window reads 300 on the day drawn after it - and both windows are put back by their own numbers, never one

number shared, because they pan together and a redraw is not the place to find out they had drifted.

**3. Why the right-button drag did nothing with a card open.** A card is drawn on a full-screen sheet that sits over the whole chart, so with a card open your right press landed on the sheet, not on the day behind it - the day you were actually looking at could not be moved at all. Measured live at a phone's width with a batch card open: a right press 200 pixels into the chart, dragged 60 to the left, left the window reading 200 exactly where it started. The sheet carried no gesture of its own, so nothing was being taken from anything else.

**4. What it does now.** The sheet answers the same right press the two windows do, and which window moves is worked out from where your hand is - the upper window for a press on the upper half of the chart, the lower one for the lower. Your own card is left alone: a right press on the card, or on any of its buttons, moves nothing, so the day cannot slide about while you are reading the card that describes it. The left button is untouched and still opens and works cards. Measured live with a batch card open: a right press 60 pixels to the left moved the modules' window from 200 to 260, the same press on the people's window moved that one from 260 to 310, the two windows came back into step at 310 on the next frame, and the same right press on the card itself moved nothing.

**5. Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. Every tap still opens exactly what it opened before. Your stored data was put back from a copy taken before the measurements and compared: byte for byte the same, and your three saved days still read 24, 4 and 24 pans.

**6. And the rule that stands on it was proved rather than assumed.** Nine faults were put back in all and each was watched failing by name, then restored byte-identically: with the putting-back left out entirely the test fails reading that the press threw the windows back to the start; with only the modules' window put back it fails the same way; with the two windows collapsed into one number it fails; with the putting-back done before the measuring it fails reading that the shelf was capped short of where the fourth day ends; with the class name matched dot and all it fails (this is the fault that was in the first draft of the fix, and the test caught it); with the sheet never wired it fails reading that a right press over an open card did not move the day; with the sheet always panning the upper window it fails on the lower press; with your own card not spared it fails reading that a right press on the card dragged the day behind it; and with the left button let through it fails the older drag test, which says the left press is the one that opens cards. One more thing was found and fixed on the way: the tests' own stand-in screen did not move a box when a box it sits inside was scrolled, which is the opposite of what a window does, and that gap had been hiding whether the measuring happened before or after the sliding. The suite is 1220 passing with none failing.

**7. No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

**23 Sep 2026 - engine v174 (no database step). Two things you asked for: the modules' window no longer lets the day's writing come up between the module titles when you pan it right, and the Your scenarios card now shows four days at a time with the rest reached by sliding the list. Both inside More -> Scenario planner.**

**What you asked, in your own words.** "the windows when panning right, some of the chart writing shown in between module titles" - and "now i had 8 scenario, we have to make the scenario just shown 4, the rest shown by slider". You were right about both, and neither was your phone.

**1. The writing between the titles, and why panning right brought it there.** The module titles on the left of each window are pinned: they stay put while the day slides under them, which is the whole reason you can read which row is which at any hour. What was wrong was the ORDER they were stacked in. Every B-number, every band of your hands and every line down the bars was drawn at or above the title's own level, and when two things sit at the same level the one drawn later wins - and the day is always drawn after the titles. So nothing had to go wrong for a number to appear over a name; the day simply took its turn on top of it. Measured live on your own day at a phone's width, with the chart panned 25 pixels: nine B-numbers and six bands of your hands were the topmost thing at their own position inside the 156-pixel title column.

**2. What it does now.** The three things that can be stacked are written down once, on the chart itself, as three

levels: the titles lowest, the clock strip above them, and the readings you take highest. The clock strip has to outrank the titles because the titles scroll under it, and a reading has to outrank the clock strip because the reading's own top line lands on the strip. Everything the DAY draws - its marks on your bars, the B-numbers and the names on them, and a module's own note - is deliberately under the titles, so no pan position can bring one of them up over a name. Measured live on your own day across six pan positions, 1,400 points sampled inside the title column: not one of them reads as the day, while the same eight B-numbers and five bands are still drawn exactly where they were, under the titles where they belong.

**3. And the clock you take is refused where the day is not in front of you.** The same fault had a second half. Pressing the clock strip over the title column could still read a minute and draw the hairline and its number across the titles - because the strip's own ruler scrolls with the day, so once you have panned, the strip's left edge has travelled behind the titles and the arithmetic no longer knows where the titles end. It now measures where the day actually starts rather than trusting the strip's edge. Measured live with the day panned 25 pixels: pressing at 140 pixels across - inside the titles, which end at 183 - reads nothing and draws nothing, while the same press 17 pixels further on reads 4:26 am as it always did. On a day that is not panned nothing about this changes at all.

**4. The shelf now shows four days and slides to the rest.** On your eighth saved day the card grew a row per day, and a shelf of eight days plus the buttons under it ran past the bottom of your phone. It is four days at a time now, with the list itself sliding up and down for the rest, and a line under it that says so: measured live with eight days, "Showing the first 4 of your 8 days. Slide the list itself up for the rest." The headings stay put as the list slides, so a row you have scrolled to is still under the heading that names it. A card with four days or fewer is exactly the card it was before - no scroller, no height, no note.

The height is the part worth being careful about, and it is measured rather than assumed. A long day's name wraps to a second line at a phone's width, so the fourth row is not always as tall as the first three, and a shelf capped by arithmetic would have cut the bottom off the fourth day. Measured live on the eight days: three rows of 59 pixels, then a fourth of 73 because its name took three lines, and the shelf stops at exactly 290 pixels - the fourth row's own foot. The fifth row begins at 290, so nothing of it shows. That is the reading this was built from: capped by arithmetic it would have been 236 pixels and the fourth day would have been cut in half.

**5. Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. Every tap still opens exactly what it opened before. Your stored data was compared before and after every measurement: 13,888 characters, byte for byte the same, and your three saved days still read 24, 12 and 24 pans. The eight-day card used for that measurement was put on a copy of your data and taken off again, with your own file restored byte for byte and nothing left behind.

**6. Every rule that stands on it was proved load-bearing, not assumed.** Eight faults were put back in, one at a time, and each was watched failing by name, then restored byte-identically. The ladder: with a B-number put back above the titles the test fails reading that panning right puts the day's writing between the module titles; with the title cell taken off its own level it fails reading that the levels can no longer be moved in one place. The reading: with the guard taken out it fails reading that a reading was taken with the pointer over the module titles. The shelf, four ways: capping it off the last day instead of the fourth fails reading where the fourth day ends; writing no height at all fails the same way; measuring it as a multiple of one row instead of reading the fourth row's foot fails reading 40 pixels where the fourth day ends at 220; and taking the slide away fails reading that the days past the fourth are unreachable. The suite is 1217 passing with none failing.

**7. No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

**23 Sep 2026 - engine v173 (no database step). Two faults in the "Minutes from one batch to the next" box, both on a day that runs more than one batch: on a module that runs only one, the box did nothing at all and said nothing about why; and on a module whose batch two had been nudged closer by hand, a smaller pace could send that batch in front of batch one, and the day then drew both on the same minute. All inside More -> Scenario planner.**

**What you reported, in your own words.** "similar problem happen. 1st module, time between each batch, set to

21, but no effect, or the chart dont workout" - and then, when asked which of the two you were seeing, "both batch start the same time". You were reading a 12-pan day, and the first module of that day is "Mixing by hand in the tub (set the minutes)": one batch, and the same pace it has always had. Both halves of your report were real, and neither was your phone.

**1. On a module that runs one batch, the box does nothing - and now says so.** A pace is the minutes between one batch and the next. A module that runs a single batch in the day has no second batch for a pace to sit between, so whatever you type there cannot move anything. That is what your first module is: measured on your 12-pan day, "Mixing by hand in the tub" runs one batch of four pans at a pace of 20 minutes, so a 21 typed into it had nothing to act on and the chart was already telling the truth. The fault was that the box said nothing at all about that - it looked exactly like a box that was broken. It now explains itself in your own words: measured live on that very module, the line under the box reads "This module runs one batch in the day, so there is no second batch for a pace to sit between yet. How many batches it runs in the day is the box above that sets it - make that two and these are the minutes between them." Make it two and the line goes.

**2. And on a module that does run more than one, a smaller pace could put batch two in front of batch one.** This is the "both batch start the same time" half, and it is the more serious of the two.

A saved day remembers the exact minute each batch starts, one module at a time. A module that carries that list is re-spaced at the pace you type, so a batch you have nudged by hand keeps the minutes you nudged it by rather than being pulled back onto the rhythm. That is right for a batch you nudged LATER - and exactly wrong for one you nudged EARLIER, because its deviation from the rhythm is a negative number, and adding a negative to the smaller rhythm lands it in front of the batch above it.

Measured on your own first module, in the shape that does it. A nudge of batch two by hand with the box still reading 87 leaves the module carrying the times 4:00 am and 4:21 am - and that batch is then 66 minutes earlier than the rhythm, so its deviation is -66. Typing 21 sent it to minute -45: three quarters of an hour before your day begins. The day cannot draw a batch that starts before it does, so both batches were drawn on minute 0 - both at the same time, exactly your words. This is also where the "no effect" reading of the chart comes from on a two-batch module: the box reads 21, and the two bars sit on top of each other and will not come apart.

**3. What it does now, and it is your own app's rule rather than a new one.** Two rules, and both were already the app's own. A hold is only ever a HOLD - every other press on this screen already obeys that, so a batch can be held back off the rhythm and can never be pulled in front of it. And no batch is left at or behind the batch above it: where a hold has become impossible in the smaller rhythm, the batch takes the rhythm rather than a minute that is not its own. Measured live on that same module, through the same card: typing 21 leaves the two batches at 4:00 am and 4:21 am, drawn 21 minutes apart with the box reading 21, and the second batch is no longer able to overtake the first.

**4. And a third fault, found while proving the second: the Auto box did not save.** This one you had not reported and would have met eventually. "Minutes from one batch to the next" carries an empty box with the word Auto in it, meaning each batch starts the moment the one before it ends. Emptying the box moved the flag and nothing else.

The zero that means Auto was being written through the same writer every other box on this screen uses; that writer refuses any number below the box's own minimum; and this box's minimum is 1 - so the refusal came back BEFORE the save and before the repaint. The choice was made in memory, nothing on the screen moved, and the next reload threw it away. Measured on your own mix with two batches: emptying the box left both batches 30 minutes apart and the screen did not move at all; the card reopened with its empty Auto box drawn over the very same unchanged chart; and a reload forgot the choice altogether. The zero and the save are now written directly, so Auto is an answer the app keeps: measured live, emptying the box redrew the two batches 20 minutes apart - the module's own cycle, which is what Auto means - and a reload kept it.

**5. Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. Every tap still opens exactly what it opened before. Your stored data was compared before and after every measurement: 13,888 characters, byte for byte the same, and your

three saved days still reading 24, 12 and 24 pans.

**6. Every rule that stands on it was proved load-bearing, not assumed.** Six faults were put back in all and each was watched failing by name, then restored byte-identically. The hold rule: with the clamp taken out the test fails reading that a batch pulled closer than its line carried a hold it does not have. The forward sweep: with the sweep removed the test fails reading that a batch was left in front of the batch above it. The one-batch note: with it gated off the test fails reading that a pace box with nothing to space does not say so. The Auto half, three ways: with the 0 written through the writer again the test fails reading that Auto did not write the 0 that means Auto; with the save and the repaint left out it fails reading that the Auto press saved and repainted nothing; and with the list not deleted it fails reading that Auto left a list of times behind, so the cycles no longer decide. One fault in the tests' own stand-in screen was found and fixed on the way - it had no firstChild, so the "did the screen repaint" assertion was comparing nothing with nothing and could never fail - which is the same lesson that file already carries about a stand-in being more forgiving than the real browser. The suite is 1215 passing with none failing.

**7. No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

**23 Sep 2026 - engine v172 (no database step). Four things: handing one stretch of work to somebody else no longer moves the people's rows, their names or their jobs; the "Minutes from one batch to the next" box reaches your batches again, and says so when your oven is what is holding them apart; the Your scenarios card is a table you can run your eye down; and the question a card asks before deleting something is drawn over the card that asked it instead of behind it, with the row of buttons at the foot of a card given the shape it never had. All inside More -> Scenario planner.**

**What you reported, in your own words.** First, and the largest thing in this version: "i have 4 persons, 1st Jien, 2nd Wei, 3rd and 4th. I saw Jien is heavy loaded, so i click one of Jien session, and select to switch that session to Wei, where he is free. BUt what happen is Jien disapper, and Jien name chage to Wei, and the originally Wei sessions disappeared". Then: "im now on my 36loaf scenario, pls check why im lowering the minutes from one batch to the next, cannot reflex even i reduce it to 13, look like something prevent it from loweriing below 25". And, separately: "can you make the Your Scenario, reorganised it to a better visual clarity, details in table, action button below?" And then, while that was being built: "please check the process of delete a module and the process of deleting a scenario, it dont look right".

**1. Handing one stretch of work to somebody else moved everybody on the chart.** This was your report, and it was the largest thing in this version.

All three of the things you saw were real, they were one fault, and none of it was your doing.

A day that is sharing them out has nobody written on it. The rows you see are worked out fresh every time the chart is drawn: the first job nobody is on goes to person 1, the next to person 2, and a new row appears only when everybody is busy. That is why a shared-out day reads as perfectly sensible people - and it is also why its row numbers mean nothing at all. They are not yours and they are not on the modules.

Your names, though, are kept by number, and every day you have shares them. So the moment one stretch was given to Wei, a row for Wei existed before the free jobs were handed out - the packing began at Wei's row instead of at the first row, every invented number moved up by one, and the row that had been person 1 became person 2. Jien was person 1, so Jien was gone; the row he had been on now wore Wei's name; and a fifth row appeared at the top with what was left over. Exactly the three things you saw.

**2. What that press does now, and what it was measured doing.** Before a stretch changes hands, the arrangement already on the screen is written down: each job is given the person the chart is already showing at it - one name for a module whose batches all sit on one row, one name per line for a module worked as lines, and a note on the single stretch where a module's batches really are split between two people. A stretch you had already placed by hand, including one you had handed back to whoever is free, is left exactly as it is. From that moment the numbers on the rows are facts rather than the packing's invention, and your names go on meaning what you see.

Measured live on your own day at a phone's width, going through your exact press. The card read "Move this

slot off Jien" and its button read "Move it to Wei". Before the press the chart drew four rows - Jien with five jobs on him, Wei with one, and two more below them. After it, the same four rows were in the same order with the same names: Jien had lost the one stretch you moved and nothing else, Wei had gained that one stretch and nothing else, and the two rows below were untouched. The People box stopped saying the day was sharing them out and read "One to a module", which is what the card had told you it would do before you pressed it. And the chart held there: the page was read again from scratch and drew the same four rows, the same names and the same jobs - where before this fix that very re-read was what moved them.

**3. And one sentence on that card was describing a day that was not in front of you.** Found by measuring the fix on your own day rather than on a made-up one. The card carried a note explaining that the press settles the whole day, and it said so in the words "This day is sharing them out" - on a day whose People box plainly read "Your own", because one stretch on it had been placed by hand. Two claims, about a day that was making neither. The note may now say only what is true of the day on the screen: the part that says the press settles the whole day is always said, and the part about the box stopping saying it is sharing them out is said only when the box is saying it.

**4. The box that would not move, found and fixed.** Two different things were true at once, and only one of them was a fault.

The fault: a saved day keeps, for each module, a list of the times its batches start. A list is read as the whole truth about where every batch sits, so a module that carries one ignores the box entirely - the box could be set to 13, or to 87, or to anything, and the batches would not move. Measured on the engine, not reasoned about: a module with the list 0, 25, 50, 75 and the box at 13 still ran 25 minutes apart; set to 87 it still ran 25 minutes apart; with no list at all, 13 ran 13.

How the list got there: the press that moves one batch's time wrote the module's whole list back onto it, and so did the day card's "work the day backwards", which does it for every module in the day at once. So one press of one button used to freeze the pace box of the entire line - which is why it looked like something was preventing it, rather than like a box that had quietly stopped working.

**5. What changed.** Three things, all small. A press that moves a single batch no longer writes a list onto a module that never had one - batch one of a module IS the module's start time, so writing that one number is the same answer the pace already gives, and leaving no list behind is what keeps the pace alive. A module that already carries a list has it re-spaced to the pace you type, with every batch keeping what you did to it by hand: each batch moves by the minutes the rhythm moved, so a batch you deliberately held off a collision is still held off by the same minutes after. (23 Sep 2026, added at v173: this re-spacing was unbounded in v172, and a batch that had been nudged EARLIER by hand carried a negative deviation that the smaller pace then added on, putting that batch in front of the one above it - the day drew both on the same minute. See the v173 entry above: a hold is now only ever a hold, and no batch is left at or behind the batch above it.) And the box now carries a live note under it, so a pace the oven cannot reach says so where you typed it instead of doing nothing.

**6. The other half of it, which is not a fault and now says so.** A module that takes one batch at a time cannot start two of them closer together than one batch lasts - that is your oven, not the app. Measured on your own "Mixing the dough in the tub": its cycle is 20 minutes, so typing 13 still runs the batches 20 minutes apart, and the note under the box says exactly that, names the 20 minutes, and names both ways to run closer: give the module a second production line under "How many production line you have", or switch on "Allow multiple production line" if those minutes are the dough's time and not the machine's. Measured live on your own frozen day: typing 13 turned that module's stored list from 1, 88, 175, 262 into 1, 14, 27, 40, and the day then ran 1, 21, 41, 61 - 20 minutes apart, its own cycle. Typing 40 ran the day 1, 41, 81, 121, and the note went back to being a plain explanation.

**7. The Your scenarios card is now a table.** One row for each saved day: the day's name, the pans a day it turns out, and how many modules it has, with the pencil at the end of its own row and Save changes and Save a copy below the table rather than mixed into it. The day the planner is currently plotting is filled in and carries a line down its left edge, and every name starts at the same place so the days read as a column.

**8. And one measurement that made the table worth having.** At a phone's width the card has 315 pixels to spend, and two headings were taking 170 of them - "PANS A DAY" on one line claimed 91 pixels to sit over a two-digit number and "MODULES" claimed 78 to sit over a one-digit one - which left the name 92 pixels. Every name you have is wider than 92, so every row wrapped, each to a different number of lines, and the card read as three ragged paragraphs. The heading is written over two lines now instead, which hands its slack back to the name: measured at 375 by 812 the name goes from 92 pixels to 138, all three rows come back to the same 59-pixel height instead of 59, 72 and 59, and every name starts at the same x. The mark that said which day was open used to be a dot typed in front of its name, which pushed that one name 17 pixels right of the other two; it is a line down the row's left edge now, which costs the row nothing.

**9. Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. Every tap still opens exactly what it opened before - a row opens that day, and the pencil beside it opens the rename card without opening the day as well. Your stored data was compared before and after: 13,888 characters, byte for byte the same. The one press in this version that does write to your day is the hand-over above, and it writes only what the chart was already showing - the same people, on the same rows, at the same jobs. Its own round trip was measured on your day, starting from your day as it stood and putting it back byte for byte afterwards: 13,888 characters before it and the same 13,888 after it. Your three saved days are untouched, and each still asks for the pans a day it always asked for - 24, 12 and 24 - and still carries the same 9, 8 and 8 modules. One reading in the table is worth naming, because it is not the same number: "Pans a day" in the table is what each day's line turns out, which is exactly the number the app already puts in a day's own summary line. For two of your days that is the 24 you asked for; for "My sister proposal 21/9/2026" the one tub passes 4 against the 12 you asked it for, and the shelf says 4 for the same reason its summary line already does - so the two screens cannot disagree.

**10. The delete button opened a question you could not see.** This was the fault you found, and it was not a wording problem or a small one: the question was being drawn underneath the card that asked it.

Both of these deletes are asked from inside a card - Delete this module from the module card, Delete this scenario from the card the pencil opens - and the card stays on the screen behind the question, which is deliberate, so you can see which module you are about to lose. What was wrong is the order the two are drawn in: the layer the question goes on sat at 60 and the layer of a card sat at 65, so the card was painted over the question every time.

Measured on your own day at a phone's width, not reasoned about. With the question open, the topmost thing at the question's own Delete button was an input belonging to the module card underneath, and the topmost thing at the question's own centre was a label from that same card. So the sentence could not be read, neither button could be pressed, and the only thing you saw was the screen dimming a second time with nothing appearing on it. Tapping anywhere did what the card underneath does instead, and the question sat there unanswered until something else happened to close the card, at which point it surfaced on its own - a delete box appearing when you had stopped asking for one. Both flows were the same fault, and both are measured fixed: at the question's Delete button the topmost thing is now the Delete button, at Cancel it is Cancel, and at the centre of the box it is the box.

The question's layer now sits at 72, above every card and above a toast, and only the app-password lock outranks it. Nothing about the flow changed otherwise: Cancel leaves the module and leaves the card open where it was, and Delete takes the module out and closes the card with it - both driven and watched on a live day, a module going from eight on the line to seven with the day answering without it, and a saved scenario going from three to two with the shelf redrawn and no name left standing on the card. And the question now names the module in quotes, as every other delete in the app names what it is about. Unquoted, a module called "The proofer again" read as one sentence with the name lost inside it: "Delete The proofer again?".

**11. And the row of buttons at the foot of a card had no shape at all.** The same report's other half, and a smaller thing, but the same kind of thing: a class used in 26 places across 12 files that never had a rule of its own.

Without one, its buttons flowed as plain inline blocks - flush against each other and hard against the left edge of

the card. Measured on the module card at 375 by 812: "Delete this module" and "Done" were 0 pixels apart, so a thumb reaching for Done was touching the button that deletes the module, with 88 pixels of dead card to their right. The scenario card was the same shape the other way up - its Delete stacked directly above its Save at exactly the same left edge - and both read as two buttons run together rather than as a row of choices.

The row is now one rule, written once: the card's own edge, 8 pixels between the buttons, and a line's worth of air above them. wrap is part of it rather than a detail, because the module's three buttons need 437 pixels and have 303 - a row too wide for its card has to go onto a second line, not off the card's edge. It is the shape the label printer's own sheet already carried in full and the shape three other cards had each patched in by hand, a little differently every time; those three patches are gone, because one answer in one place is what stops a fourth being added. Measured after, on the same card: Duplicate this module on a line of its own ending at the card's edge, then Delete this module from 94 to 264 and Done from 272 to 345 - 8 pixels apart, Done sitting where a thumb expects it at the card's own right edge.

**12. Every rule that stands on it was proved load-bearing, not assumed.** Nineteen faults were put back in all, and each was watched failing by its own name before being restored byte-identically.

For the hand-over: with the arrangement left unwritten the test fails reading that the day has stopped being able to say what it is, which is the fault put back; with a person written onto a module without the crew that carries it, the same test fails with your own three symptoms - five rows where there were four, renumbered 2, 3, 4 and 5; with the People box allowed to read a crew of zeroes as an arrangement, the hand-back test fails reading that a day which has not moved no longer says it is sharing them out; and with the card's note gated off, it fails reading that the card does not say what the press settles.

For the pace: with the re-spacing taken out the test fails reading "the stored list did not follow the pace she typed"; with the note about the oven taken out, "a pace the machine cannot reach is not explained"; with every batch pulled back onto the rhythm instead of keeping its own offset, "a batch held off by hand was pulled back onto the rhythm"; and with the press left writing a list again, "Mixing the dough in the tub came out of the press carrying a list of times, so its pace box is dead".

For the question: with the layer put back under the card it fails reading "under the card that asked it"; with it put above the app-password lock, "drawn over the app-password lock"; with the module's name left unquoted, "does not name the module in quotes"; with the destructive button put first, "answered by the safe button and then the destructive one"; with Cancel made to run the delete, "Cancel deleted the module"; and with the card left open behind the delete, "the card stayed open after the module went".

For the foot row: with the rule declared a second time it fails reading "declared 2 times"; with the buttons put back to inline blocks, "flow as inline blocks"; with the 8-pixel gap taken away, "sits flush against"; with the row back at the left edge, "not laid out against the card's own edge"; with wrap taken away, "cannot wrap"; with one card patching its own row by hand again, "lays its own foot row out by hand"; and with the foot's buttons reordered, "does not carry its three buttons". The two faults found while measuring the note on your own day are put back the same way: with the note always claiming the day is sharing them out it fails reading that the card claims a box that is not saying it, and with the note taken off the card altogether it fails reading that the card does not say what the press settles. The test suite is 1215 passing with none failing.

**13. No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

**23 Sep 2026 - engine v171 (no database step). Three things you asked for while looking at the day chart: a right press and hold moves a window's day about without opening anything, the clock balloon rides two inches clear of your pointer, and the batch card is brief. All inside More -> Scenario planner.**

**What you asked, in your own words.** First: "And can i drag the module window up/down, left/right by right click and hold? Dont let this action open up the card". Then, while that was being built: "put the clock balon 2 inches higher than. cursor". And: "the batch pop up, make it as brief as possible".

**1. A right press and hold now takes hold of a window's day and moves it, both ways at once.** Press the right button anywhere on either chart window, keep it down and move: the day comes with your hand, sideways and up and down together, exactly as if you had taken hold of the paper and pulled it. The window under your

hand moves and the other one comes with it sideways, so the two can never disagree about a minute - that part was already there and is untouched. Measured live at a phone's width (375 × 812): a right press on the modules' window and a 60-pixel drag moved its day from 0 to 60, and the people's window to the same 60. The window marks itself while your hand is down and unmarks the moment you let go.

**And it opens nothing, which was the other half of your sentence.** The gesture is the RIGHT button, and a browser fires no ordinary click at all for a right press, so a press for panning can never be read as a press for opening. The ruler's own press, which drags the clock reading, now ignores the right button too - one press, one gesture. Proven live, not assumed: a right press on a bar dragged the day and left the card shut, and a left click on that very same bar opened it. Your own hand is on the safe side of it either way - a finger cannot set the right button at all, so nothing here is reachable from your phone and the way you touch the chart is exactly as it was. The browser's own menu, which a right press would otherwise put on the screen in the middle of the gesture, is stopped over the chart.

**2. The clock balloon rides two inches clear of your pointer.** It used to sit 8 pixels above your finger, which on a phone is under it. It is lifted two CSS inches now - 192 pixels, and named in the unit you used so the number is one this can be held to rather than a feeling. Above your pointer wherever the window has the room, and where it does not, the same two inches BELOW it - which is the side that matters, because the clock strip you press with a finger is at the top of the modules' window, and a balloon that came to rest at the window's top would be lying on your hand. Measured live at 375 × 812: the full 192 pixels of clearance in both directions, and never nearer than 175 pixels anywhere on the chart. It is still held inside the window you are looking through rather than to the day, so a tall day cannot carry it off the screen.

**3. The batch card is brief, and its own reading is no longer cut off.** What left the card: the paragraph under each pair explaining what a move means in that module's own terms, and the second line that repeated the two times the line above it already reads out. What is on it now: one line of four readings - which batch, how it sits against the line, the two times, and what it costs your hands - then the two pairs, five minutes and one minute, and the way back where there is one. Measured at a phone's width, the card came down from 338-366 pixels tall to 265-288. The explanations are still true and are still there, on the press itself as its accessible name, and every press still answers in words when it lands.

**And one fault found while measuring that card, fixed here.** Four readings on one line ran off the card's right edge - "Batch 2 on the line 8:51 am -> 8:5" - with the end time, the clock you opened the card for, cut off. A reading is set never to be squeezed, on purpose, so a time can never be shrunk to nothing; a line one word too wide therefore overflows rather than shrinks. The card's readings now WRAP to a second line instead, which only ever happens where the line would otherwise be clipped. The same fault was possible on every card that shares that line, and it is fixed on all of them by the one change.

**Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. Every tap still opens exactly what it opened before. Your stored data was compared before and after: 13,561 characters, byte for byte the same, and your three saved days still read 24, 12 and 24 pans.

**Three rules now stand on it, and every one was proved rather than assumed.** One says a right press and hold drags a window both ways and that a right press over a bar opens nothing, while a left press on the same bar still does. One measures the balloon against the palest bar and pins the two inches of clearance, above or below, with the reason the window's own top is not good enough written into the test. One pins that a card's readings wrap and that they still refuse to be squeezed - the two settings asserted together, so neither can be changed alone. Each was proved by putting the fault back and watching it fail by name, then restored byte for byte: no downward fallback for the balloon read "a balloon with no room above her pointer sits -50px below it, not the two inches she asked for"; no upward preference read "the balloon sits 232px above her pointer, not the two inches she asked for"; and removing the wrap read "a card's readings are back on one line, so a row too wide for the card is clipped at its edge again". The test suite is 1209 passing with none failing.

**No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

CHANGELOG and the guide both carry all of this. Both PDFs rebuilt and read back with PyMuPDF, the app's own More screen reads Engine v171, and your phone is untouched by nothing here.

**23 Sep 2026 - engine v170 (no database step). The frame around every bar is drawn quietly now, so a block of work still reads as a block without every bar being a box drawn round it. Nothing else moved. All inside More -> Scenario planner.**

**What you reported, in your own words.** "the border look like over emphasized". You were right, and it was my own v169 line being too loud. Nothing about the day, the bars or your data was wrong - the frame was simply drawn heavier than it needed to be.

**What was too much.** The frame went in at a line one and a half pixels wide, drawn at full strength in the bar's own solid colour. Round a pale bar that is not an edge, it is a box drawn on: every bar on the chart was outlined that way, so the day read as a page of boxes rather than as a day with edges on it.

**What changed, and it is only the frame's own two numbers.** The line is one pixel wide rather than one and a half, and the whole frame is laid down at 55 parts in 100 rather than at full strength. The colour is untouched and is still the bar's own tone - the colour your hands already wear inside that bar, and, for a person, the colour of their card's stripe - so the frame still says at a glance which module a bar belongs to. Measured live on your own day at a phone's width: the frame reads the bar's own colour, a 1-pixel line with the layer at 0.55, on a batch bar, on a lane bar that shares its row, and on a person's time slot alike.

**Your B-numbers and your people's names are untouched,** and no line crosses a label, exactly as at v169: the frame is still drawn below them.

**One number softens every frame together.** The strength is declared once, on the chart's own wrapper beside the marker's transparency, so the modules' window and the people's window and all sixteen tones are all softened by the one edit and cannot be tuned apart. Lower it towards 0 for a whisper, raise it towards 1 for v169's line back again - it is one number either way, so say the word if this is still not the weight you want.

**Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. Every tap still opens exactly what it opened before, and the frame still takes no taps at all. Your stored numbers were compared before and after: 13,561 characters, byte for byte the same, and your three saved days still read 24, 12 and 24 pans.

**And the rule that stands on it was extended rather than left standing on the old numbers.** The frame test now carries your report in its own name, pins the 1-pixel line, pins the layer's strength and refuses a strength of 0 or 1 - so a later edit cannot quietly put the shout back - and refuses a second declaration of it. Each was proved by putting the fault back and watching it fail by name: the 1.5-pixel line back made it read "the frame is no longer a 1px line"; full strength back made it read "frame is 1: one at 1 is v169's shout back again"; and the strength declared twice made it read "the frame's strength is declared in more than one place, so the two windows can disagree". Each was restored exactly as it was afterwards. The tests are 1,201 passing with none failing.

**No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

**23 Sep 2026 - engine v169 (no database step). Every batch bar and every person's time slot now wears a frame in its own colour, so a block of work reads as a block and not just as a patch of colour on the paper. And the clock's lines have gone to the back of the day while every bar became see-through, so the ruler reaches inside them the way you asked instead of sitting on top of them. All inside More -> Scenario planner.**

**What you asked, in your own words.** First: "Can the batch and person time slot having a more highlighted frame?" Then, while that was being built: "its scale should be at background, the marker should be ontop, with some degree of transparency". Those two together are this release, and the second one turns round the way the ruler was drawn at v165 and v166.

**Every marker now wears a frame, and it is drawn in its own colour.** A batch bar and a person's time slot are the very same shape in the app - one rule frames both, so a batch and a slot cannot end up looking like two different things. The frame is the bar's own solid colour: the same colour the hands inside that bar already wear,

and, for a person, the same colour as the stripe down the side of their card. That matters more than it sounds - a frame in a colour of its own would be one more thing to keep in step, and it would drift the first time a tone changed. Nothing about the bar's shape moved for this: a batch bar is still 16 pixels tall and a person's slot still 11, measured, and the frame is a line laid inside the bar's edge rather than a border added round it. A border was tried and thrown away - it would have squeezed the inside of the bar, which is where your hands and the cycle shades are placed, and an inner shadow was no better because those same shades are painted over it.

**The ruler is at the back and the markers are over it, as you asked.** Since v165 the lines were drawn as a layer of their own on top of everything. They are now the paper's own ruling again - the surface the bars are drawn on - and every bar is a wash over it rather than a solid block. The bar is 72 parts in 100 opaque, which is the "some degree of transparency" you asked for: enough that the clock reads through it, not so much that the day looks faded.

**What that cost, honestly, and what was done about it.** A line seen through a bar loses about a quarter of its strength, because the same wash covers both the line and the paper beside it. Left alone, the ruler would have gone faint again - and faint is the fault you reported twice. So both lines were drawn heavier: the hour line at 60 parts in 100 of full strength where it was 34, and the finer grid at 45 where it was 20. Measured through one of your batch bars, the hour line now differs from the bar beside it by 29 parts in 255 and the grid by 22, against the 20 that your own v166 rule demands. The honest other side of that: on the bare paper between the bars the ruling now reads about twice as strong as it did at v168. That is the price of drawing it behind the markers, and it was paid deliberately.

**And no line crosses a label.** Your B-numbers and the name of the person on a stretch of their day are laid above the frame, so thickening the frame cannot put a line through a number you have to read.

**Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. Every tap still opens exactly what it opened before - a batch its Batch card, a cycle its own card, a person's stretch the hand-over card; the frame takes no taps at all. Your stored numbers were compared before and after: 13,561 characters, byte for byte the same, and your three saved days still read 24, 12 and 24 pans.

**Three rules now stand on it, and every one was proved rather than assumed.** One says the ruling is the paper's own background and that no second copy is drawn over the markers - two grids, one in the gaps and one across a bar, is how a reader ends up seeing two different grids. It also pins the transparency to one number declared once, so the modules window and the people's window can never be tuned apart. One says every batch bar and every person's slot wears a frame in the tone its own hands or card already wear, and that no bar grew a border or a shadow of its own. One is the arithmetic above: measured, not trusted, that both lines still differ from a marker by at least 20 parts in 255. Each was proved by putting the fault back and watching it fail by name, then restored exactly as it was. Five faults were put back in all, and one of them is the point of the whole release: with the grid's ink back at v166's strength, the test fails reading "differs from the marker beside it by only 9.8 parts in 255, which is invisible on a phone - the fault she reported twice at v165 and v166". The tests are 1,201 passing with none failing.

**No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

*Softened at v170:* the frame described above went in at a line one and a half pixels wide and at full strength, and you reported it as "the border look like over emphasized". It is now a one-pixel line laid down at 55 parts in 100, in the same colour as before. Read the v170 entry above for the rest.

**23 Sep 2026 - engine v168 (no database step). The extra slider between your two chart windows is gone, so each window's own sideways bar is the pan. The modules window is taller. The "People at once" row is out of the people's window, and an overlap is still shown in red. All inside More -> Scenario planner.**

**What you asked, in your own words.** First: "since the both windows have their own slider, additional slider is redundant. Remove that" - so the slider that sat between the two windows is removed. What you asked for when the two windows were built still stands and is untouched: the two windows stay in step, so any minute is in the same place on both, whichever one you drag. Then: "can you increase the modulle window further?" - the

modules window is taller. And: "im thinking of remove the people at once?", settled by "you can just show when there is overlapping highlight in red box, like previously have" - the row is out, and the overlap is still shown in red.

**The extra slider is gone, and the two windows still move as one.** Measured on your day at a phone's width, 375 by 812: nothing sits between the two windows any more but their own two edges - the chart is the modules' window and the people's window, in that order, with nothing in between. Dragging one window's day sideways moved the other to the same pixel, and the window under your finger took no write back at all, so a drag is never fought by the window that is following it. On the two windows being able to reach the same place: they could not quite, and it was one pixel. The ruler's last hour sits at the very end of the day, and its line was being drawn one pixel past it, which made the modules window 2,461 pixels wide against the people's 2,460. Both windows now read 2,460 and both pan to 2,139. That last hour's name is also written to the left of its own line now, where a ruler with an end on it has always put it, and it reads in full inside the day - measured, all 25 of the ruler's names sit inside the day, the last of them ending 3 pixels short of its end.

**The modules window is taller, and here is exactly what that buys.** It takes the larger half of the chart's ceiling now: 64 of the 84 parts where it had 50 of the 80, and 590 pixels where it had 470. On your phone at 812 tall this changes nothing you will see - your day's modules already fitted the window exactly (403 pixels of modules in a window that was already wide enough for them), so the taller window is room there rather than a difference on your screen. Where it shows is a shorter screen: at 667 tall the old window held 333 pixels and made your day scroll, while the new one holds 427, so your whole day sits in front of you with nothing to scroll. It is also the room the next two module rows will need as you add them.

**The "People at once" row is out of the people's window.** You were right that the day already says it without that row. An overlap is still shown in red, in the places that were already saying it: the red outline on the colliding bar in the people's window, the colliding stretch of that person's own row, the red line in their card, and the collisions written out in words under the diagram - "two jobs at once", or how many. Nothing about an overlap is hidden by the row going, and it is one line to bring back when there is a second pair of hands to weigh up.

**Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. No bar changed colour, and every tap still opens exactly what it opened before - a batch its Batch card, a cycle its own card, a person's stretch the hand-over card. Your stored data was compared before and after: 13,561 characters, byte for byte the same, and your three saved days still read 24, 12 and 24 pans.

**Four rules now stand on it, each proved load-bearing rather than assumed.** The ones from v167 still hold, with the slider's part of them narrowed to what is left: the two windows are adjacent with nothing between them, and a drag on either moves the other while the window under the finger takes no write back. Three are new. One asserts that the "People at once" row leaves nothing on the chart while a collision is still marked in red. One asserts the ruler's last name is drawn inside the day, and that the ruler's own track clips, so the two windows cannot disagree by even a single pixel. One asserts the modules window holds the larger half of the chart's ceiling, 64 of the 84. Every one was proved by putting the fault back and watching it fail by name, then restored byte-identically.

**No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

**23 Sep 2026 - engine v167 (no database step). The day chart is now two windows sharing one horizontal slider - the process above, the people below, one ruler's lines down both - and the blank space under your people's rows is gone. This release also carries two smaller faults you reported while it was being built.**

**What you asked for, in your own words.** First: "the window for person stay over size", which you confirmed means the empty space under the people's rows. Measured at a phone's width, 375 by 812: the chart was one window, fixed at 650 pixels tall, the day's own content was 473, and your people's rows ended at 457 - so 177 pixels of empty window sat under them. Then your proposal, which was better than a tweak: "instead of consider then one window, why not create 2 windows, and let the ruler sync in the 2 windows. As the business grows, the

persons will grows, processss might not, we need to have a better way to manage, what you say?" And on how the two should sit: "can the 2 window share the horizontal slider, place between the 2 windows, make the 2 windows as close as possible", with the slider going straight into the people's bars underneath it.

**What is on the screen now.** Two windows, one above the other, with a single slider lying between them and touching both - measured gap of 0 pixels either side. The upper window holds the clock and every module, batch and cycle; it scrolls up and down on its own when you add more modules. The lower window holds your people's rows and nothing else, and it scrolls on its own too, so as your people grow the second window grows and the first one does not have to. The ruler's lines reach down both. The slider between them is the only horizontal control: drag it, and both windows move together; drag either window's own day sideways, and the other follows and the slider follows with it. Measured on your day at 375 by 812: the slider reads a travel of 2,139 pixels, a drag on one window moved the other to the same pixel, and the window under your finger received no write back at all - the echo is dropped rather than sent, which is what would make a phone stutter mid-drag. On a day short enough to fit, the slider dims and cannot be dragged, because a control you can grab that does nothing reads as a fault.

**The blank space.** It is gone. The chart is no longer forced to a fixed 650 pixels; it is now exactly what its two windows need. Measured on your day: upper window 403, slider 20, lower window 70, totalling 493, with your people's last row sitting exactly on the window's floor - 0 pixels of empty window under it, where there were 177. The height ceiling from v157 survives as a ceiling: the chart is never taller than it was, it is only allowed to be shorter when your day is shorter.

**The first of the two smaller faults: a deleted saved day left its name on your card.** In your words: "why the deleted scenario stil listed?" You were right, and it was reproduced rather than explained. Deleting "My sister proposal 21/9/2026" left the card still offering "+ My sister proposal 21/9/2026 · Mix by hand, no chiller", dressed exactly like a day still saved, while your storage held only two. The cause was that the card offered each ready-made day by name whenever that name was not in your list, so deleting one put the name straight back. Now the card offers one plain row, "+ Add a ready-made day", which names nothing; opening it lists whichever ready-made days you do not already have, and one tap puts that day on your own list, where it is yours to change like any other. Proven live: after deleting that day the card named it nowhere, and the chooser handed back exactly that one day.

**The second: on a computer, a person's name card was cut.** Hovering a person's name on the chart opens a card of what they do that day. In the new lower window that card was taller than the window it opened into - measured 180 pixels of card inside a 69-pixel window, 96 pixels of it below the window's own bottom, with no scroll position that could show it whole. It is now laid on the screen rather than inside the window, so it reads in full wherever the name is, and it is pulled back from the screen's edges rather than running off them. The "People at once" row's card, which sits in the same window, is placed by the very same function, so the two rows cannot behave differently. Measured on a computer at 1280 by 900: both cards fully on the screen, with 8 pixels to spare at the foot.

**Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. No bar changed colour, and every tap still opens exactly what it opened before - a batch its Batch card, a cycle its own card, a person's stretch the hand-over card. Your stored data was compared before and after: 13,561 characters, byte for byte the same, and your three saved days still read 24, 12 and 24 pans.

**Three rules now stand on it, each proved load-bearing rather than assumed.** One asserts the two windows are adjacent with the slider between them, and that a drag on either moves the other and the slider while the window under the finger takes no write back. One asserts that a ready-made day you have deleted leaves no name standing on your card, with the offer row itself excluded from the check. One asserts that both a person's card and the tally row's card are placed beside their name and pulled back from the screen's edges. Every one was proved by putting the fault back and watching it fail by name, then restored byte-identically.

**No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

*Added at v168:* the single slider shared by the two windows, described above, has since been removed - each window now carries only its own sideways bar, and the two still move as one. Read the v168 note at the top of this file for what is on the screen now.

**23 Sep 2026 - engine v166 (no database step). The ruler's lines are now drawn strongly enough to be seen on a phone - including across your batches, your cycles and your people's occupied slots.**

**What you reported.** In your words: "At ver165, you still fail to reveal the ruler at person's time slot". You were right, and this release is that same fault finished rather than a new idea.

**What v165 got right, and what it got wrong.** v165 did draw the ruler over the bars - that part worked, and it is untouched. What it got wrong was the strength of the ink. The grid line was drawn at about one part in six of full strength, and over one of your slots it came out as a colour that differs from the slot by about ten parts in 255 - a hairline the eye cannot find on a phone in normal light. So the ruler was drawn, and it was invisible, which on a screen is the same as not being there. That is why you came back with the same words a second time.

**What changed, and it is only the ink.** Both lines are drawn in a strength measured against the bar they cross rather than against the cream paper between the bars - the hour line at about a third of full strength and the finer grid at about a fifth. The lines themselves are in the same places as before, at the same intervals for the scale you are reading at, and the ruler at the top of the chart still draws none of them over its own ticks. Nothing moved; the day simply reads as one ruled sheet now, so the edge of a marker has a line beside it wherever you look.

**And a line never crosses a label.** Your B-numbers on the bars, and the name of the person on a stretch of their day, are now laid over the ruling rather than under it. A gridline drawn through a B2 is a number you have to guess at, and the whole point of the ruling is to make the day easier to read, not harder.

**Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. No bar changed colour, and every tap is untouched - the ruling takes no taps at all. Your stored numbers were compared before and after and are byte for byte the same, and your three saved days still read 24, 12 and 24 pans.

**About the height of the planner window, which you also raised.** That is not in this release. You are right that it is a real thing, and what you proposed - two windows, one for the process and one for the people, sharing one ruler - is the honest fix rather than another tweak. It is written up as its own piece of work, so it is decided on its own rather than half-done here.

**No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

*Changed at v169:* the ruler's lines are no longer drawn over the bars, as described above. The ruling is the paper's own background again and every bar is a see-through wash over it, so the clock reads through a marker rather than across it. The goal is the same one named here - the clock must hold inside a slot you are working - reached the other way round. Read the v169 note at the top of this file.

**23 Sep 2026 - engine v165 (no database step). The ruler's lines are now drawn over your batches, your cycles and your people's occupied slots. Before this the lines stopped dead at the edge of every bar, so inside a slot you were working there was no ruler at all.**

**What you asked.** In your words: "i requested a ruler to draw into person's line time slot, maybe you cant visual my complaint". Asked how the lines should reach that slot, you said: "Just like the ruler draw over the batches and cycles". You were right that I had not pictured it, so I went and measured instead of answering.

**What was actually happening, and it was worse than "the lines are only on some rows".** Your person's row did carry the ruler's lines - it has since v160. But every marker, batch and cycle is a solid coloured bar, and a solid bar sits ON TOP of the lines behind it. So the ruler vanished wherever a bar was. Measured on your own day: of one module row's ruler lines, 5 fell inside your batches; of the slots on one person's row, 14 fell inside them. Every one of the 14 was covered. Pointing at a ruler line inside one of your slots, the topmost thing on that pixel was the slot itself. That is why the ruler looked present on an empty stretch of the day and absent wherever you were actually working.

**And it was the same on the module rows.** So the lines were not reaching over your batches and cycles either - what you were seeing was them showing in the gaps between bars, with your eye joining them up. This release makes that true for real rather than only in the gaps.

**What changed, and it is one thing.** The ruler is now a layer of its own, drawn OVER the bars - over the batches, over the cycles and over your people's occupied slots alike. The lines themselves are the same lines as before: the same hour line, the same step for the scale you are on, at the same positions. Nothing about the ruling moved; it is only no longer hidden. The whole day now reads as one ruled sheet, and a slot is ruled exactly like the empty day beside it.

**Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. No bar changed colour, and every tap on a bar is untouched - a batch, a cycle and a person's own stretch are still tapped and still open the same cards. Your three saved days were read before and after: One baker day, No fridge 1 person and My sister proposal 21/9/2026 each still on the target they had, stored numbers untouched.

**Two rules were proved load-bearing rather than assumed.** One test now stands on that there is exactly one ruling - if the lines were ever drawn both under the bars and over them, the translucent grid would read darker in the gaps than across a bar. Another stands on that the ruling is drawn at a height above the bars, that it lets every tap through, and that the ruler's own strip at the top turns it off so it cannot put lines between the ticks you read the clock off. Each was proved by putting the fault back: removing the ruling's height made the first fail, and putting a second ruling back on the track made the other fail.

**No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

*Changed at v169:* the lines described above are no longer drawn over the bars as a layer of their own. The ruling is the paper's own background again and every bar is a see-through wash over it, so the clock reads through a marker rather than across it. What has not changed is why: the lines must reach into a slot you are working, and the ruler's own strip at the top still draws none of them over its ticks. Read the v169 note at the top of this file.

**23 Sep 2026 - engine v164 (no database step). A number you type into the cycles of a module now stays where you put it. Before this, typing into a second box put the first box back to what it was when the card opened, while the first box still showed the number you had typed.**

**What you asked.** In your words: "why some parameter that i save can change by itself", and then, when I had guessed wrong about where: "it was on stretch and fold module, the cycle1,2 , loading, unloading. It is not like you explain". You were right and my first answer was not. I stopped explaining and went and did it on your own fold module until I could see the number move on its own.

**What was really happening, and it is exactly where you said.** Inside a module's card there is a list of cycles, and each cycle has its own boxes: the name, Minutes, Load, Unload. The card does not redraw itself while you type, and that is on purpose - it was your own instruction at v142, because a redraw would throw the box out from under your finger. But the code was keeping a copy of the whole cycle list from the moment the card opened, and every box wrote that old copy back. So the first box you used was still carrying the numbers as they were when the card opened, and the moment you touched a second box, the first box was put back to those old numbers. The screen kept showing what you had typed, the day behind it kept the old value, and the two disagreed. That is what "changing by itself" was.

**Measured on your own day, so it is not a theory.** Your fold module runs four cycles. Cycle 1's Minutes was 31; typing 40 into it stuck. Then touching cycle 2's Unload put cycle 1's Minutes straight back to 31, while the box under your finger still read 40. The same thing happened on one row: Load 5 stuck, then typing 44 into the Minutes beside it put Load back to 0 while its box still read 5. Renaming a cycle stuck only because nothing else was waiting to be undone.

**The fix, and it is small.** A box now changes only its own value, on the list as it stands at the moment you touch it, instead of writing back the copy from when the card opened. Nothing else in the card can be dragged back with it. The comment beside the code names your words and the exact numbers above, so the next person to touch it cannot put the copy back.

**Nothing of yours is rewritten.** Not one module, batch, start time, cycle or saved scenario changed, and nothing here blocks a sale or reads an order. Your three saved days were read before and after and compared byte for byte: One baker day, No fridge 1 person and My sister proposal 21/9/2026 each still make what they made, their stored numbers untouched.

**A test now stands on it.** It opens your fold module's card, types into cycle 1's Minutes and then cycle 2's Unload, and asserts the first number survives the second box - the exact fault. It also covers two boxes on the same row, and a cycle renamed after a number was typed. Proved load-bearing: with the fix taken back out it fails, reading "the second box she touched put cycle 1's Minutes back to where it was when the card opened".

**No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

**23 Sep 2026 - engine v163 (no database step). The clock above your people's rows has been taken back out. The clock is drawn once again, at the top of the chart where it has always been, and the ruler's lines still run down through your people's rows exactly as they did.**

**What you asked.** In your words: "i dont want the clock, i just want the ruler to draw into person row". That was said after v162 had put a second clock above your people's rows, so this release takes that second clock back out. Nothing else about the chart moves.

**The ruler's lines were never the thing missing, and they are still there.** The lines that run down your people's rows came in v160 and they are untouched: every row of the day carries the faint grid, so a marker's edge still lands on a line you can follow up the chart to the top. That is the part you asked for, and it is the part that stays.

**What comes back.** The chart window is about 35 pixels taller again, because the block pinned at the foot of the chart no longer carries a clock row inside it. On your own day at a phone's width the foot block goes back from 105 pixels to 70.

**One rule is kept, and it is worth naming.** There is now a test whose whole job is that the clock is drawn once and that no clock is ever drawn above the people's rows again. It was written because the second clock came from a shared builder, so putting it back is one line - and a thing that can come back by accident should have something standing on it.

**What has not moved.** Not one module, batch, start time, cycle or saved scenario of yours changed, and nothing here blocks a sale or reads an order. Your three saved days are the same three days, each on the target it had, their stored numbers byte for byte the same.

**No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

The operations guide (section 23) carries the same change, and the tests are 1188 passing with none failing.

**23 Sep 2026 - engine v162 (no database step). The times along the clock are written a second time, standing directly above your people's rows, so a marker at the foot of the day can be read without tracing it 368 pixels up to the top.**

**What you asked.** In your words: "why cant you draw the ruler into person's lines?" I went and measured it rather than answering from memory, because the answer turned out to be "half of it already is". The ruler's *lines* have reached your people's rows since v160 - every row of the day carries the faint grid, so a marker's edge already lands on a line you can follow up the chart. What had not reached them is the *clock*: the times written along the ruler, which lived only at the top of the panel. On your own day the top of the chart stood 368 pixels above your people's rows - the whole height of your eight module rows - so reading the clock and reading a marker meant looking in two places far apart. Asked which of the two ways to close that gap you wanted, you chose this one: "yes, draw the clock above the people's rows".

**The clock is now drawn twice.** Once at the top of the panel where it has always been, and once more standing directly above the people's rows, inside the block that is already pinned to the foot of the chart - so it rides the same pin as the rows it was asked for, and the day scrolling can never leave it behind. The foot one sits on the first person's row with no gap at all between them, measured on your own day.

**The two clocks cannot disagree, because they are the same clock.** They are built by one and the same

function, from the same minute, with the same step and the same labels - so the top of the chart and the foot of it can never say two different things about where a minute is. Measured on your day at your usual reading: 97 ticks each, every label the same, and the 8:00 am tick at exactly the same pixel on both, whether the day is scrolled or not.

**And the foot clock is read exactly as the top one is.** Point at it, or drag your finger along it, and the same hairline follows with the same reading in the same words and to the same minute, snapped the same way. It wears a crosshair under a computer's pointer, as the top one does. Two clocks that look alike now behave alike; there is no second way to work the foot one.

**What it cost, measured rather than promised.** About 35 pixels of the chart window. On your own day at a phone's width the block at the foot grew from 70 pixels to 105, and your day still fits inside the window with nothing to scroll - so those 35 pixels came out of empty space under your bars, not out of your day. At a wide screen the window is 720 tall holding 705 of content, and the day still fits.

**One fault was found by testing on the real app, not by a test.** The code that finds a clock's own track asked the wrong kind of list for the answer, and a real browser's list does not have that question to ask - so the screen failed to open at all with "This screen couldn't load", while every test stayed green, because the test's stand-in list is more forgiving than a real one. That is the standing lesson about stand-ins, and it is written into the code beside the line so the next person to touch it cannot repeat it. Fixed, and the tests now cover the wiring that hid it.

**What has not moved.** Not one module, batch, start time, cycle or saved scenario of yours changed, and nothing here blocks a sale or reads an order. Your three saved days were compared before and after: One baker day makes 24 pans still, No fridge, 1 person makes 24 pans still, and My sister proposal 21/9/2026 makes 4 pans still, each on the target it had, their stored numbers byte for byte the same.

**What is not in it, so you are not left looking for it.** The other way to reach the markers - moving the clock times around rather than drawing a second one - stays available if you ever want it instead. And the per-batch person, letting one batch go to somebody else while its neighbours stay, is still a release of its own because it needs a stored field and a one-time step in Supabase.

**No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

The operations guide (section 23) carries the same change, and the tests are 1188 passing with none failing.

**23 Sep 2026 - engine v161 (no database step). Your people's rows are the app's own white now, and a tap on one of their markers moves THAT STRETCH and nothing else - the batch it belongs to is untouched.**

**What you asked, and it was two things.** In your words: "I want to have white background for person's line", and "The reassign job to next person is not whole day, it is that slot only". Asked what one tap should take with it, you told me: "we dont change the batch. Say a labour slot belongs to person1, clicking that slot, will offer to swap it to others, this basically to balance work load".

**Your people's rows take the app's own white.** They wore the cream band that every soft strip in the app wears, which reads as a highlight sitting under the markers rather than as the plain paper the day is read against. They are now the same white as the panel behind them and as every module's name cell, so the foot of the chart is the same paper as the rest of it. What still marks the block as the foot of the day is untouched: its own line above it, its own shadow, and the hairline between one person's row and the next.

**A tap now moves one stretch, not the whole job.** Until this release, tapping a marker handed over the whole module on that line - every batch of it. Your fold loop folds three times in one batch, so a tap anywhere near it took three of your markers with it. That is not what you asked for and it is not what balancing a day needs. What changes hands now is the single stretch your finger is on: the one batch, and which part of that batch. Its neighbours stay exactly where they were, and the batch itself is never touched.

**The card says so before you press it, with the number.** If the batch you tapped has more than one stretch on that person's row, the card counts them and says only the one you tapped moves. You are told what is about to

happen rather than finding out from the chart afterwards.

**And what you can hand a stretch to.** The list is the other people who have a row on this day, named as you have named them - plus "whoever is free". That last one is not a person; it is what the day does when nobody is named, and without it a stretch on a module nobody was put on could be handed to somebody and could never be handed back. Pick it and the press under the list changes to say so, before you press it.

**Your module's own card keeps up.** A hand-over is now one stretch of a module rather than the whole of it, so "Who is at this module" is the module's default and no longer the whole answer. The card now carries a note naming what has been handed on, and to whom, with the count - so a card reading Person 2 while a marker of that module sits on Person 1's row cannot happen. On a day where you have handed nothing over, there is no note at all.

**One thing the earlier note got wrong, said plainly.** v160's entry said that moving the person on one batch of a module would need a new stored field on the module and a one-time step in Supabase. The stored field was indeed needed - it is there now - and the Supabase step was not. Your planner travels inside the app's own settings record, so nothing had to be run.

**Nothing of yours is rewritten.** The only thing that changes is the one stretch you hand over with your own press. Your day was read before and after: One baker day still makes 24 pans, and its one person's row still carries the same 32 stretches across the same clock. No module, no batch, no start time and no cycle moves by being read, and there is nothing to run in Supabase.

**Measured on the real app rather than assumed.** A stretch was tapped on your own day and handed over: the toast named the module and the clock, the marker left one person's row and arrived on the other, and the module's other three batches stayed where they were. Tapping in a gap still opens that person's own card, exactly as it always did.

**Measured rather than promised.** The suite is 1187 passing with none failing. The two tests that stood on the module-level move are now three that stand on the single stretch - that only the one key is written and the module's own person, its crew and its times are untouched; that a second hand-over keeps the first; that a stretch handed back to whoever is free survives as its own answer; and that the chart puts exactly the tapped stretch on the other row. The app's own More screen reads Engine v161.

**23 Sep 2026 - engine v160 (no database step). In the Scenario planner, the ruler's lines now carry on down through the people's markers, and a tap on a stretch of somebody's day hands that job to another of your people.**

**What you asked, and it was two things.** In your words: "can the ruler extend down to peoples marker area?", and "can the personX marker be click to change it job to personY, by a drop down person selector". Asked which way you wanted the ruler to reach the markers, you chose **a faint grid down every row** - the clock itself stays where it has always been, at the top. Asked what the tap should do, you told me: "click on the person's occupied time slot, a drop down list, list the other people available".

**The ruler's lines reach the markers.** Every row of the day has always drawn a faint line on the hour, so a marker could be traced up to an hour and no closer. That grid now carries on down through the module rows and through the people's rows, so the edge of a marker always lands on a line you can follow up the chart to the clock. On your day the lines are 36 pixels apart at the widest reading, 24 at the standard one, and 12 and 16 at the two close ones - so the grid you read a marker against is the ruler's own step everywhere the ruler can be read at all.

**Why the grid has a step of its own.** At the closest reading a single minute is 3.2 pixels wide. A line every 3.2 pixels is not a grid, it is a wash of grey. So the grid takes the smallest step that is both at least the ruler's own - which is what makes every grid line also a tick on the ruler, so the two can never disagree about where a minute is - and at least twelve pixels of your day. The clock strip at the top keeps its own ticks and is untouched; a module you have switched off keeps the grid rather than losing the only thing its dimmed bars are read against.

**It costs nothing on the screen.** The grid is painted, not drawn one line at a time, so your day is exactly as fast to open and to scroll as it was - which matters, because the ruler alone draws one tick for every minute at the

closest reading.

**Tap a stretch of someone's day and hand it over.** Tap a person's marker on the timeline and a small card opens naming the job it is about - the module, its line where it has one, and the clock it runs from and to, in the same words the marker says under your pointer. Under it is a list of the other people on this day, and one press - "Move it to ...", named with whoever you picked - hands that job over. The two rows swap it the moment you press, and anything that now collides is drawn red, exactly as before.

**What the card tells you before you press it.** A module carries one person for all of its batches, so if the module you tapped runs four batches on that line, all four move together - and the card says so, with the number, rather than letting you find out afterwards. The list offers only people who have a row on this chart, because somebody who is not standing anywhere on your day is not an answer the day can give. If you have only one person on the whole day, the card tells you that instead of opening an empty list.

**The card goes with the job.** The moment you press, the job leaves that person's row and the card closes with it, so what is left on the screen is the day as it now stands. A tap on the marker again - now on the other person's row - opens a fresh card naming whoever holds it. A card left standing would have gone on offering to move a job off somebody who no longer has it, and the press under it would have done nothing.

**The tap is read by the minute under your finger, not by the marker.** A person's marker is 11 pixels tall in a 34-pixel row, and at the widest reading one minute of work is 1.2 pixels wide - so a fingertip lands beside a job far more often than on it. The tap takes the whole height of the row, and it takes the nearest stretch of work within about six pixels, which is what makes the narrow end of the scale usable at all. A tap where that person is not working is not swallowed: it still opens the person's own card, which is what every tap on a person's row did before.

**Nothing of yours is rewritten.** The only thing that changes is the one job you hand over with your own press. No module, no batch, no start time, no cycle and no saved scenario moves by being read, and there is nothing to run in Supabase - the whole of this release is drawn from the planner you already have on your phone.

**What is not in it, so you are not left looking for it.** Moving the person on one batch of a module, leaving its other batches where they are, cannot be done without a new stored field on the module and a one-time step in Supabase. It is a release of its own whenever you want it. And a clock repeated above the people's rows was the other way to reach the markers - you chose the grid, and the other way stays available.

**Measured rather than promised.** The grid's spacing at each of the four readings, its reach and the card's own list are all covered by tests, as is the card closing the moment the job moves: the suite is 1186 passing with none failing, six tests new for this release and one of those standing on the card going with the job. The app's own More screen reads Engine v160.

**23 Sep 2026 - engine v159 (no database step). Every card that opens over a screen can now be pushed aside by its own title bar, and the Scenario planner's chart window is one fixed height that no longer changes as you add people.**

**What you asked.** In your words: "whenni click on batch to adjust the start time the pop up windows should be allow to drag" - and then, while I was working on it: "the chart window became shorter and shorter as no. of people grows. the window height should be fix".

**What you settled, in three answers.** You asked about the batch card and told me it should be **every** card, not only that one. You told me the position should **always reset** - a card opens where cards have always opened, and nothing is remembered. And you asked for **a faint grip line** on the title bar so it reads as something to hold rather than something to discover.

**The batch card sat over the very bars it moves.** The card that adjusts a batch's start time opens over the day it is about, and on a phone it covers the bars you are deciding between. That is what the drag is for: push it down out of the way, keep working, and it stays where you put it until you close it.

**It is every card, not only the batch one.** All of the cards that open over a screen share one title bar, so all of them now move the same way: the batch clock, a module, a person, an order, a customer, a product, a supplier,

a unit and the rest. A card that looked the same but refused to move is a difference you would only find by trying it twice.

**A card always opens where cards have always opened.** The drag is a way to push a card aside while you work - it is not a setting and it is not remembered. Open the next card and it is where every card has always been.

**The grip is what says so.** There is a short faint bar near the top of every card's title bar, and the pointer over that bar becomes a hand. Measured, the bar is 34 pixels wide.

**A card is held inside the screen, and that is deliberate.** A card can be pushed until its own edge comes close to the edge of the screen, and no further. This matters more than it sounds. A card that could be dragged past the edge would hand the screen behind it something to scroll that it never had - and you could then scroll your own close button off the top, where no finger reaches it. Holding **the whole card** inside the screen means no button on it can ever be parked out of reach. On the phone a card is already nearly the full width, so what you get is vertical travel, which is the direction you want.

**Measured rather than promised.** On your One baker day the batch card for the mixing module is 337 pixels tall. Dragged down by 300, it moved by exactly 300 and its whole box stayed on the screen, and the screen behind it gained nothing to scroll. Dragged by 900 - far further than a finger can reach - it moved by 8 pixels and stopped, because 8 pixels is all the room it had. A module card, 784 pixels tall, moves 8 pixels; a person card, dragged 2000 pixels across and down, stopped with its whole box still inside the window; a wide card on a phone, 347 pixels wide in a 375-pixel screen, moves 8 pixels sideways, which is all the room it has. A drag on the card's close button is a press on that button and nothing else: measured, the card does not move and the cross still closes it.

**The buttons keep working after a drag.** Pressing the plus-five-minutes button on a batch card that has been dragged aside moves the batch and leaves the card where you put it - the card redraws its contents without going back to the top.

**A drag on the day itself still scrolls the day.** The title bar asks to be handled by hand; nothing else on the screen does. Dragging the chart - or the page - moves the page exactly as it did before, and the page does not move while you are dragging a card.

**The chart window is one fixed height now.** This is the second thing you asked for. The panel showing the day used to be a rubber band: its height was whatever the day's rows added up to, held between a floor and a cap. Measured at 375 pixels with one person it was 473 pixels tall and with a taller people list it was 648 - so the window grew as you added people, and once it had grown to its cap the pinned people list at its foot carried on growing into the days above it. At the same phone size your eight-module day left eight rows of modules visible with one person and four with a crowd. The panel is now **one height** - 80 percent of your screen, up to 760 pixels - and it does not change whatever you add. On your phone it is 649 pixels at 375 by 812 and 720 pixels at 1280 by 900. Your own eight-module day draws 473 pixels inside it, so **all eight module rows are visible at once**, and even your tallest people arrangement - seven person rows, which draws 648 pixels - still fits in the window with nothing to scroll.

**Nothing of yours was rewritten.** Your three saved days were compared byte for byte before and after this release: One baker day, No fridge 1 person and My sister proposal 21/9/2026 still stand on the targets they had, and not one module, batch, start time, cycle or saved scenario changed. Nothing here blocks a sale and nothing here reads an order.

**No database step.** The whole of this is drawn from the scenario already on your phone, so there is nothing to run in Supabase.

**22 Sep 2026 - engine v158 (no database step).** The Scenario planner's people rows come down to one line each and hold still, everything about a person moves into a tip on their own name, the "PEOPLE" heading is gone, and the clock ruler now reaches a single minute where you are zoomed in the closest.

**What you asked.** In your words: "can i have all the peoples cards noted in tool tips as well?", "i dont need the PEOPLE title", and "make the ruler resolution to 1min". Every one of those is below. Where a thing left the

screen you told me where it should go instead: every person's own notes into the tip on their own name, and the ruler stepped so a minute is drawn where a minute can be drawn.

**The people rows are one line each, and their height can no longer move.** A person's row used to print its own notes as text on the row - how much work they are carrying, how many places they have to be, and a red line when two of their jobs collide. That made it the only part of the chart taller than the bars beside it, and the only part that changed height as you worked: measured at 375 pixels, a row was 56 pixels on a clean day and 70 pixels when someone had a collision, so the whole strip measured 91 pixels one moment and 105 the next.

**The strip sits pinned at the foot of the window,** which is what you asked for at v154 - and a pinned block that grows and shrinks as you scroll is a block that moves the very thing you are reading it against. The notes have left the row, so a person's row is **35 pixels** now, **on a clean day and on a day with a collision alike**, and the strip measures a flat 70 pixels either way. Those 35 pixels are what the row is: the same name cell the modules wear, and nothing else.

**Everything about a person is in the tip on their own name.** Point at a person's name on a computer and their own box opens beside it, carrying all of it: who they are (the whole label, including " (with 2, 3)", which the row shortens), how much work they are carrying, how many places they have to be, the red collision line in the words it already used, and **What they do today** - their jobs, each with its clock, up to six of them and then a count of the rest. It is the same list their own card carries, from the same builder, so the tip and the card cannot drift apart and say different things. The tip opens from the **title** alone, exactly as the module tips do since v156: pointing anywhere else on the row does nothing, and the tint that used to follow the pointer across the whole row is gone. That was your own v156 instruction - "only the note should shot as a too tip" - now applied to the last rows on the chart that had not had it.

**Told rather than hidden: the tip is bottom-anchored, because the last row sits on the bottom edge.** A module tip is centred on its row, which is what keeps a first or a last module's box off the edge of the window. The people strip is different - its last row **is** the bottom edge - so a centred box would be cut in half. A person's tip now sits on its row's own bottom line and grows **upward** into the day. Measured at the window's tightest, on a person with a collision and six jobs, the tallest possible tip is 196 pixels tall and it clears the clock strip above it with room to spare.

**The widest-person trap, measured rather than promised.** A person's tip is taller than a module's, so its lines have to be allowed to fold - and a folding box inside a 156-pixel name cell collapses to the width of its widest single word: measured live at 66 pixels wide and 923 pixels tall, one word to a line. The box is now told to be as wide as its own content, and the rule carries a note saying why, because the fault looks like a styling detail and is not. Measured after: 272 by 180 pixels, sitting on its row's bottom line, inside the window.

**The "PEOPLE" heading is gone.** It named what every row under it already names - each one wears a person glyph - so the strip is now its rows with nothing over them. The strip keeps its line and its shadow, so it still reads as the foot of the day.

**The tally row's three facts are in its own tip, and on a phone they are unreachable.** The **People at once** row is the one you asked for: the most hands the day needs at once, the busiest stretch by name, and the minutes paid for twice. Those three facts are printed nowhere else in the app, and that row has no tap of its own - so once they are in a tip, a phone has no gesture that can open them. **You chose this, with the consequence in front of you** ("tip only, no card"), and it is written here rather than left to be found: on your phone those three facts are now unreadable. On a computer nothing is lost. Giving that row its own card is one line whenever you want it, and this note is the place that says so.

**The collision is not lost by the row giving up its red line.** Under the day there is still a note for every collision, in full: who it is, **two jobs at once** with the minute it happens, both jobs named on lines of their own, and the sentence saying what to do about it - move one along the day, or combine with another person and accept it. The red outline on the bar itself is unchanged. So the row's red line leaving costs nothing you were reading.

**The ruler now reaches a minute where a minute can be drawn.** The day is drawn at four scales, from 1.2 to 3.2 pixels to a minute, and a minute hairline is physically undrawable at the two wide ones - so the tick step

follows the scale, which is what you chose ("a minute where it can be drawn"). **Wide** ticks every 30 minutes, 36 pixels apart, which is the ruler you had. **Standard** ticks every 15 minutes, 24 pixels apart. **Close** ticks every 5 minutes, 12 pixels apart. And **Closest** ticks every **single minute**, 3.2 pixels apart. The hour is still solid and labelled at every one of the four, and the half hour is still the dashed tick it always was - the one you judge a 15-minute pass by. Everything finer is a new thin hairline at reduced weight, so a ruler at Standard does not read as a ruler made of dashes. The count is honest: at Closest your 24-hour window draws 1441 ticks, and a full redraw takes 2.8 milliseconds against 2.3 at Wide - half a millisecond for the lot, so the ruler is drawn as elements rather than as a painted pattern.

**And the cursor reads the exact minute.** The hairline that follows your pointer down the day used to snap to the five minutes a batch's buttons step by. That was stricter than the screen it reads: a batch can be nudged by **one** minute, so the cursor could not name a time the chart was already able to set. It reads to the minute now - measured, a pointer 101 minutes into the day reads **5:41 am** where it used to read 5:40 am - and it still refuses to name a time outside your day, and still hangs its label to the left at the far end of the evening.

**What has not moved, measured rather than promised.** Your three saved days were compared before and after: **One baker day** still reads 24 pans across 8 modules, **No fridge, 1 person** still reads 24 pans, and **My sister proposal 21/9/2026** still reads 4 pans, each with the target it had. Their stored numbers were compared byte for byte and are untouched - the whole of this release is drawn from the scenario already on your phone. Nothing here blocks a sale and nothing here reads an order.

**No database step.** Not one stored field is added or changed, so there is nothing to run in Supabase.

CHANGELOG and the guide (section 23) both carry all of this, both PDFs rebuilt and read back, the app's own More screen reads Engine v158, and the tests are 1174 passing with none failing - eight of them new: that no person row prints its own notes and the tip carries them all, that the tip and the person's own card agree word for word, that a collision changes nothing about a row's shape, that the tip is a sibling of the name and not inside it, that the two tip rules both survive as file text, that the tally's three facts are in its tip and off its row, that the "PEOPLE" heading is gone, and that the ruler's step follows the scale at all four stops.

**22 Sep 2026 - engine v157 (no database step).** The Scenario planner's list of controls is much shorter and the modules window is taller: the module tags have moved into the tip, the scale is two presses around the stop it is on, the people are one box, and "The line" has left the row for the module's own card.

**What you asked.** In your words: "i want to make the modules window taller. The module tag, can you put them in the tip tips? Make the scale with+ -, People: make it drop down option, The line: this name is very unclear,, we just need the button, WORK BACKWARD. The idea is to make this list shorter and make room for a bigger modules windows." Every one of those six is below, and where a thing left the screen you told me where it should go instead: every tag into the tip, the line's own settings into the module's card, and the two paragraphs above the chart removed.

**Two paragraphs left the day.** The two sentences that used to stand between the "The day" heading and the first bar are gone: the one teaching how to read the chart, and the one naming the moment your day hangs from. At 375 pixels they measured 330 pixels between them, 210 and 120, and they were the whole of what you scrolled past before you reached a bar. With them gone the first bar sits 278 pixels below the heading where it sat 707. Nothing is lost by it: how to read the chart is section 23 of the operations guide, and the button that moves the day now says its own name in the row under the chart, so the move is still findable without a sentence pointing at it.

**The modules window is taller, and now it fits.** The window has a floor of 500 pixels and a cap of 760 where before it had only a cap of 560: the floor is what makes it a window rather than a box that resizes itself every time you edit a module, and the raised cap is what lets a long day use the room the rest of this release gave back. On your seeded day at 375 pixels the window is 616 pixels tall where it was 536, and the day's own drawing is 616 - so before this release the window hid 80 pixels of your day and now it hides none of it.

**The tags are off the row and into the tip.** Every tag a row used to wear is gone from the row: how many of a

module you have, how many of its lots are in it at once, whether it waits above, whether it runs itself, and whether you have asked for more passes than a day holds. Point at the module's own title and the tags are the first line of the box, joined with a dot, with the two notes under them. Measured at 1280 wide on your day, the box is 42.6 pixels for the two notes alone and 58.9 with the tag line, and all nine boxes measured inside the panel left and right.

**The tip is now taller than the row it belongs to, and here is why that is all right.** Three lines of tip is 58.9 pixels against a 45-pixel row, so the box no longer fits inside its own row the way v156's did. What keeps it inside the panel is that it is centred on its row: half of it goes up into the clock strip's row above your first module, and half of it goes down into the rows below your last one. Every tip was measured inside the panel at both ends of the day, and nothing on the chart moves while one is open.

**The scale is two presses.** The four stops are kept exactly as they were, with the same pixels behind them: Wide, Standard, Close, Closest. What changed is that they are no longer four buttons. There is a round minus and a round plus, and the stop you are standing on is the word between them, so you can see where the dial is without opening anything. At either end of the dial the press that would go past the end is off rather than doing nothing when you tap it.

**The people are one box.** Three chips became one drop-down. What the box says when it is closed is the arrangement in force right now - Sharing them out, One a module, One to a line, Your own, or Combined when you have combined two people yourself - so the box names what it has done rather than hiding it. The four lines inside it are the arrangement in force and the two things it can do to your day, One a module (a person for each job, one job each) and Share them out (as few hands as can cover the day), and Combine two people on the end.

**"The line" left the row, and its name with it.** You said the name was unclear and that its place is the module's own card, and it is there now. The control row reads Scale, People, Walk the day and nothing else. On the module's card, under the words **Waiting on the module above**, is the same sentence the old card opened with - which of your modules wait on the one above, by name, and which keep the times you placed - and the same two presses it always offered, Chain every module and Take the waiting off. One module's own three answers, As the one above finishes, Never before the one above finishes and Its own time, are untouched and still on that same card, which is where they have been since v154.

**The button alone.** You said we only need the button, so the label that used to stand beside it is gone. It read "Your day backwards · first batch out 9:09 am", which at 375 pixels wrapped to two lines: 54 pixels of a phone screen spent explaining a button that already says what it does. What is left is the button, **Work the day backwards**, in the row under the chart.

**What the row came back to.** At 375 pixels your control row is 222 pixels tall where it was 305, so 83 pixels came back to the chart. Together with the two paragraphs that left and the taller window, the run from the "The day" heading to the first bar is 278 pixels where it was 707, and the day itself is not scrolled to at all: the window is 616 and the drawing is 616.

**Nothing on a phone is left without a way to read it.** A tip opens where there is a pointer that can hover at things, and a phone has not got one - measured at 375 pixels with a touch phone emulated, no tip opens at all. Everything a tip carries, the tag line and the two notes, is written out word for word at the top of the module's own card, in the row's own tag colours, and that card is what a tap on the row has always opened. There is nothing new to learn and nothing new to find.

**What has not moved, measured rather than promised.** Your three saved days were compared before and after this release. One baker day still reads 24 pans across 8 modules, its first batch out at 9:09 am and its day finishing at 1:30 pm. No fridge, 1 person still reads 24 pans. My sister proposal 21/9/2026 still reads 4 pans. Not one module, batch, start time, cycle or saved scenario of yours changed in this release. Nothing here blocks a sale and nothing here reads an order.

**No database step.** The whole of this is drawn from the scenario already on your phone, so there is nothing to run in Supabase.

The app's own More screen reads Engine v157. The tests are 1166 passing with none failing, eight of them new: that no tag is left on any module row and that every tag the tip carries is the same tag the module's card carries word for word, that a module you have two of is carried the same way, that the taller tip is still held inside the panel it is drawn in, that the scale is exactly two presses around the name of the stop it is on and that each press is off at its end of the dial, that the people are one drop-down whose closed text is the arrangement in force, that the line's own group has left the control row for the module's card, that the window is taller and wider-capped and the two paragraphs above it are gone, and that the day-backwards group is the button and nothing else.

**22 Sep 2026 - engine v156 (no database step). The module's notes now open as a tip over the day when you point at its title, and the row keeps its height while you read them.**

**What you asked.** In your words: "i prefer not to expend the module with mouse hoover. Remain the height. When mouse hoover to the module title, only the note should shot as a too tip." That is a correction to v155, and it is a fair one: v155 brought the notes back by making the row taller, which moves every row below it the moment you point at it, so the bar you were reading is no longer where you left it. Reading a number should never move the thing you were reading it against.

**The tip, and where it comes from.** Point at a module's own title and its two notes open in a small box beside the name: the time the module starts and how many batches it runs, then what a batch costs in minutes and in your own hands. It is drawn over the day rather than in it, so nothing on the chart moves while it is open - not the row it belongs to, and not one row under it. Point away and it is gone.

**Measured rather than promised.** On your One baker day, with the tip open on the mixing module, the row reads 46 pixels before and 46 pixels after, the row underneath starts on the same pixel it did, and the day's own total is 527 pixels either way. The box is 43 pixels tall, which is inside the row it belongs to - that is deliberate, because it means a tip on the first or the last module of a long day cannot be cut off by the edge of the panel. The whole box measured inside the panel on the first module and on the last one.

**Only on a computer, and only on the title.** The tip opens where there is a pointer that can hover at things, which a phone has not got: on your phone pointing is a finger and a finger is a tap. There the notes are written out word for word at the top of the module's own card, which is what a tap on the row already opens, and that is unchanged from v155. Pointing at the row anywhere other than the title does nothing, because you asked for the note and not for the row to react.

**What has not moved.** Everything v155 did to the row is kept: a row is still exactly as tall as the bars it draws, the name still takes the row's first line with the badges on the line beneath, the name column is still 156 pixels, and your whole day still fits the window. Your three saved days were compared byte for byte before and after: One baker day still reads 24 pans across 8 modules with its first batch out at 9:09 am and its day finishing at 1:30 pm, No fridge, 1 person still reads 24 pans, and My sister proposal 21/9/2026 still reads 4 pans. Nothing here blocks a sale and nothing here reads an order.

**No database step.** The tip is drawn from the scenario already on your phone, so there is nothing to run in Supabase.

This change history and the operations guide (section 23) both carry all of this, both PDFs rebuilt, the app's own More screen reads Engine v156, and the tests are 1158 passing with none failing - one of them new: the tip is off the row, laid out absolutely so the row cannot grow, centred on the row it belongs to, opened by the title alone, and its two lines are held to a height that fits inside a row.

**22 Sep 2026 - engine v155 (no database step). Every module row is now as tall as its own bars, so your whole day fits the window with no scrolling; the two note lines come off the row and come back when you point at it on a computer, and they are always written out on the module's own card.**

**What you asked, and you were right.** In your words: "The window for scrolling module became too small, why not reduce the height of each module, the notes can just shown up upon mouse hoover." You were right that the rows were too tall, and the cause was not the bars. On your One baker day the eight module rows measured 912 pixels of drawing inside a window about 536 pixels tall, so only about four of them could be seen at once and the rest had to be scrolled to. The bars themselves need only 45 pixels; it was the words beside them that

made the rows tall. The name column is 138 pixels wide, and a name like The rests and the stretch and folds wraps to seven lines in it, with each of the two note lines wrapping to two more.

**Every module row is now the height of its bars.** A row measures 46 pixels on its own day, and the whole of it - the clock strip at the top, your eight modules and your people at the foot - now measures about 502 pixels inside the 527 the window gives it, so the day is read in one look and nothing is scrolled to. Every number in it is the number it was.

**The two notes leave the row, and two things bring them back.** On a computer, pointing at a row brings both notes back and tints the row, and the row grows taller to hold them rather than covering what is beside it, so a note is never cut in half. On your phone there is no pointing, so a reveal that needed one would be a reveal your phone could never reach; the notes are written out instead at the top of the module's own card, which is what a tap on the row already opens. Nothing new to learn and nothing new to find. On your mixing module the card now opens with what the row used to whisper: the time it starts and how many batches, then what a batch costs in minutes and in your own hands.

**The name takes the row, and the badges drop below it.** This was the real damage of the old row, and it is worth naming. Your longest module name is 273 pixels wide, and it was sharing its line with a small badge, which left just 35 pixels of the name to read - about two words of it. The name now has the row's first line to itself and the badges sit on the line beneath, and four of your eight module names now read whole where one did before. The name column is 18 pixels wider as well, from 138 to 156, which is the difference between one and three more of your names reading whole. A name still too long for the row is shortened with three dots, and the whole of it is on the module's card and under the pointer on a computer.

**A fault found while building it, worth naming.** A module drawn as two production line and set to wait on the module above it could not be drawn at all: the drawing asked for the waiting badge's words in a place where they did not exist, and that is not a blank space, it is an error that stops the screen. Two taps on your own module card would have found it. None of your saved modules is drawn as two production line, which is the only reason you never met it. It is fixed, and a test now stands on it so it cannot come back.

**What has not moved, measured rather than promised.** Your three saved days were compared byte for byte before and after this release. One baker day still reads 24 pans across 8 modules, its first batch out at 9:09 am and its day finishing at 1:30 pm. No fridge, 1 person still reads 24 pans. My sister proposal 21/9/2026 still reads 4 pans. Not one module, batch, start time, cycle or saved scenario of yours changed, and the change is in how a row is drawn, not in what any row says. Nothing here blocks a sale and nothing here reads an order.

**No database step.** The row is drawn from the scenario already on your phone, so there is nothing to run in Supabase.

This change history and the operations guide (section 23) both carry all of this, both PDFs rebuilt, the app's own More screen reads Engine v155, and the tests are 1157 passing with none failing - three of them new: one on the notes being carried by the row and printed by the card, one on the name taking its own line ahead of its badges, and one on the module drawn as two production line that waits on the module above.

**22 Sep 2026 - the documents, not the app (no engine version, no database step). The guides and this change history print their own italics now, so the stray asterisks are gone.**

Every one of these documents is built from a markup in which one asterisk on each side of a word means "print this in italics". The builder stopped understanding the single asterisk at some point and reads only the pair, so a word meant to lean quietly came out wearing its own asterisks instead: *'why no button for work backward', Cutting and packing, status New*. Not one sentence read wrongly, which is why it went unnoticed for so long. There were 116 of them in this change history, 44 in the operations guide, and 9 more across the money and profit guide, the alert guide and the safety guide. All of them now use the pair of underscores the builder does understand, so they print as the italics they were always meant to be.

The asterisks that were meant to be read stay exactly as they are: the one marking a required field on the order form, and the fill-in lines on the weekly note sheet. Those are marks, not markup.

Nothing in the app changed. More still reads Engine v154, every screen is exactly where it was, and no test

moved. All five documents were rebuilt and read back to check: this change history prints no asterisk at all now, and the only two left in the operations guide are the two that are meant to be there.

**22 Sep 2026 - engine v154 (no database step). A module now says how it takes its start - three answers where there was one switch - so the packing begins the minute the cooling ends; every batch's own card offers the move again; and the people's rows are held below the modules so you can read a slot against them.**

**What you asked.** Your three reports, in your own words: "the cut and packing batch din follow the earlier batch end", and then your own correction, "cutting and packing sit below cooling down, so cutting and packing batch start should follow cooling down batch end". "The delta t disappeared, before this we have it. I want each batch start time to be adjustable, like the 1st module. Just need to show delta on the batch 1st offset only, then following module of that step dont have to show the delta because it follow the previous module tightly." And "I want to freeze the persons card, so that by scrolling thru modules i can see exactly where that slot of that person tie up to and searching for opportunity to move some batch start time to reduce the number of person needed." You also set the rule the first of them had to be built by: "the behaviour has to base on configuration, not a hard wired".

1. **What was actually wrong, and it was words against behaviour.** The old switch was called **Waits for the module above**, and its hint promised a follow - *2nd module 1st cycle should follow 1st module 1st cycle completion*. What the day really did with it was a **floor**: a batch could be pushed later than the module above, never pulled back onto it. So a packing step sitting 8 minutes after the cooling ended could never come back onto it, however the switch was set. Your report was right and it was the most useful kind: the sentence on the card and the arithmetic under it disagreed.

2. **A module now says how it takes its start, in one of three answers.** Open any module and the card asks **How this module takes its start**, with three choices: **As the one above finishes** (batch 1 of this module starts the minute the module above finishes its own batch 1, batch 2 on batch 2 - a tight follow with no gap); **Never before the one above finishes** (a batch here can never start before the module above has finished that same batch, but a later time you set yourself stands - this is the old switch, kept exactly as it behaved); and **Its own time** (the module above has no say over this one). Each one carries a sentence saying what it means before you tap it.

3. **The choice is on the module, not on a module's name.** This was your rule and it is the way it is built: nothing in the day knows a module by what it is called. Set **As the one above finishes** on your packing and it lands on whatever really sits above it - cooling, or anything else you put there, in whatever order you keep your modules. There is a test that renames every module of a day and confirms the same answer, so this cannot quietly become a rule about cooling.

4. **Nothing of yours was re-laid by being read.** A saved module keeps the behaviour it already had: a module that was waiting on the one above it opens on **Never before the one above finishes**, and a module that stood on its own opens on **Its own time**. A new module arrives on **Never before the one above finishes** too, because **As the one above finishes** on a module you have not yet told what feeds it would move your day the moment you added it. Your three days were compared before and after: 24 pans on One baker day, 12 on the seeded line, 4 on your sister's, and not one start time of yours changed.

5. **Every batch's card offers the move again - this is your second report.** A batch below batch 1 of a module set to wait used to be **refused** the two button pairs, with a sentence telling you to go and change the module instead. Every batch at every module now offers **Five minutes at a time** and **One minute at a time**, and the last paragraph of this note says plainly what a move means in each place so the two cannot disagree.

6. **The offset is read out on batch 1, and only on batch 1 - exactly as you asked for it.** On batch 1 of a module that is not the first, the card reads how this module takes its start, in words: **follows the module above**, **never before the module above**, or **on the line**. The batches below batch 1 do not repeat that, because they ride the module above at this module's own pace - there is no offset of the module's to read on them. A batch you have **held** on purpose is the exception, and it is a deliberate one: it reads **delta t = +5 min** on its own card,

because a batch that is off the line must never look like one that is on it.

7. **A move below the first module is a hold, and a hold can only ever put a batch later.** That is not a limitation of the buttons: a batch cannot start before the dough it is made of exists. What makes it safe rather than a wall is that a hold is an **offset measured from where the line puts the batch**, not a time - so move the module above it, or change how many batches it runs, and the held batch comes with it, keeping its place in the queue. The card of a held batch carries **Back onto the line**, which takes the hold off, and taking a hold all the way off now says so - *Cutting and packing is back on the line* - where it used to say *held 0 minutes behind*, which is a sentence about nothing happening.

8. **A press on batch 1 of a module that follows moves the whole module.** A module set to **As the one above finishes** has no start time of its own to write: it begins where the module above ends, whatever a stored start says. So a press there writes the module's own hold instead, and **every** batch of it comes with it - which also keeps a press on batch 1 meaning the same thing on every module, rather than being a button that appears to do nothing on one of them.

9. **The people are held below the modules - this is your third report.** The module rows scroll inside the day's own panel and the **People** rows and the **People at once** total are pinned at its foot, so scrolling down through eight modules never takes the person you are reading away from the bar you are reading it against. This is the move the whole report was for: hold a slot in view, look for the batch you can move to empty it, then go and move that batch. The red now-line of the walk through still runs over the people's rows, because that line is the day happening rather than a reading of it.

10. **One fault found while testing, and it is worth writing down.** The test that stands in for the browser kept an attribute a real browser also exposes as **dataset** - and the chart reads which batch you tapped off that. So every simulated tap had been reading as **batch 1**: a whole class of taps the tests could not see, which is exactly why "the delta t disappeared" could be true on your phone while a test suite said all was well. The stand-in is unforgiving now, the way the browser is. Two tests written against it failed immediately and both were real: the wording of a hold taken off, and the day's own card opening where a batch's card should.

11. **No database step, and it is on every phone.** Nothing to run in Supabase. The planner travels with the app's own settings row, so this new per-module answer goes up with everything else the planner holds and comes down onto your other devices the same way - the choice you make on your phone is the choice your tablet reads.

12. **What has not moved.** One baker day still reads 24 pans across 8 modules with its four stored rhythm times untouched; the seeded line still makes 12 with the chiller as the wall; your sister's line still makes 4. v152's backwards press and v153's button on the row are unchanged, and neither this release nor those reads a single order or blocks a single sale.

The CHANGELOG and the guide (sections 23 and 26) both carry all of this, both PDFs are rebuilt, and the app's own More screen reads Engine v154. The tests are 1154 passing with none failing - 14 of them new: 8 on the model (the three answers and how a saved module is read through them, the packing landing on the cooling's end, the follow being a choice on the module and not a rule about a name, a switched-off module stepped over, a module above with fewer batches answering for the batches it has, a hold riding the module above and never pulling a batch earlier, and her three days reading the same times) and 6 on the screen (the three pills on the module card, the packing landing on the oven, the pairs on every batch, the offset read on batch 1 alone, a press on batch 1 of a following module moving the whole module with its way back, and the people's rows held below the modules). No database step - nothing to run in Supabase.

**22 Sep 2026 - engine v153 (no database step). The day's own row works your day backwards - one press, and it is a button that acts instead of one that only tells you where to tap.**

**What you asked.** Your words: *'why no button for work backward, this is to reposition the batches latest start time'*. You were right, and it was about a control that only talked: v152 put the move behind a tap on one bar and left the row carrying a chip that named the moment and then told you which bar to tap. A thing shaped like a button whose whole effect is to send you somewhere else reads as a dead control, and that is what it was.

1. **The button does the work now.** The row under the chart reads **Your day backwards · first batch out 9:09 am**, and the one pressable thing beside it is **Work the day backwards**. One press and the day is worked back on the spot - no card in the way, the same seven modules pulled to their latest starts, and the same message saying what moved and what it cost your hands.
2. **It is the card's own press, not a second one.** Same writer, same words, same snapshot, same way back - so the two doors cannot drift apart. Press from the row, then open the card on batch 1 of *Cutting and packing*, and the card is showing you the day the row just made, with the way back on it.
3. **Pressed on a day already worked back, it says so and writes nothing.** This was your own choice and it is the rule the button lives by: a button that comes and goes reads as a fault, so it stays in the row on every day. On a tight day the press answers *Every module of your line is already as late as it can go - there is nothing left to pull back*.
4. **The way back is beside the button that moved the day.** While there is something to put back, the row carries **Put my start times back** next to **Work the day backwards** - the pattern Stop beside Start already uses. The card keeps its own copy, so the undo is either one tap away or three, and never missing. It is still one-shot and still lasts only while the screen is open: leave the screen and the day you pressed is the day you have.
5. **A line with no chain says so, rather than drawing a dead button again.** A day of a single module is not a chain - its first batch is simply that module's own start time - so the button is drawn, carries no moment in its label, and says in words that there is no module above the last one to work back through. The fault you found is the one thing this release refuses to repeat.
6. **The sentence above the chart now names the button.** It used to tell you to tap batch 1 of *Cutting and packing* and stop there, which was the only door v152 had. It now names both: the button in the row does it in one press, and tapping batch 1 opens the same move with the whole chain laid out before you press it, which is the one to read first.
7. **One fault found while building it, and it was in this release's own new code.** The undo works off a snapshot of the day kept in the screen's own memory and never stored, and v152's card took it from the right place. The row's new press first took it from the app's saved state instead, which put that snapshot into your data - and a reload would then have offered you a way back to a day you had already left. It was found by measuring the stored data after a press rather than by reading the code, it is fixed, and there is now a test that fails if the snapshot ever reaches your data again. Nothing of yours was changed by it.
8. **What has not moved.** Your three shelf days were compared byte-for-byte before and after: *No fridge, 1 person* still reads 24 pans, *My sister proposal 21/9/2026* still reads 4, and *One baker day* still reads 24 pans with the first batch out at 9:09 am and the day finishing at 1:30 pm. A press followed by the way back leaves the stored data byte-for-byte identical to what it was - measured, not promised - and the snapshot is gone when the screen is. Nothing here blocks a sale and nothing here reads an order.

CHANGELOG and the guide (section 23) both carry this, both PDFs rebuilt, the app's own More screen reads Engine v153, and the tests are 1140 passing with none failing - five of them new: the row press moving the day exactly as the card does, a finished day keeping its button and saying it is done, the way back sitting beside the button that moved the day with the card offering it too, a line with no chain saying so instead of drawing a dead button, and the one that pins the snapshot out of your saved data. No database step - nothing to run in Supabase.

**22 Sep 2026 - engine v152 (no database step). The planner can work your day backwards - the day hangs from your first batch, and every module gets its latest start.**

**What you asked.** Your words: *'how to make the calculate backward works?'* The backward count already existed on More -> Production line (section 24), where it counts back from the minute the first pans must be at the oven. The Scenario planner had none: it could only ever push a day later, never pull it earlier. So the planner has one now, on the planner's own day.

**The moment your day hangs from is your own last module, not the oven.** The anchor is the end of your first

batch at the last switched-on module of your line - whatever that module happens to be. Add, remove or reorder a module and the anchor moves with your line. On your own One baker day that module is Cutting and packing, so the day hangs from 9:09 am.

**Where it is, and how you find it.** Tap the B1 over the first bar of that last module - the card you already get by tapping a batch - and it is now the day's own card. It reads the module, the end of your first batch and your day's finish, then every module with the time it sits at and the time it could sit at. On your line it reads: Mixing the dough in the tub 4:01 am to 4:43 am, The rests and the stretch and folds 4:21 to 5:03 am, Oil the pans and weigh the dough out 6:24 to 7:06 am, Into the proofer 6:39 to 7:21 am, Dimple and top 7:24 to 8:06 am, The proofer again 7:30 to 8:12 am, The oven swap and the bake 8:00 to 8:42 am, and Cutting and packing 8:57 am with a tick. A tick means that module is already as late as the line allows. So the control is findable rather than hunted for, the day card also carries a chip reading First batch out 9:09 am with the sentence saying which bar to tap.

**The rule is measured off your own day, not subtracted from the anchor.** This is the one place I departed from what I first planned, because the arithmetic was wrong on your own numbers. Counting backwards is not the sum of the cycle minutes: on your 24-pan line the eight modules' minutes add to 853 while the chain you actually run spans 1107 minutes, because your modules do not sit end to end - the fold loop and the proofer are wider than the few minutes a batch takes in them. Take 853 off your anchor and the mix is handed a LATER start than the one it already has, which is the opposite of what you asked for. So the calculation measures the room between each pair of modules as your stored times really stand, and pulls each module back by that room. On your line it gives the mix 4:43 am, and the 42 minutes it gains are the slack your day really has.

**One press, and a second press does nothing.** Pull them back to their latest start moves the seven modules that can still come later and leaves every other one exactly where it is. A module that is switched off is stepped over, exactly as the day already steps over one. A press never moves a module EARLIER than it already stands, and pressing it again finds nothing left to take.

**It is a real change, and the way back is on the same card.** Said plainly rather than quietly: this rewrites the start times of your switched-on modules and saves them into your scenario. It is not a preview. So while you are on that screen the card carries a second button, Put my start times back, which puts every module the press moved exactly where it was and then goes. It is one-shot, and it lasts only while the screen is open, because it is never stored - the same reason the walk through's own clock is not. Leave the screen and the day you pressed is the day you have.

**Below that are the same two pairs of buttons you already know, and here they move the whole day.** Five minutes at a time and One minute at a time, each with an earlier and a later button. Press one and every module moves together by that amount, so the day keeps its shape: on your line all eight modules move five minutes and every gap between them is the number it was. That is the move for reading what an earlier or a later start does to the whole day without re-drawing it.

**A tighter day can need another pair of hands, and the app says so instead of hiding it.** This is the honest cost of taking the slack out. Squeeze the day to its shortest span and two of your jobs can land in the same minute, so a press can raise the people the day needs - on your One baker day from one person to two. Nothing blocks and nothing is refused: the toast reports it in words, the People area below shows the collision in red, and the button under it takes it back.

**The clock you point with now follows your finger, and the clock strip stays where you can read it.** Your four asks of that day chart, all in this release. The reading follows the pointer DOWN the day as well as across it - it used to sit at the top of the chart, which is the right place only while the top of the chart is on screen, so pointing at a bar four rows down meant reading a clock a screen away. It rides 8px clear of your finger rather than under it, because a reading under your own finger hides the bar you are holding it against - and near the top of the chart, where there is no room above, it drops below the pointer instead.

*Lifted at v171:* the 8 pixels above your finger described here are two CSS inches now, because 8 pixels is under a fingertip rather than clear of it. The rule itself is unchanged and this entry still describes it: above your pointer

where the window has the room, and below it where it does not. Read the v171 entry above for the rest.

The chart is a panel of its own now: the clock strip is pinned at the top of that panel and the rows scroll under it, so the hours stay beside the bars you have scrolled to. The panel is capped, so it never grows past what you can see at once.

**And one fault found while building it.** The red now-line of the walk through was being drawn in the chart's name column - 138px, about an hour and a half, to the LEFT of the minute it was naming, because its position was measured when the chart was built and a chart that is not yet on the page has no position at all. Measured off the clock strip's own column instead, it now stands on the minute it names. Nothing about the walk through changed except that the line is where it says.

**What has not moved.** Your three shelf days were compared byte for byte before and after: No fridge, 1 person still reads 24 pans, My sister proposal 21/9/2026 still 4 pans, and One baker day still reads 24 pans across 8 modules with the first batch out at 9:09 am, the day finishing at 1:30 pm and the stored rhythm of 4:01, 5:28, 6:55 and 8:22 am untouched. Not one module, batch, start time, cycle or saved scenario changed, because nothing moves except by your own press. Nothing here blocks a sale and nothing here reads an order.

CHANGELOG and the guide (section 23, cross-referenced to section 24) both carry all of this, both PDFs rebuilt, and the tests are 1136 passing with none failing - nine of them new: four on the calculation (the exact latest start of every module on One baker day, that it is measured off the day rather than subtracted from the anchor, that a switched-off module is stepped over, and that a press never moves a module earlier nor does anything a second time) and five on the card and the drawing (the card opening on batch 1 at the last module and nowhere else, one press leaving the day's own numbers alone, the way back putting every start time exactly where it was, the step pairs moving the whole day with its shape kept, and the chip signposting the moment the day hangs from). No database step - nothing to run in Supabase.

**22 Sep 2026 - engine v151 (no database step). The module card answers to you, every batch is numbered, and the day calls your people.**

**What you asked for, in your own words.** Eleven things in one message, all of them inside More -> Scenario planner: the scenario's name first; the module card asking only what you alone can answer; a B number over every batch; moving a batch at any module the same way you move one at the first; naming Person 1 and Person 2; a marker that reads as a person; and the day calling you a minute before each job. It all ships together, as you asked. One of the eleven was not a wish but a fault: a person's card could not be opened at all before this, which is why naming Person 1 and Person 2 had no way in.

**The scenario's name comes first.** The card at the top of the planner now reads What this scenario is called, then My day starts at, then Pans a day you want from it. Going to that screen to rename a scenario used to mean reading past two numbers you did not come for.

**The module card no longer asks you two things you do not think in.** Which job of your line is this has gone. How you place each batch's start time has gone as well, and you said an alternative is coming - so the batches' stored times are kept exactly where they are, and nothing reads differently for it. What taking those two blocks out costs you is written at the end of this note, honestly.

**A new module now arrives ready to use.** One cycle of 20 minutes with all 20 of them as your hands at the start, Pans in one batch 1, waits for the module above switched on, allow multiple production line switched on. Before this a new module arrived at 15 minutes with both switches off, and you had to type five things before it was a module at all.

**Batches follow the one above, unless you say otherwise.** Minutes from one batch to the next and How many batches it runs in the day both carry an Auto. Auto is written as an empty box with the word in it, so you can always tell a number you set from a number the day worked out. The batch count is a box on the first module - that is where your day's number comes from - and on any module that has a number of its own. A module on Auto says its number out loud and names what it is following, and beside it is one button, Give it its own number, which keeps the count it is running now so taking the wheel never moves your day a minute. Auto spacing is not new behaviour: the day has always read a module with no pace of its own as one batch following the one before

it end to end. It has a name now.

**How many production line do you have - the number is unchanged, the words are yours.** You said it yourself: "that 2 refer to 2 production line, if the meaning is for that purpose, a better remark needed for that. no functional changes needed." So the behaviour is exactly what it was, and the remark is rewritten around production line instead of the word lines. It also says plainly what happens when Allow multiple production line is on: then the module is one production line with one person, and this number is not in force. And Let its batches overlap is called Allow multiple production line now, everywhere on the card and in the note under the day.

**Duplicate this module.** At the foot of the module card, beside Delete, there is now Duplicate this module. The copy lands directly under the one you copied and carries its cycles, its numbers and its person exactly - a copy is a copy and none of the new-module defaults reach it. You asked for this because you can add modules as you wish now, and two ovens of yours are two of the same thing.

**Every batch is numbered, on every module and every production line.** A small B1, B2, B3 above each bar, in a band the row grew for it, on every module including one drawn as two production line. The number carries the batch it names, so tapping B3 opens batch 3's own clock - a number you use to tell one batch from another is no use if it is not also a way to pick one.

**Move a batch at any module, not only at the first.** Tapping a bar or its number opens that batch's clock, and the card now says the batch number and its delta t beside each other. On the first module a move is the batch's own start time, which is what it always was. On every later module a move is a delta t - a hold measured from where the production line puts that batch - so the batch rides the chain: move the module above it and this batch comes with it, keeping its offset. A delta t can only ever hold a batch back, because a batch cannot start before the dough it is made of exists, and the tag over the bar shows delta t = +5 whenever there is one. When your batch is held back on purpose the card also offers Back onto the line, which takes the hold off, so a nudge is never a one-way door.

**Your people have names, and the card that was unreachable opens.** This is the fault you reported: "in the person card, now person card is not accessible". The person's row had no handler of any kind, so no gesture anywhere opened their card. Tapping a person's row below the day now opens it: a box for what you call them, a switch for whether the day calls them, and the jobs they are on today. The name is kept in the app's settings rather than in the scenario, so you type it once and every scenario uses it - person numbers start again at 1 in each scenario, so the name you give person 1 shows wherever person 1 is working, and the card says so. The card's own heading and its "Call them a minute before their next job" line follow the name as you type it, so the card never goes on calling them Person 1 while the row behind it already says their name. The box itself is never rebuilt while you are typing in it, which is what keeps your cursor where you left it.

**A person's marker is theirs now, not the module's.** The People rows used to be tinted by the module each stretch of work came from, so one person's row was eight colours and read as nothing in particular. Each person now has their own colour, with their name on the stretch where it is wide enough - so a row reads as one worker's day and two rows are two people. A collision still wears its red outline over the top.

**The day walks, and it calls you.** A Start the day now button in the Walk the day row, with Stop beside it while it runs, and a red now-line that follows the real clock down the chart. The call comes one minute before each job a person owns - one minute and not the minute of it, because your fold is a one-minute job and telling you to fold as you should already be folding is telling you too late. A call card appears over everything: their name, their own colour, the job, and the clock time it is for, with one OK. OK takes it away. If you do not tap it, the next call simply replaces it rather than queuing behind it, which is your own rule. Each person's call is a different note, so two people called in the same minute are two sounds. The card and the guide both say the limit plainly: it runs only while this screen is open and awake, and a phone that locks or a browser that closes stops it. It is a call for the day you are walking through, not an alarm that survives the app being shut.

**The call names the cycle only when you have named it.** A module's own name can be a mouthful ("The rests and the stretch and folds"), and the thing you can actually go and do is the cycle inside it, so the call carries the cycle's name as a line of its own when that cycle has one. An unnamed cycle gives no line at all. That last part is

a fault this release nearly shipped: the line was drawn through the same number-formatting helper the rest of the chart uses, and on a cycle with no name that helper answers "0", so the card went out printing a bare 0 above the clock. There is a test on it now, and it is written so it fails if the line ever comes back.

**Your day start is never rewritten by it.** Starting the day sets the run's own minute zero to the minute you press, and nothing is saved - so a 4:01 am plan started at 9 am calls at 9 am, and your scenario's own day start is exactly where you left it.

**What this release takes away, said plainly rather than quietly.** Three things are gone with the two blocks removed from the module card. The job list is gone, so the way to CORRECT a module whose name the Production line does not recognise is gone with it - a module still crosses over by what you call it, exactly as before, and every module of yours already crosses. The per-batch start boxes are gone from the card, so the one control that put every batch back on its own even spacing, Space them evenly again, is gone too; tapping a bar gives you the same five-minute and one-minute moves, but not that reset. Say the word and any of the three comes back in one line.

**What has not moved.** Your two saved lines and the seeded starting line were compared byte-for-byte before and after this release: No fridge, 1 person still reads 24 pans, My sister proposal 21/9/2026 still reads 12, and the seeded starting line still reads 12 with the chiller as the wall. Not one module, batch, start time, cycle or saved scenario of yours changed, because every new default is for a module you have not filled in yet - a stored number or a stored start is yours and is never rewritten, which is the same rule the cycles migration used. Nothing here blocks a sale and nothing here reads an order.

CHANGELOG and the guide (section 26) both carry all of this, both PDFs rebuilt, and the tests are 1127 passing with none failing - fourteen of them new: five on the model (a new module's own defaults; Auto following the one above while a number of your own is never overruled; a delta t riding the chain and never pulling a batch in front; reading a line twice still a no-op; all three seeded lines pinned batch time by batch time; and the call landing one minute before the job) and nine on the drawing and the card (a batch number on every bar including two production line, a copy carrying its source's cycles, the person's name on the row, the card's own heading following it, a named cycle on the call and no line at all for an unnamed one, and the four halves of the live clock). No database step - nothing to run in Supabase.

**22 Sep 2026 - engine v150 (no database step). The word is cycle - and a batch of four now reads as four.**

**What you told me, in your own words: "there is not steps term in this project, i want the cycle shown in batch, and the cycle labour shown visually".** You were right on the first point, and the app was the thing saying it. I measured the day chart before changing anything: nothing had disappeared. On your One baker day the fold is one batch of four cycles and its bar draws four shaded bands with your three folds on them, and the oven is one batch of two cycles with your two-minute swap on the second - 24 cycle bands and 32 stretches of your hands across the day. The feature was built and it was drawing. What was wrong was the word on it.

**Why your own two saved lines look plain.** Both the line you call No fridge, 1 person and My sister proposal 21/9/2026 have exactly one cycle in every module, because both were saved before cycles existed and were given one cycle each so the bar's shape would not move. One cycle is drawn as one plain bar. You settled that yourself, and the app agrees with you: each batch has at least one cycle, and a batch with one cycle is that cycle. So nothing of yours was rewritten - both lines are correct as they stand, and not one module, batch, start time or saved scenario was touched.

**The cycles box no longer calls a cycle a step.** It used to say "Each cycle is one step of this module", which is exactly the sentence that made you doubt the feature was there. It now says "Each cycle is one piece of this module's work", in the order you work them. The hint over the day chart says the same thing in the same word: the paler bands inside a bar are the separate cycles of that batch. Three more places where the app said step while it meant in sync now say what they mean - batch counts that differ from the module before them, and the line under the day naming any that no longer match. The batch's own card used to say "the step the day is read in"; it now says the amount the day is read in, which is what it is.

**The Production line says stage, not step.** That screen walks your bake day in order, and it called each part of the day a step. It now calls each of them a stage. The word job is left alone there on purpose: on that screen job already means a hand-job - one of the six jobs you do - and the proofer is not one of them, which the screen says in its own sentence. Calling the proofer a job would have put two contradicting sentences on one screen.

**A batch of four now really does read as four.** This is the one fault in the drawing, and it was a real one. The shades the cycles are drawn in said "a batch of four reads as four steps", but the ladder only had three shades and started again at the fourth - so on a four-cycle batch the first cycle and the fourth wore the same palest band, and the palest of the three was so faint over the pastel bars that it barely showed at all. There are four shades now, spaced evenly, and the palest has been lifted off the floor so it still reads as a wash. A fifth cycle and beyond reuse the darkest rather than falling back to the palest, so a long batch never ends on the faintest band.

**What has not moved.** Not one minute, position or labour band on the day chart. Your own saved line still reads 24 pans across 9 modules, the seeded starting line still 12 pans with the chiller as the wall, and your sister's still 4 pans in 8 modules. Nothing here blocks a sale and nothing here reads an order.

The tests are 1113 passing with none failing - three of them new, and one of them fails if a four-cycle batch is ever drawn with two cycles sharing a shade. No database step - nothing to run in Supabase.

**22 Sep 2026 - engine v149 (no database step). The way down - taking batches off, so a day that makes more than you asked for can come back to your number.**

**What you found, in your own words: "This scenario makes 36 pans, and you want 24. this does not agrees?"** You were right, and it was two faults in one card, not one. Your One baker day made 24 pans. You typed 36, pressed **Use these numbers**, and the ladder did what a ladder does - it raised all eight modules from 4 batches to 6 and the day made 36 pans, which is what you had just asked it for. Then you put 24 back in the box. The card underneath said "This scenario already makes your 24 pans" while the line directly above it said the day makes 36. Both sentences were drawn from the same card, one of them had to be wrong, and the ladder that could have settled it only ever went one way.

**The climb only adds.** Every rung of it raises something: more batches in the day, or a second machine. Not one rung has ever taken a batch away, so there was no way on that screen to bring 36 back to 24 - no amount of pressing would have done it. That half simply did not exist, and no wording change could have covered for its absence.

**There is now a way down.** When the day makes more than the number you asked for, the card is headed **The way down** and carries one rung: every module the day is waiting on, and the batches to take off each of them. On the day you were looking at it reads: all eight modules holding the day at 36 pans, 6 -> 4 batches in the day at 6 pans a batch, the same change in each of them - that takes the day from 36 to 24. The same **Use these numbers** button applies it, on the heading and at the foot of the card.

**Why it is one move and not a ladder.** The day is the **least** any module turns out, so every module sitting on that least is holding the day there. Bring one of them down alone and the day does not move at all, because one of the others takes its place as the limit. They have to come down together, and then the day lands on your number in a single move. That is why this half is a settling move rather than a rung-by-rung climb.

**The batch times you have set by hand are left exactly where you put them.** This is the one place the way down deliberately differs from the climb. Raising a count makes a new rhythm, so the climb re-spaces a module's batch times from its start. **Lowering** a count only takes batches off the **end** of the day - so if you have dragged a batch to a time that suits your morning, that time stays. Nothing on this move rewrites a minute you chose.

**When a batch is bigger than the number you asked for, it says so.** Six pans a batch and one batch a day, and you ask for three: no count of batches gets under one batch, so there is no move to offer and no button to press. The card says it plainly instead - one batch is 6 pans, so this line can never come below 6 - because a card with nothing on it and no explanation reads as broken.

**The two numbers can no longer contradict each other.** The branch that reads "already makes your 24 pans"

is now only reached when the day really does make 24. When the day makes more, you get the way down and the honest reading of the gap: "This scenario makes 36 pans, and you want 24. That is 12 pans more than you asked for."

**What has not moved.** Not one module, batch time, cycle or saved scenario. Your own saved line still reads 24 pans across 9 modules, the seeded starting line still 12 pans with the chiller as the wall, and your sister's still 4 pans in 8 modules. Nothing here reads an order and nothing here blocks a sale.

Engine 149. Tests are 1110 passing with none failing, eight of them new - four on the move itself (the 36 -> 24 settling move on all eight modules, a day already at your number offering nothing, a dragged batch time surviving the move untouched, and a batch too big to come down past) and four on the card (its heading and button, the two numbers never contradicting each other, the button actually bringing the day down, and the plain answer when a batch is too big). No database step - nothing to run in Supabase.

**22 Sep 2026 - engine v148 (no database step). A cycle's labour drawn where you work it, and the ladder's button found.**

**A cycle's labour is now drawn where you work it, and wide enough to see.** This is your own sentence from 22 September: "cycle should be drawn with its labour shown at the appropriate timing." It was being drawn at the right minute already - the fold at the end of its rest, the oven swap after its bake - but it was drawn too small to find. A one-minute fold is under two pixels at the scale the day is read in, and where it fell it landed exactly on the seam between two rests, so what you saw was the seam and not your hands. Two changes, and neither of them moves a minute:

- **The width has a floor.** A stretch of your hands inside a bar is now never drawn thinner than 4 pixels, which is the same floor the person row below already gives the same stretch of your day. So the bar and the person attending it now agree about what a one-minute job looks like - before this, the fold was 1 pixel on the bar and 4 pixels on the person row, which is two drawings of one fact disagreeing with each other. **Each end of it wears a fine light edge**, so a band that lands on the seam between two cycles still reads as a band rather than as the join.

Only the width is floored. **The position is the model's own minute**, so the picture still says WHEN: the fold sits at the end of its rest and the oven swap sits after the bake. Hover or long-press a bar and it now names the cycle and the exact minute, so the true length of a floored band is never lost - it reads, for example, "1 min of you at 4:51 am".

**The button that applies the ladder is now on the ladder's own heading.** You told me you could not find **Use these numbers**. You were right, and the reason was arithmetic: it sat at the very foot of The climb card, and a day of nine modules makes a nine-rung ladder, so on a phone the button was about four screens below the heading it belongs to. It is now on the heading as well, beside the words The climb, and it is still at the foot of the card. Both press exactly the same thing.

**When there is nothing to apply, there is no button - and the card says so.** If your scenario already makes the number you asked for, the card reads "This scenario already makes your 24 pans" and offers nothing to press, because there is nothing for it to change. That is the other half of why the button could not be found: on a scenario that already reaches your number it does not exist at all. Raise the number you want above what the line makes, and the ladder appears with its button on the heading.

**What has not moved.** Not one module, batch time, cycle or saved scenario. The day still reads the same numbers it read in v147: One baker day is still 24 pans in 8 modules, one person, 3 h 52 min of hands, 4:01 am to 1:30 pm. Nothing here reads an order and nothing here blocks a sale.

Engine 148. Tests are 1102 passing with none failing, five of them new - they pin the fold's drawn minute, the bar and the person row agreeing about a one-minute job, the oven swap being drawn after the bake, and the apply button being on the heading. No database step - nothing to run in Supabase.

**22 Sep 2026 - engine v147 (no database step). Module, batch, cycle - the three words on the Scenario planner, and the model that moved with them.**

**What you asked.** In your own words: *"let us realign our terms use in the production line"* - a module instead of a brick, batches inside a module, and cycles inside a batch, with a batch's time moved by buttons instead of a drag. The words and the model under them were changed together on purpose: renaming first and building the cycles later would have left the word **cycle** meaning two different things while you were learning it.

**The three words, and what each one is.** A **module** is the equipment plus the hands tending it - what this app has called a brick until now. A **batch** is one lot: the six pans of one oven load, one tub of dough. A module can run several batches in a day, and how many is yours to set. A **cycle** is one step of that module's work, and it is the cycle that owns the minutes and the labour. Your own sentence settled it: *"batch is not a process but cycle is."* So a batch carries no minutes of its own - the time it holds is the sum of the cycles inside it. A batch with a single cycle is drawn on the day as one plain bar, exactly the way a module has always been drawn.

**Labour now sits at its real end of a cycle: its load, or its unload, or both.** Open a module and under **Cycles in one batch** every cycle has a name box, **Minutes**, **Load** and **Unload**. Load is your hands at the start of that cycle, unload is your hands at the end, and the day draws them where they really happen. A cycle cannot start until the cycle above it ends, so a cycle is never offered a start time of its own - its position is set by its minutes.

**Two of your eight steps were in the wrong place, and both are corrected.** Your 1-minute fold was drawn two rests early; it is now the **unload of each 30-minute rest**, so the rests run back to back inside the batch - 30 minutes, fold, 30 minutes, fold, 30 minutes, fold, 30 minutes - and the fold lands at the end of the rest, where your hands really are. And your **oven swap is a cycle of its own after the 13-minute bake** rather than labour before it, so the first swap now reads 8:13 am, thirteen minutes after the pans went in. Nothing else about the eight steps moved: the tub is still 4:01 am and the first six pans are still at the oven at 8:00 am, the minute the whole day is counted backwards from.

**Moving a batch is buttons now, and the drag is gone.** Tap a batch on the day chart and a small card opens with that batch's clock and its own two pairs of buttons: **Five minutes at a time** and **One minute at a time**, each with an earlier and a later button, plus the batch's own times written out. Five is the step the day is read in and the step the time line reads out; one is the nudge for lifting a batch off a collision. This replaced dragging a bar completely, and it was your call: a bar being dragged across the day and the hairline of the time cursor were fighting over the same finger. Nothing about a press does arithmetic of its own - a pressed time and a typed time give the same answer - and when the day's own rule puts the batch somewhere else the card says which rule it was. A cycle drawn inside a bar is not a separate control: tap it and you get that batch's card, because a cycle's start is its batch's start plus the cycles above it.

**A batch that waits on the module above offers no buttons, and says why.** Where a module has **Waits for the module above** switched on, a batch's time is not its own - batch 3 of that module IS batch 3 of the module above - so there is nothing to step and nothing to mislead you. The card says exactly that, in as many words, and names the switch that would free the time. The row itself carries a **waits above** tag and the words **waits on** and the module it is waiting for, so a chained batch is never silent about why it is where it is.

**Batch counts follow the one before, and the day says when one does not.** Editing **How many batches it runs in the day** carries down to the modules after it that were running the same number, and stops dead where a module has a number of its own - it is never quietly overwritten. A standing line under the day then names each module whose batch count differs from the one before it, both numbers written out, with a plain sentence that it is a note and not a rule: the day runs on the numbers you have put in. On the seeded starting line, for instance, it reads **Wash, oil and fill - 6 batches, The fold loop - 4**. Nothing is locked by it and nothing is refused.

**The tile that said Cycle time now says Minutes a pan.** Same quantity it always was - the minutes one pan takes off the slowest module, with that module named under it - said in the words this app already uses elsewhere, because cycle now means a step inside a batch. No arithmetic moved with the name.

**A module can have as many lines as the day needs, and lots go to the free one. How many of these do you have** is that - a mixer, a sink, two ovens. At the even spacing the app works out, the lots take their turns down the lines exactly as before; where you have moved a start, a lot leaving one module now goes to

whichever line is actually free rather than the one whose turn it is. No figure on an untouched day changes because of it.

**One baker day was re-derived, and it is an hour later than the last version of it said.** Correcting where your fold and your swap really happen costs the day a work slot, so the pace one pair of hands can hold is **87 minutes** between batches, not 81: the tubs now go in at 4:01, 5:28, 6:55 and 8:22 am, the packing starts at **8:57 am**, the last batch is packed at 1:18 pm and the day ends at **1:30 pm**, which is 9 h 29 min. The 81 no longer holds, and the reason is checkable rather than a matter of opinion: at 81 the second rest's fold lands on minute 82, the very minute the second tub's dough has to be mixed, so the app puts a second person on it. It still reads 24 pans a day in 8 modules, one person, and 3 h 52 min of hands, and nothing collides. Guide section 25 carries the whole story, including what the sweep of every other pace found.

**What has not moved.** Your own saved line still reads 24 pans across 9 modules and exactly the day you asked for; the seeded starting line still reads 12 pans with the chiller as the wall; your sister's still 4 pans in 8 modules. Not one saved module, batch time or scenario was altered - the words you read and the way you move a time changed, and the days themselves did not.

**22 Sep 2026 - engine v146 (no database step). One baker day - your own bake day set on the bricks, with a latest start on every batch.**

**What you asked.** In your own words: *"i need you to create one scenario and save it as One baker day. Set the bricks for me, with latest start time each brick batch."*

**Where it is.** More -> Scenario planner. Under **Your scenarios** there is one new row, **+ One baker day**, reading **One pair of hands, set for you**. One tap puts it on your shelf and opens it. From then on it is an ordinary saved scenario, exactly like your own line and your sister's proposal: it stays in your list, it opens when you tap it, and the row that offered it is gone because you already have it. Delete it the way you delete any scenario, and it stays deleted - it is never dropped back in on you.

**The eight bricks are your eight steps, in your own order.** Mixing the dough in the tub; the rests and the stretch and folds; oiling the pans and weighing the dough out; into the proofer; dimple and top; the proofer again; the oven swap and the bake; cutting and packing. Nothing invented, nothing left out. The fold and the two proofer bricks carry no line step against them, because the Production line has no field for them - and the screen names them rather than writing their minutes onto a field that means something else.

**Every brick's first cycle is a latest start, which is what you asked for.** Open the day and the eight bricks read:

- Mixing the dough in the tub - **4:01 am** (20 minutes, all of it yours) The rests and the stretch and folds - **4:21 am** (123 minutes, 3 of them your hands) Oil the pans and weigh the dough out - **6:24 am** (15 minutes of you) Into the proofer - **6:39 am** (45 minutes, no hands) Dimple and top - **7:24 am** (6 minutes of you) The proofer again - **7:30 am** (30 minutes, no hands) The oven swap and the bake - **8:00 am**, the anchor: the minute the first 6 pans have to be at the oven (2 minutes of you) Cutting and packing - **8:27 am** (12 minutes of you)

The four batches sit **81 minutes apart**, so the second tub starts at 5:22 am, the third at 6:43 am and the fourth at 8:04 am. The last batch of the day is packed at 12:30 pm and the day is done at 12:42 pm.

**One of the eight is not where your chain puts it, and that is the whole finding of the scenario.** Your own chain would start the cutting and packing at **8:15 am**, fifteen minutes after the oven. Alone, you are still folding the last tub at 8:27, so **8:27 is the first minute that job needs nobody else**. It is not a preference: start it a single minute earlier, at any minute from 8:26 backwards, and the day asks for a **second pair of hands**. The window you can move it in without buying anybody is **8:27 to 8:33 am**, and the seeded day sits at the start of it.

**Why the batches are 81 minutes apart and not 40.5.** This is your own proofer figure, from the same morning: a batch is in the cabinet 81 minutes - 45 in, 6 out to be dimpled, 30 back in - so the cabinet could take a new batch every **40.5 minutes**, about 8.9 pans an hour. But 40.5 assumes somebody is free to carry a batch out and the next one in the moment the cabinet empties, and with one pair of hands there is no such minute: the hand-work of a single batch is 58 minutes by itself. Rather than guess, every rhythm from 40.5 upwards was swept through the app's own arithmetic, and **81 minutes is the pace one pair of hands can hold**: every

hand-window in the day then tiles exactly, each one beginning where the last one ends, and the proofer is never asked to hold two batches at once. It still works at 80.5 minutes; at 80, or at 40.5, the very same day reads **2 people**. So the two figures you have are both right and they answer different questions: the card's 40.5 is what the cabinet allows when you are not alone, and this scenario's 81 is what one baker can actually feed it.

**What the day reads. 24 pans a day in 8 bricks** - four batches of the six pans you bake. **One person, 3 h 52 min of hands** across the whole day, and a run from **4:01 am to 12:42 pm**, which is 8 h 41 min. There are **no red notes** under the day, because nothing collides: one pair of hands, no job doubled up. The brick setting the pace is the **mixing**, and seven others pass the same 24 pans - so the card says they are one limit between them, and relieving only one of them will not move the day.

**It loads onto the Production line as your own numbers.** Under **Start from a scenario** at More -> Production line, tapping **One baker day** shows you everything it would change before anything moves: minutes to mix one tub 20, oiling the pans and weighing the dough out 15, dimple and top 6, take 6 pans out and put 6 in 2, minutes the pans are in the oven 13, pans in one load 6, cutting and packing 12, the target 24 pans, and 1 person. The fold and the two proofer bricks are listed by name, because that screen has no field for them. Nothing is written until you press the button.

**Every number is yours to move.** The day is seeded, not frozen: drag a bar, open a brick and type a number, change how many pans a day you want, or move the time your day starts, and the whole thing answers again the way every other scenario does.

**What has not moved.** Your own saved line still reads **24 pans across 9 bricks** and exactly the day you asked for, the seeded starting line still reads **12 pans with the chiller as the wall**, and your sister's still **4 pans in 8 bricks**. One baker day arrives as a row you choose: not one brick, start time, merge or saved scenario of yours changed, and nothing here blocks a sale.

CHANGELOG and the guide (section 25) both carry all of this, both PDFs rebuilt, and the tests are 1090 passing with none failing. No database step - nothing to run in Supabase.

**22 Sep 2026 - engine v145 (no database step). Your real bake day on the Production line - and the same day read backwards from the oven, so no dough is ever mixed too early.**

**What you asked.** In your own words: *"work backwards, from end of process, the previous process whould have a latest start time, by going this way, we prevent preparing dough too early and prevent dough from over fermented. But we can fine tune each process to start earlier to better manage manpower utilisation."* You had corrected your own step order that morning, after checking it with your sister: *"i think my self confused, i double check with my sister and now understood the right flow. we cannot mix the dough too early, it will over fermented."* And you had already worked your ceiling out yourself and asked about it: *"6pan sit in it for 81min, proofer able to hold 12pans, so effectively the poofer can produde 6 pan/40.5min? So the proofer became the bottle neck?"* Yes - 40.5 minutes exactly. This release gets that figure onto the screen out of your own numbers rather than out of a guess.

**The flow on the Production line was not your day.** It read mix, then the chiller, then washing and oiling the pans, then weighing the dough out, then the oven. Against your day that put the chiller where you have none, transposed the oiling and the weighing out, and - the worst of it - had no fold in it at all. The map now runs your order, top to bottom: mix the dough in the tub; the rests and the stretch and folds; oil the pans and weigh the dough out; into the proofer; dimple and top; proofer again; the oven; then cutting and packing. The proofer is drawn twice because your day uses it twice, with the dimple in between.

**The wall was a chiller you have not bought.** The screen watched a chiller of 12 trays over 5 hours and made that your ceiling - 12 pans a day - which is a cap on a cabinet that is still only a what-if on your planner, kept switched off. The station you actually own is the proofer, and it is the honest ceiling: a batch sits in the cabinet for the whole 81 minutes (45, then the dimple, then 30), so 12 pans means one batch every 40.5 minutes and no faster. Change **Pans your proofer holds at once** to 18 and the screen answers 27 minutes instead, and says which of your limits is the wall now. It never tells you to buy one: at one pair of hands a bigger cabinet buys you nothing, and the screen says so.

**Your day, read backwards.** A new card beside the flow, **Your day, backwards**, gives every step the last moment it may start, worked back from the oven - the end of the process, which is where you asked to start from. For the first six pans to be standing at the oven at 8:00 am, the dough goes into the tub at 4:01 am:

- Mix the dough in the tub - 4:01 am The rests and the stretch and folds - 4:21 am Oil the pans and weigh the dough out - 6:24 am Into the proofer - 6:39 am Dimple and top - 7:24 am Proofer again - 7:30 am The oven, bake and swap - 8:00 am

That is **239 minutes** from the tub to the oven and **254** from the tub to the pans coming out, which is your own arithmetic and nothing else. One batch is **46 minutes of your hands**: 20 mixing, 3 folding, 15 oiling and weighing, 6 dimpling, 2 swapping.

**The clock is a field, not a rule.** The minute the first batch must be standing at the oven is yours to set - it arrives at 8:00 am - and every start time below is worked back from it. Move it and the whole day moves with it.

**Five minutes early is allowed, and nothing is locked.** Every step also carries a five minute band above it: a step may begin up to five minutes early to suit your hands, which is the room you asked for to manage the people. It is advice drawn on the screen and it is never a gate - no time here stops you baking, hides an order, or overrules your own eye. You watch the dough; the timetable watches the clock.

**Five numbers changed what they were asked, and each one is named here rather than changed quietly.**

Your saved numbers move across by themselves the first time your phone opens this version, so you will see these the moment you load it - and every one of them can be typed back in one tap if your own timing says otherwise:

- **Minutes to mix one tub of dough:** was 6, now 20. The old box asked how long it took you to load a mix; this one asks how long the whole mix takes, which you gave as twenty minutes for one tub. **Pans one tub of dough makes:** was 28, now 6. The old box asked what your mixer's bowl holds. Your cycle is one tub of dough into the pans of one oven load - six - and the mixing and folding are spread over that tub, so this is the number the chain needs. **Oil the pans and weigh the dough out, for 6 pans:** was two jobs under two names (your washing and oiling read 20, and the weighing out was never timed), now one job at 15. Your own count of the hand-work has it once, which is the proof they were one job: with the two merged, the app's 46 minutes a batch is your 46 to the minute. **Minutes to dimple and top 6 pans:** was 8, now 6 - your one minute a pan. The 8 was measured with the topping in it. **Minutes to take 6 pans out and put 6 in:** was 4, now 2 - "2min covers out and in", as you put it. One turn of the oven is the baking and the swapping together, 15 minutes as before.

Everything else of yours is untouched: your two pairs of hands, your 5 hours, your 24-pan day, your 12 pans, 6 pans a bake, 15 minutes a turn, the 30-minute cooling - and your 12 minutes of cutting and packing, which you had timed and which is still yours.

**A step with no minutes on it is still named, never counted as free.** Cutting and packing is the one job that arrives with an empty box - every other step of the day comes seeded with your own figure - so on a fresh phone the flow says **not timed yet** and the backwards card says in a sentence that it is not counted in the 254 minutes, rather than quietly treating your time at the table as free. You had already timed yours at 12 minutes for 6 pans, so on your phone that box holds your 12 and the card's heading reads **After the bake** rather than **Still to time**. A heading left reading **Still to time** over a figure you had given would read as the screen not having noticed it. Either way the step is drawn on the backwards card as sitting **after the bake, outside the 254 minutes**, because those minutes happen while you are waiting on the next batch. Nothing was reset.

**One thing on the join between the two screens reads differently, and it is a correction.** When you tap a saved scenario under **Start from a scenario** on the Production line, the list of numbers it would change is now the corrected list: **oiling the pans and weighing the dough out** is one row where it used to be two, because it is one job under two names, and a **retard** brick is named as something the scenario said but the line has no field for rather than quietly writing a tray count onto a field that no longer exists. The bricks themselves are untouched, and nothing is written until you press the button. What did **not** ship is a button on the Scenario planner to push a planned day's own start times onto the bricks: your bricks are timed from a day that starts at 3 pm, and the bake day here is a morning that has to be ready by 8 am, so writing those times in would have

moved every saved brick. The timetable is drawn, and it is not pushed anywhere yet.

**What has not moved.** The Scenario planner, brick for brick. Your own line still reads 24 pans across 9 bricks and your sister's still 4 pans in 8 bricks, and both are stored exactly as they were - not a brick, a start time or a merge has changed. The chiller bricks stay where they are, switched off, because the fridge is still worth planning for.

CHANGELOG and the guide (section 24) both carry all of this, both PDFs rebuilt, and the tests are 1084 passing with none failing. No database step - nothing to run in Supabase.

**21 Sep 2026 - engine v144 (no database step). The People row counts the places one person is in, a line of a brick counting as a place - and a collision is written so it can be read.**

**What you asked.** In your own words: *"fix the hats count and the collision sentence. And what is the concept of hat, i dont understand, why name them by numbers?"*. Both were real faults, and the question was fair - "hat" was my word, and it was a poor one. It is gone from the app and from this book; what it meant is said plainly below.

**What a "hat" was.** It was my shorthand for **the places one person has to be in the day**. You put it better yourself: *"one person would not need to wear too many hats"* - one worker asked to stand at six different jobs in one day is being handed six hats. That is the thing the People row is there to count, so that is what it now says: **in 6 places**.

**The count was wrong.** Each person's row read **at 6 bricks** - the number of different bricks, not the number of places - so a worker on **line 1 and line 2 of the same fold** counted as being in **one** place. That is exactly the doubling-up the number is supposed to show you, and it was the one thing the row said nothing about. A line of a brick is a place of its own now, so that same worker reads **in 2 places**. On your own line the row has gone from **at 6 bricks** to **in 7 places**, because your fold is drawn as two lines. The number only appears when it is more than one, exactly as before.

**The collision sentence was unreadable.** It was one long sentence - *Person 1 is at X and Y at the same time, ...* - and your bricks have long names: *Mixing by hand in the tub (set the minutes)* is forty characters, twice in one sentence, with the word *and* between them lost in the middle. It is now three parts on four separate lines. Line one, who and when: *Person 1 - two jobs at once, 4:00 pm to 4:05 pm*. Lines two and three, each job on a line of its own with a bullet in front of it: *Mixing by hand in the tub (set the minutes), line 1* and *Mixing by hand in the tub (set the minutes), line 2*. Line four, what to do about it: *Move one of them along the day, or combine with another person and accept the collision*. Two lines of one brick still say which line they are, because without that they read identically. And if a day has more than three of these, the last block **counts the rest** instead of quietly stopping - a list that stopped at four read as though the day had four problems when it had nine.

**Why the people are numbers.** Because a number is what the app stores: who is on a brick is a number, combining two people is joining two numbers, and the same person on three bricks is the same number in three places. That is what makes "put person 1 on both lines" work in one tap. It also has nothing to do with you - **Person 1, 2, 3** are places in the day, not names of people, and the app never shows them to a customer. Naming them (Mei, Kak, Adik) is a small change on top and I have not built it - say the word if it would help you plan.

**What has not moved.** The day itself. Not one minute, one pan or one figure. Your own line still reads **24 pans across 9 bricks**, the starting line still **12 pans with the chiller as the wall**, your sister's still **4 pans in 8 bricks**. Only the words on one row and the shape of one red note changed.

CHANGELOG and the guide (section 23) both carry all of this, both PDFs rebuilt, and the tests are 1062 passing with none failing. No database step - nothing to run in Supabase.

**21 Sep 2026 - engine v143 (no database step). A time cursor on the day chart: point at the chart and a line follows your pointer down the whole day, reading out the clock, so you can line two bricks up against each other.**

**What you asked.** In your own words: *"i need a time x axis cursor on the time chart"*.

**Where it is.** The day chart in **More -> Scenario planner**. One change, in one screen.

**How to use it.** Run the pointer across the chart and a fine line follows it, down through every row - the ruler, every brick, every person, the total. A little brown label at the top of the line reads the clock at that point: 5:05 pm. Move the pointer and the reading moves with it; move the pointer off the chart and it goes away.

**On a phone.** A finger cannot hover, so on a phone the **clock strip is the handle** - the one band along the top of the chart where the hours are written. Put your finger on that strip and slide it left or right and the line follows, reading out the time. Everywhere else on the chart your finger still scrolls the day the way it always did, so you have not lost the swipe.

**Why it only counts in fives.** The line reads the clock to the same five minutes a bar snaps to when you drag it. That is deliberate: the cursor can never name a time the chart could not also set, so you are never shown a time you cannot ask for.

**At the ends of the day.** Point before the day starts or past the end of it and the line goes away rather than naming a time that is not in your day. At the far end of the evening the reading hangs to the left of the line instead of the right, so it is always written out in full and never half cut off.

**Two bricks lined up.** This is what the line is for. Point at where a brick starts, read the time off the label, then point at where the next one starts and read that - no counting squares. It reads the chart and it writes nothing: no brick, no time and no number changes because you pointed at something.

**Nothing you have has moved.** Not one figure. Your own line still reads 24 pans across 9 bricks, the starting line still 12 pans with the chiller as the wall, and your sister's still 4 pans in 8 bricks.

CHANGELOG and the guide (section 23) both carry all of this, both PDFs rebuilt, and the tests are 1061 passing with none failing. No database step - nothing to run in Supabase.

**21 Sep 2026 - engine v142 (no database step). Typing a number into a long card no longer throws the card - the box keeps your cursor and the card stays where you were reading.**

**What you asked.** In your own words: *"why the display jump and it is very confusing"*. It was a real fault, and this is it fixed.

**What was actually happening.** The brick's card is a long card - fifteen boxes, a box per line, a row for every cycle - and **every single number you typed into it was rebuilding the whole card underneath your finger**. That is the jump you saw: the box you were typing in was thrown away and a brand new one put in its place, so the cursor went with it and the keypad shut. On a phone that reads as one digit per tap, on a card that had moved somewhere else. It was worst on the three boxes that change how tall the card is - typing a 6 into *How many of these do you have* grew the card by a line's worth of boxes on every keystroke.

**What it does now.** Typing a number writes the number and **leaves the card alone**. The box keeps your cursor, the keypad stays open, and the card stays exactly where you had scrolled to - checked box by box: the brick's own numbers, the people box, and every line box under *How many of these do you have*. Three boxes still change the card's shape, because their answer really does add a row or move a time - **How many times it runs** (one more row per cycle), **How many of these you have** (one more line box), and **Minutes in, when its first pass starts** (cycle 1's own time). Those three now redraw **once, when you leave the box**, instead of once per digit.

**And it is not only the brick's card.** The jumping was in the shared pop-up that every screen in the app opens, so the same fix covers all of them: any card long enough to scroll keeps its place while you work down it, and a card you open fresh still starts at the top where it should.

**Nothing you have is affected.** This changes only when the screen redraws, never what it redraws: every figure, every scenario and every saved line reads exactly as it did in v141. Nothing new to run in the database.

**21 Sep 2026 - engine v141 (no database step). A brick you have two of is now drawn as two lines, and each line carries its own person - so one worker stands at one job instead of wearing every hat in the day.**

**What you asked.** In your own words: *"i want is just better drawing.. Since now one brick having 2 lines, each line can have its own person, and few line can have a combine person sharing to different lines. This will make training of worker easier, because one person would not need to wear to many hats".* So this is the **drawing**, and **who is on each line** - nothing about how the day adds up has moved.

**A line, and why a brick you have two of is two of them.** Everything on this screen was already worked out as if a brick you have two of were two streams taking the lots in turn - lot one and lot three on one, lot two and lot four on the other. The screen was drawing those two streams flattened into a single row. Now each one is a row of its own: the brick's own name, badges and minutes on the first line, and a row per further line underneath, tied together by an accent down the left and a dotted seam between them.

**Who is on each line is yours to set.** Open the brick and, once *How many of these do you have* says 2 or more, you get a box per line - **Line 1 - who is on it, Line 2 - who is on it.** 0 means whoever is free. Put the same number on two lines and that is one person covering both, which is exactly what you asked for; if those two lines really do need them in the same minute, their row in the People list goes red and names the minute and which line collided with which. The line boxes and the brick's own *Who is at this brick* are the same number, so the two can never disagree.

**The People list now counts the places one person has to be.** Each person's row reads how many different places that one person is in - and every job on it names its line ("Stretch and fold in the tub, line 2"). That number is the whole point of this release: it is the one that tells you when one person is being asked to be in too many places at once.

*(This entry was first written as "counts hats", as **at 3 bricks**. Both the word and the count were wrong, and **v144** fixes them: the row now reads **in 3 places**, and a line of a brick counts as a place. See the v144 entry at the top.)*

**Your starting point in one tap. One a line** (it was *One a brick*) gives every line its own person, so a brick you have two of takes two people rather than one. It says *One a line* only when a brick in the scenario really is drawn as lines; otherwise it reads exactly as it always did. Its name and the message it leaves say which of the two it just did.

**What has not moved.** With no lines drawn and nobody set - which is every brick you have - the day is computed exactly as before, and every figure you have already been given reads identically: your own scenario still 24 pans and exactly the day, the seeded line still 12 pans with the chiller as its wall, your sister's still 4 pans in 8 bricks. Two of a brick still buys cycles out of the pans you planned; it does not bake pans you did not plan.

**One thing it deliberately does not do.** With **Let its cycles overlap** switched on, a brick's lots no longer take turns, so there are no lines to draw and the brick goes back to one row with one person. The people you set per line are **kept**, not thrown away, and the brick's card says so - switch overlapping back off and your lines come back, each with the person you gave it.

**Nothing to run in the database.** This is a screen and a drawing; no SQL, no new table.

**21 Sep 2026 - engine v140 (no database step). A brick can now be told that its cycles may overlap - one switch, with the rule for when to use it written beside it - and the two lots are drawn as two bars.**

**What you asked.** In your own words: *"i also want to allow each brick cycle to overlap, this will improve the visualisation"*, and then, as I was building it: *"allow overlap button and the overlapping criteria"*. So it is a **button** in the brick's card, **Let its cycles overlap**, and the criteria sit under it in plain words rather than in a rule you would have to work out.

**What a brick did before, and still does unless you switch this on.** A brick held its own cycles apart: lot two could not begin until lot one was out of it. That is the right rule when the dough is physically *in* the thing, and it is what stops you folding dough that is still in the mixer. Where you switch it on: when the minutes in a pass are the **dough's** time and not a machine's - dough resting between folds, a second bin on the go. Then every cycle sits exactly where you put it.

**How you can tell it is on.** The brick's row wears a **2 at once** tag naming how many lots are in it together, and

the row grows taller so the overlapping lots are drawn as separate bars in their own lanes instead of painting over each other. A brick that is not overlapping looks exactly as it did - same bar, same height, same place.

**Three things it deliberately does not do.** It never lets a lot start before the dough exists: a brick that waits for the brick above still waits, so switching this on is not a way around the chain. It does not put pans in the day that you did not plan. And it does not quietly double you up - if two of your own cycles need the same person in the same minute, that person's row still goes red and names the minute.

**And it is not the same as a second brick. How many of these do you have** means one lot in per brick you have - with 2 of them, lot three may start while lot one is still folding, but lot four waits for lot two. **Let its cycles overlap** means every lot at once. The switch is the wider of the two.

**One correction to v139, written down so it never looks like a bug.** v139 said a brick whose passes come round faster than one pass takes "now has its passes held apart instead of drawn overlapping - that day could never be worked". That is still what happens **with the switch off**, which is how every brick arrives. With it on, those passes are drawn overlapping, in lanes, which is the whole point of the switch.

Nothing else in the app moved, and both of your scenarios read exactly as they did - the sister's line still 4 pans in 8 bricks. Tests are 1048 passing with none failing, and the guide (section 23) carries all of it.

**21 Sep 2026 - engine v139 (no database step). The scenario planner now builds a real line: a brick can be two of them, every pass in a brick has its own start time you can drag or type, and a brick can be set to wait for the brick above it - lot 10 waits for lot 10.**

**What you asked.** In your own list: "1. allow me to set the same brick to 2 or more, if that module became bottle neck.", "2. in a brick, let say we might need 8 cycles, each cycle should be able to have its own start time, drag-able.", "3. 2nd brick 1st cycle should follow 1st brick 1st cycle completion, and so on.", "4. Each brick, cycle10 only starts after one brick earlier cycle 10 ends, etc.", and "5. objectives are: less man, less manhours thru better planning, shorter hours." Then, as I was building the screen: "from your layout, the stretch and fold should have 2nd , 3rd brick as the person who do that process and start another cycle without having to wait for thee 1st cycle to finish" - which is what **How many of these do you have** does: a second fold starts the next lot without waiting for the first to finish.

**What changed. Everything below is inside More -> Scenario planner. Nothing else in the app moved, and every figure you have already given reads exactly as it did.**

**Every pass in a brick is drawn on its own, and dragging one moves only that one.** Before this version a brick's passes were a single rhythm - a gap and a count - and every pass sat where that rhythm put it. Now each pass is **its own bar on the timeline** and can be dragged by itself: cycle 3 moves and cycles 1, 2 and 4 stay where they were. The toast when you let go names the one you moved - "Cycle 3 -> 4:15 pm" - so there is no doubt which pass changed. For a small nudge, a number beats a careful finger: the brick's card now has a **row per cycle**, each with a box for the exact minute and the clock reading beside it, so a five-minute correction is typed rather than dragged. And **Space them evenly again** puts every pass back on the even rhythm from the first one, once you are done experimenting.

**A brick can be two of them - or more.** A new box in the brick's card, **How many of these do you have**, and the row wears a **2 of them** tag when you raise it. What a second one really buys you, and what it does not: it lets the brick hold **twice as many passes in a day**, and it runs at **twice the pans an hour** - but it does **not** change the pans you planned, because a second mixer standing idle does not bake anything. To use it you raise the passes. That is the honest reading of buying one: it removes a wait, it does not invent work. With **2 of them**, that brick's own passes are allowed to overlap, because two machines can run at once - which is the whole point of having two.

**A brick can wait for the brick above it - and that is the chain you described.** A switch in the brick's card called **Waits for the brick above**, and it does exactly your points 3 and 4: lot 1 of this brick cannot start until lot 1 of the brick above has finished, lot 2 waits for lot 2, and so on down the line. Turn it on for the bricks that really are one line and a slow pass upstream holds every later lot behind it - which is precisely why a second fold, or a second mixer, is worth buying. Two things worth knowing. **A chained bar cannot be dragged**, because where it

sits is decided by the brick above: those bars are drawn faded and wear a **waits above** tag, so nothing on the screen looks like a control that does nothing. To move a chained lot you move the cycle it is waiting on, and the whole line moves with it - which is the lesson the screen is for. And **if two bricks have different numbers of passes**, the screen says so instead of guessing: passes past the end of the brick above follow its *last* pass, and the row names it. **The chain arrives switched off**, so nothing you have already measured moves until you turn it on yourself.

**The line, in one tap - and your three objectives as three numbers.** You named the goal as three things, so the screen now says all three in one line under the day: **2 people · 13 h 40 m of hands · a 16 h 20 m day**. **People** is the most hands needed at once (*less man*), **of hands** is every person-minute of the day added together (*less manhours*), and **the day** is first job to last (*shorter hours*). Drag a cycle, move a start, add a second brick, and all three answer again at once - so a change either improves all three or you can see which one it cost. Beside the timeline a chip reads **3 waits above**, or **Not chained**: tap it and it explains the rule, names which bricks are waiting, shows those same three numbers, and offers the whole thing in one tap - **Chain the whole line** or **Take the waiting off** - so you never set eight switches by hand.

**The climb can now offer a second one, not only more passes.** The ladder used to relieve the wall by raising that brick's passes, and stopped where a day could not hold more of them. That stopping point is exactly where a second brick is the real answer, so a rung can now read **2 of them** as well as **more passes**, with the same before-and-after pans figures. A rung that adds a second one says what it costs in as many words - **that is a second one to buy** - because that rung is money where the others are time. A brick that needs both is offered both, in order.

**One arithmetic change worth writing down, so it never looks like a bug.** A brick whose passes come round *faster* than one pass takes - a 5-minute gap on a 28-minute fold - was always a drawing of something that cannot happen: the same hands cannot start pass 2 while pass 1 is still on them. Those passes are now held apart to one at a time, when the brick is the only one of it and is not chained. None of your own bricks is built that way and every figure you have given reads exactly as it did, but if you ever type a gap shorter than a pass, the screen will hold the passes apart rather than draw a day you cannot work.

**21 Sep 2026 - engine v138 (no database step). The scenario planner is yours to build in: bricks you make, move, give to a person and drag along the day, a ruler and a start time, scenarios you save and name - and the scenario you choose can now be loaded onto the Production line.**

**What you asked.** In your own list: *"1st i cannot change the time scale, say i want a strat at 8am. And i need an time scale ruler", "I need to move the sequence of the lego brick", "I want to be able to create, edit and delete lego brick", "i want to be able to same and name a scenario. Edit or delete", "how to have person ??, I would want one person for each module, And a Total person showing overlapping persons time slot if any", "The lego brick should be able to restart? if choosen to, and how, can it be drag tomove the start time of each brick", and "The fold brick should show 28min, and its person should show 2 minutes, when i adjust the brick until minumum total person overlap, that will be best scenario for manpower planning."* Then, as I was building it: *"can allow to save a scenario? scenario and production line has some function overlapping, the chooses scenario should be able to be loaded to production line"*.

**What changed. Everything below is inside More -> Scenario planner, plus one card on More -> Production line. Nothing else in the app moved.**

**The day starts when you say, and there is a ruler to set it by.** One box at the top sets **the time your day begins**, and every clock reading on the screen is counted from it - so *"say i want a strat at 8am"* is one field rather than ten start times to retype. A ruler now runs under the timeline: a tick every half hour with the o'clock ones named, so a bar can be read against the clock instead of guessed at. And the **time scale** is yours: four chips - **Wide, Standard, Close, Closest** - change how wide a minute is drawn, from the whole day on one phone screen to close enough to tell a 15-minute bake from the one before it. The hour lines on the timeline follow whichever you pick, so grid and ruler always agree.

**A brick is now a thing you own, not a row you read. + New brick** adds one, and it arrives **switched off** so an

empty brick cannot become the wall and answer your day with a nonsense figure. Open any brick and you can rename it, change its picture, change every number in it, and - this is new - **delete it** or move it **up Earlier / down Later**, because the list order is the day's order: the timeline is read top to bottom and a tie for the wall goes to the earlier brick. So *"move the sequence of the lego brick"* is two buttons, and the day re-sorts around it.

**One person for each module, and the Total person you described.** Every brick can be given to a person: **0** means whoever is free, **1, 2, 3...** names someone. Two chips do it in one tap - **One a brick** gives each job its own person, **Share them** hands them all back to whoever is free. Under the timeline there is now **one row per person**, each showing the jobs that person is at and how much work they carry, and where two of their jobs overlap the bar wears a red outline with a sentence naming **both jobs and the minute they clash**. Nothing blocks a plan that cannot work - the whole point is that it is *shown*. The **Total person** row stacks person 1, person 2, person 3 and the rest on top of each other, which is what you asked for: *"with that i can see, which time period utilise more manpower"*. It reads out the most people needed at once, the busiest stretch by name, and how many minutes are being worked two at a time. Bring the bricks closer together and that row flattens - that is the manpower planning.

**Combining two people, which was your own example. Combine two...** asks two questions: the person who keeps the job, then whose bricks move to them. Person 1 then person 3 makes the row read **Person 1+3** - *"to generate person 1\_3"*, in your words - and whatever then collides is exactly what that combination cannot pay for. Give one brick back in its own editor and the **1+3** label prunes itself the moment person 3 has a brick again, so a label can never claim somebody who is not really there.

**A brick restarts, and a brick can be dragged.** Same number, two ways to move it. Type the gap and the number of runs and the brick restarts itself later in the day - your fold, four times. Or take the bar on the timeline with your finger and **drag it**: the start moves in 5-minute steps, the row's times update as you drag, and letting go will not open the brick's card. Drag to move, tap to open. Your pans do not change when you drag; only your people do.

**Your fold brick now reads the way you described it. It holds for 28 minutes**, it is **2 minutes of you**, and it restarts 30 minutes after the last fold, four times. One honest note: if your phone's fold brick still reads **30** in *minutes one pass holds it*, that is the copy your phone saved from the first version of this screen. Type **28** into that box and it is right. I have not overwritten it for you, because your numbers are never changed by an update - that is a rule of this app, not an oversight.

**Save a scenario, name it, open it, rename it, delete it. Save changes** keeps the one you are in; **Save a copy...** makes a second named one and carries on in it - which is how you keep the line you have and the line you are thinking of buying side by side. **Your scenarios** lists everything saved: tap to open (the row you are in wears a filled dot), or tap the **pencil** to rename or delete it. Deleting a saved scenario never touches the bricks you are working on.

**Your sister's line is in there too, built from her own words.** Next to your saved scenarios there is one more row - **+ My sister proposal 21/9/2026** - and one tap puts it on the shelf and opens it. It is the line she wrote out, and it has **no mixer and no chiller in it**: everything mixed by hand in one big tub, folded in that same tub for 2 minutes every 30 minutes three times, tipped into four pans, an hour in a proofing cabinet, then dimple and oil in the pan, half an hour of rest, the oven, and cooling and packing. As she described it that is **4 pans a day in 8 bricks**, and the thing stopping it is **mixing by hand in the tub** - not the oven. Bring the bricks closer together and the Total person row will show you where the three folds collide with filling the pans.

**Two of its numbers are mine, not hers, and both say so on the brick.** She never gave a mixing time - she mixes by hand, so there is no machine time to give - and I have seeded it with the length of your own mixer pass so the brick is not blank. The **Into the pans** brick is worked out from **your** measured 3 minutes a pan, which is 12 minutes for four. Everything else is her own numbers, and every number on every brick is editable. The tub mixes **4 pans in one go**, so if her tub is bigger, change the batch on that brick and the whole day re-reads itself.

**One repair under this.** Opening a saved scenario used to leave the box at the top of the screen still reading the

scenario you were in *before* it - the name, the start time and the pans you want. It now repaints with the scenario you actually opened, so the top of the screen and the row you just tapped always agree.

**And the chosen scenario loads onto the Production line.** A new card at the top of More -> **Production line - Start from a scenario** - lists everything you have saved. Tapping one shows you **every number it would change before anything moves**: what it was, what it would become, and which numbers it would leave exactly as they are. So that the two screens can agree without retyping, every brick's card now carries one more line - **which step of your line is this** - where you say whether it is the mixer, the wash, weighing the dough out, the topping, the oven, the cooling or the retard. A brick's **minutes** and its **batch** are what cross over; a brick that is not a step of the line (your fold, loading the chiller) is named as left out rather than silently dropped. **How many pans you own** and **how many hours you will bake for** stay exactly as they are, because a scenario does not know them and the screen says so instead of filling them with a nought. Nothing is written until you press the button, and the button counts the numbers that would really move - so a scenario you have already loaded says *"nothing here changes"* rather than promising eleven.

**21 Sep 2026 - engine v137 (no database step). More -> Scenario planner: build your line out of modules, watch the day as a timing diagram, and raise the pans you want until something stops you.**

**What you asked.** "can i have a apps function that visualise the flow... something for me to optimise line balancing", and then, once we were talking about it: a screen you could **configure**, a **timing diagram**, and - this was the important one - *"not optimised for mobile, for a clearer interface"*, so you could "toase about all possibility to configure the production flows". You described the parts yourself: a module around the oven, a module around the mixer, a module around a fermentation, each with its **own cycle time**; the folds as a **loop**; ingredients going in "from time to time, not one shot"; and the fridge and the bins as **half-finished goods** sitting between steps. And you set the rule for the whole exercise: *"alll this are design, and require proofing, and before the proofing, the fugures have to be right first, at least logically."*

**What changed.** A new screen, **More -> Scenario planner**. It is the one screen in the app that is **wider than the rest** - up to 900px instead of 540 - because a day laid out across a timeline needs the room. Everything else about the app is exactly as it was.

**A module is a Lego brick: a machine and the pair of hands that tends it, counted as one thing.** Each module has its own cycle time - how long one pass holds it - how many pans that pass deals with, how many minutes of you it takes, and how often it repeats. Your fold loop is just a module that repeats more often, so a loop needs no special anything. Ten modules come seeded: the mixer, the fold loop, wash/oil/fill, load the chiller, retard overnight, unload, dimple and top, the oven, cool and pack - and the fridge, which arrives **switched off**.

**Scenario 1 is the line without a fridge, and the fridge is a switch, not an absence.** It sits in the module list switched off and reads *"not in this scenario"*, so the screen answers without it. Switch it on and it appears. That is deliberate, because you said it plainly: *"if the output fulfilled my need now, then fridge would not be the 1st thing to buy for me"* - so the buying question and the design question had to be the same screen, one tap apart.

**The day is a timing diagram.** One row per module running across the day, with the clock along the top every two hours. A **solid block is you standing there**; a pale one is the module running without you - the retard and the bake have no solid block at all because they need nobody. Below the modules is a row **per person**, carrying the jobs that person is at. There is one person on your seeded day, doing 4 h 56 min of work. Swipe sideways; the day is wider than the phone.

**The climb - this is the part you asked for.** You described the method yourself: *"i will slowly increase the output needed per day until 1st limitation hit, tweaking the 1st limitation and expecting the output to grows, until the next limitation surface"*. So: type how many pans a day you want, and a ladder appears. Each rung names the one module that stops you at that number, exactly what to change about that module, what the day becomes, and which module becomes the wall next. **One tap puts the whole ladder into the modules.**

**And the ladder has to be honest, which is where it earns its keep.** Your seeded day makes **12 pans** and it is the chiller that stops it. Raise the target to 36 and the ladder climbs to **24** - then stops, and says why in plain words: *"A 12 h pass fits 2 times in a day, and it is already running that often. So this one cannot be run more*

*often - it has to take more pans at once.*" Twelve trays, twice a day, is twenty-four pans and no amount of wanting makes it more. **Thirty-six pans needs eighteen trays.** It is the same trays-to-a-day arithmetic from the Production line, now holding this screen to account - and it means this screen can talk you **out** of a purchase as well as into one.

**Every figure is yours to move.** Tap any row - or any brick in the "Not in this scenario" list - and one card opens with that module's numbers, each label naming its own unit: minutes one pass holds it, pans in one pass, minutes of you, minutes from one pass to the next, passes in the day, minutes in when the first pass starts, people this pass needs. Nothing here reads an order and nothing here blocks a sale; it is a planner you type into.

**One more honest note.** A day can only hold so many passes of something: a 12-hour retard fits twice in 24 hours, and the screen will not count a third however many you type. Your number is kept exactly as you typed it - the row just wears an *"a day's limit"* tag and counts the passes that are really there.

**20 Sep 2026 - engine v136 (no database step). The Production line now opens with a flow map: your bake day top to bottom, in the order it actually happens.**

**What you asked.** "can you do a flow map so that i can visualise better, make a intuitive one". The four bars answered *how fast* each part is, but nothing on the screen showed the **order** - that the dough is mixed first, rests overnight, and only then goes pan by pan through the oven and back round again. Numbers side by side don't tell you what happens next.

**What changed.** A new card called **The flow** now sits at the top of the screen, above the bars. It is the bake day written down the page, one step under the next with a numbered dot and a line running between them, so your eye follows it like a recipe:

> Weighing in and loading the mixer -> Retard overnight in the chiller -> Wash, oil and > fill -> Weighing the dough out into pans -> Dimple and top -> The oven swap and bake -> > Cooling and packing -> The day.

Each step carries its own cost: the hand steps say **minutes a pan**, the chiller, the oven and the pans going round say **pans an hour**, and a step you have not timed says **"not timed yet"** rather than quietly reading as free. The last node states the day the line actually makes. The four bars you already use are still there, unchanged, directly below the map.

**The step that sets your pace is the red one.** The same red the bars use, so the two agree. Whichever step is the wall wears a **"the slow one"** tag and the one sentence that explains why - at your current numbers that is the chiller, holding 12 trays; fill in the three untimed steps and the mark moves to the wash, and the traffic goes to your hands instead.

**One honest note about the hand steps.** The six hand jobs are one shared pool - you do them in turn - so no single one of them has a pace of its own. The map therefore shows each as **minutes a pan of work** and never as a rate, and when the wall is the hands the "slow one" mark goes on the heaviest job while the sentence beside it says the minutes are shared across your pairs of hands. Only the chiller, the oven and the pans are given a rate, because those are the only three that are not you.

**20 Sep 2026 - engine v135 (no database step). The mixer box on the Production line now explains itself, and its number is right: 25 kg is about 28 pans, not 25.**

**What you asked.** "Most dough in one mix, in pans - What is this?" Fair question, and the fault was mine. Your form asked for your mixer in **litres or kg** ("Most dough your mixer takes in one go - look for the bowl size on the mixer or in the manual"), and the Production line asks for it in **pans**, because every other number on that screen is in pans so it can be compared to the rest. I converted your answer without telling you, and never checked which unit you had meant. So the box read 25 and quietly meant "25 pans" - a figure you had never given.

**What changed.** You told me your mixer takes **25 kg** of dough. One pan takes about **900 g** of dough - that is what bakes down to the 800 g you sell, the rest being water that bakes off and oil that goes in the pan - so 25 kg is about **28 pans**. That is now the figure the box is seeded with, and the hint under it says the whole thing in one line, so the conversion is on screen rather than in my head: *"Your mixer's bowl size, counted in pans: one pan takes about 900 g of dough - that is what bakes down to your 800 g - so a 25 kg bowl is about 28 pans."*

**One caveat, said plainly.** A 25 kg mixer means the bowl holds 25 kg, but a wet focaccia dough climbs the hook and will overflow a bowl filled to the brim, so what you can actually mix in one go is probably less. The hint says so too - lower the number if you never fill it. You know your mixer better than the arithmetic does.

**If your box still reads 25, type 28.** The figure the app remembers for you is the one you last typed, so your own phone keeps showing 25 until you change it - the new seed only reaches a fresh phone. This is the one small thing to do by hand.

**What it affects.** Two things, and both are small at your scale: how many mixes a day takes (the day's pans divided by this), and - once you time "Minutes to weigh in and load one mix" - the mixing minutes divided across each pan. At 28 pans a day of 36 is two mixes.

**20 Sep 2026 - engine v134 (no database step). Every time box on the Production line now says what it wants: minutes.**

**What you asked.** "the field input unit is not shown?" You were right, and it was worse than a missing label - it was six missing labels. Every other box on that screen carries its unit in its own name ("Minutes per bake", "Pans per bake", "Hours you'll bake for"), so the pattern was already there and the time boxes did not follow it. You saw "Wash, oil and fill 6 pans" above a box holding 18 and had to guess whether that meant minutes or seconds.

**What changed.** All six hand-work boxes now lead with the unit, so the box and its number read as one thing:

- Minutes to wash, oil and fill 6 pans Minutes to dimple and top 6 pans Minutes to take 6 out and put 6 in Minutes to weigh in and load one mix Minutes to weigh the dough out into 6 pans Minutes to cool and pack 6 pans

The two card headings say it as well - the times you measured are "each for 6 pans, in minutes", and the rest of the kitchen work is "the hand-work around the bake, in minutes".

**Nothing else moved.** No number changed, no arithmetic changed, no figure on the screen reads differently. This is the labels only, so if you had already worked out what went in each box, nothing you typed has been affected.

**Nothing to set up, and no database step.** Open the app and the boxes say minutes.

**20 Sep 2026 - engine v133 (no database step). The Production line now counts the whole bake day, not three jobs of it: weighing in and loading the mixer, weighing the dough out into pans and cooling and packing each have their own box to time, and any job you have not timed is named on the screen rather than quietly counted as free.**

**What you asked.** "no time allocation for handling the mixer? and weighing the dough, no other process that need manpower, or should still need some resources?" You were right, and it was a real hole in the screen. It was timing three jobs - wash oil and fill, dimple and top, and the oven swap - and treating those three as the whole day's work. The mixing, the weighing out and the packing were not in the arithmetic at all, so the day it promised you was a day nobody could actually have delivered. Worse, the plan I wrote for this screen had named cooling and packing; the screen I built dropped it.

**What changed.** "The rest of the kitchen work" is a new card on the Production line with three more boxes to time, the same shape as the three you already timed: weighing in and loading the mixer, weighing the dough out into pans, and cooling and packing. The list of jobs now reads all six, each with its minutes a pan or the words "not timed", and every pair of hands is placed against all six and not just three.

**A job you have not timed is named, never assumed free.** Leaving a box blank is allowed - you may already have that time inside your wash, oil and fill figure, and there is a line under the box saying so. But a blank box does not silently count as nothing: the screen lists the jobs still untimed and tells you plainly that the day above looks longer than it really is until you time them. A time with nowhere to go - minutes on the mixer but no batch size for it to spread over - is treated the same way, rather than being divided by nothing.

**The mixer is timed per mix, not per pan.** Its minutes are spread over the batch it makes, because that is the honest reading and it means a bigger mixer really does cost you less work for every pan. Twenty minutes of weighing in is one seventh of a pan's work in a 25-pan mixer and half that in a 50-pan one.

**And the answer changes when you time them.** This is the part worth knowing before you do. At your own numbers with the three new boxes filled in - 20 minutes a mix, 3 minutes to weigh out 6 pans, 12 minutes to cool and pack 6 - the whole day's hand-work is 8.3 minutes a pan, so one pair of hands does about 7.2 pans an hour instead of the 12 the screen used to claim. Five hours then makes **36 pans, not 60**. The wall moves too: it stops being the chiller and becomes your own hands. Every number on the screen is yours to correct, and the day figure moves the moment you type.

**Nothing to set up, and no database step.** Open the app and the three new boxes are there, blank and waiting to be timed once.

**19 Sep 2026 - engine v132 (no database step). Your customer's own track page now keeps up with the order: anything you change about an order reaches their page when you save it, and an order you have just added has a page behind its link from the moment it exists.**

**What you asked.** "add new order and edit order din sync?" They were, on your own phones - an order is one record, so an edit made on one phone is on the other within seconds. What was NOT in step was your customer's track page, and that was a real defect. Their page was only rewritten when the delivery day moved, the tracking number changed or the courier charge changed - so if you corrected the items, the price, the address or the customer's name and saved, their page went on showing the order as it used to be: the old items, and the old total, in the very message you had already sent them the link in. Adding an order wrote no page at all, so a link you sent straight away could answer "not found".

**What changed.** Every way of saving an order - adding one, editing one, the Note / tracking box, and deleting the courier's Delivery & fuel line on Money - now hands your customer's page the whole new version of the order, and the page itself decides whether anything it shows has moved. Where the app used to hold a short list of the things it believed that page showed, it now compares the entire row, so a figure on that page can never be left out of the decision again.

**An added order has its page at once.** Adding an order publishes your customer's page there and then, instead of waiting for the first time you happen to change its status, so the track link in your confirmation message always has something behind it.

**A save that changes nothing writes nothing.** The page compares itself with what you last published, so saving an order without changing anything they can see writes no page at all - and if a write is ever refused (a phone with no signal, a SQL script not yet run), the next save tries again rather than counting it as done.

**Nothing to set up, and no database step.** Open the app and it is already in step.

**19 Sep 2026 - engine v131 (no database step). A Production line screen: your bake day drawn as a row of bars, the station that is holding you back named and made the only red thing on the screen, and a ranked list of the changes you could make with what each one actually buys you.**

**What you told me.** "can i have a apps function that visualise the flow, bottle neck highlight and something for me to optimise line balancing, so that i can have an optimise manpower usage?" Then, when I asked how it should help you decide: "Both - ranked levers and what-if", and the numbers should come from "A planner I type into". You asked for all four levers to be yours: **People, Pans and trays, The task times, and Days and hours.**

**The line, drawn.** Open **More -> Production line** and your day sits on one screen: one bar each for **your hands, the chiller, the oven and your pans**, every one measured in the same unit - **pans an hour** - so they can be compared honestly. The shortest bar is the one setting your pace. It wears a red wash, is named in a sentence just underneath, and it is the only red thing on the screen, so your eye lands on the one thing that matters. Each bar carries a plain line saying what that station is: "1 pair of hands · wash, oil, fill, top, oven swap", "12 trays of dough over 5 hours", "6 pans every 15 min".

**What your own numbers already say.** The three tasks you timed for every 6 pans - 18 minutes washing, oiling and filling, 8 dimpling and topping, 4 for the oven swap - are 30 minutes of work for 6 pans, which is **5 minutes of work for every pan**. One pair of hands therefore finishes **12 pans an hour**; your oven does **24**. And the chiller, holding your trays of dough overnight at one tray to one pan, turns 12 trays through a 5-hour day at only **2.4 pans an hour**. So the screen names **the chiller as the wall, not the oven** - your day comes to **12 pans, 48**

short of the 60 you want - and it tells you that **one more pair of hands buys nothing at all** while the trays are already full.

**The third pair of hands buys nothing.** That is the manpower answer, and the screen says it in words rather than leaving you to work it out: once the chiller is out of the way, the **second pair doubles the line** (12 pans an hour to 24) and the **third buys nothing**, because the oven stops at 24. Spending on help before fixing the wall is money standing around.

**Where the hands go.** With two or more pairs the screen splits them by how much work each job is, so two pairs reads "1 on the wash, oil and fill, and 1 on the dimple and top", with the oven swap taken by whoever is free. Every job finishing at the same moment is the whole trick to balancing a line, and it is the same arithmetic at two pairs of hands or at six.

**What to change, ranked.** Six moves are scored by the pans they actually buy on a day like yours and sorted best first - **one more pair of hands, six more trays, six more pans, a wash 10% faster, one more hour, one more pan per bake**. Each row says what the day becomes and which wall is left standing after it. At your numbers only **six more trays** gains anything (+6 pans, taking the day to 18) - and the other five keep their place and say so in as many words ("No help - the chiller still sets the pace"), so nothing on the screen ever looks like a button that does nothing.

**A planner, not a gate.** Nothing on this screen reads an order, nothing here blocks a sale, and nothing here asks you to keep anything in step with it - the numbers are yours to correct right on the screen, and the screen's answers move as you type. It is seeded from the kitchen form you filled in, so it says something true the first time it opens. The planner shares between your phones like the rest of your settings, so both of you plan from the same numbers.

**One number worth settling once.** Your form answered **8** trays where you have also told me **12**. The tray count is your day's hard ceiling, so it decides everything above - the field says so, and it is yours to set.

**19 Sep 2026 - engine v130 (no database step). The courier charge can now be changed from Edit as well, and deleting the courier's line on the Money screen takes the charge off the order with it.**

**What you told me.** "q2, yes make deleting it clear the charge too", and "edit a courier related now only at one button, we should make edit an order able to alter details for courier."

**Deleting the courier's line now clears the charge.** A charge you paid has two halves: the **Delivery & fuel** row in your books, and the charge on the order. The Money screen only shows you the row, so deleting it used to take that half alone - the order still wore the charge, and the next Save in the order's own courier box quietly wrote the row straight back. Now the row and the order go together: the amount, who bore it and the COD mark all come off. The confirmation says so before you agree, naming the order, and the row is labelled **Courier (order #A3F9C2)** so you can see which order you are deleting it from. Deleting an ordinary expense still touches no order at all.

**The courier charge is under Edit too.** The amount, who paid the courier, how they settle it and how you paid are now asked in the Edit order pop-up as well as in the **Note / tracking** box - one shared set of controls, so the two forms cannot word it differently or save a different charge. Edit's own order total moves as you type the amount, so you can check the figure before you save rather than after.

**Nothing was taken away.** The Note / tracking box keeps all of it and is still the quick way in; Edit simply gained the same questions, so an order can be corrected end to end without opening a second box. Nothing changes on its own: a charge is only written when you save, and saving an unchanged form writes an unchanged charge.

**19 Sep 2026 - engine v129 (no database step). Still to collect now counts the courier charge the customer pays with the order, so the row promises the same money their own message asks them for.**

**What you told me.** "q1, it should reflex rm72." RM72 was RM64 of bread plus the RM8 the courier was charging, and the Money screen was showing RM64 - the items alone.

**What it does now.** When the customer bears the courier charge and pays it with the order, **Still to collect** counts what they will actually hand over: the items plus that charge. So the figure on your Money screen is the

same RM72 their own message asks them for, and you are never left counting a charge you are about to be paid.

**A COD charge stays out of it.** When the courier collects the charge at the door, that money was never yours to collect - so it is not in this row either, exactly as it is not in the total you ask them for.

**A charge you bore is not in it.** Nothing changes there: the bread is what they owe you, and the postage is your own cost.

**Your takings do not move.** What has already come in - Cash in, TNG in, the net, every journal - still counts the items alone, because a charge the customer pays is pass-through: it arrives and leaves in the same breath, so it was never profit. Only **Still to collect** changed, because that is the one figure about money still on its way to you.

**19 Sep 2026 - engine v128 (one database step - run supabase/courier\_cod.sql BEFORE you deploy). A courier charge the customer bears can now be a Courier COD charge: the courier collects it at the door instead of it sitting in the total you ask for.**

**What you told me.** "courier charges can be collect, that means customer pay courier upon collect." A charge the customer bears was always being added to the total you ask them for. That is right when they pay it with the order - but when the courier collects it at the door, asking for it by TNG as well means the same money is asked for twice, once by you and once by the courier.

**What it does now. The Note / tracking box asks how they settle it.** Under **Who paid the courier**, answering **The customer paid it** now opens a third box, **How they pay it**, with two choices:

- **With their order (in the total)** - the charge sits inside the total you ask for, exactly as it always has. Nothing you have already recorded changes, because this is where that box starts. **COD - the courier collects it on delivery** - the charge comes out of the total you ask for, and is named everywhere as payable to the courier when the order reaches them.

**Nobody is asked for the same money twice.** The confirmation, the payment reminder and the shipped message all quote the items alone as the total, and name the charge underneath as COD with the instruction spelled out - "Courier charge: RM 8.00 - COD, pay the courier when your order reaches you". The customer's track card says the same. COD is the word the couriers themselves use for a parcel the receiver pays for; it is also called a reverse charge, but COD is what your customers will understand.

**What you see.** The box tells you what the customer owes as you type, and it now reads differently for COD - "the customer owes RM 30.00 - items total RM 30.00, plus RM 8.00 collected by the courier on delivery" - so what you are asking for and what the courier is asking for are never confused. The Edit pop-up's order total follows the same rule. And the order's row is tagged **Courier RM 8.00 · customer · COD**, so you can see at a glance which charges the courier is collecting rather than looking for that money in your tin.

**Your books are untouched, either way.** A charge the customer bears was already pass-through, and COD does not change that - it still never reaches your takings. Only a charge you bore yourself becomes a Delivery & fuel expense, exactly as before. (v129 above changes where that charge is counted under **Still to collect**.)

**One database step.** Run **supabase/courier\_cod.sql** in the Supabase SQL editor BEFORE you deploy this build. The app publishes the whole tracking row in one call, so if that column does not exist yet the call is rejected as a whole and the customer's tracking page stops updating for EVERY order - not only the ones with a COD charge.

**19 Sep 2026 - engine v127 (no database step this time). Deleting a courier charge really deletes it - the tag comes off the order and the charge leaves the customer's track card.**

**What you told me.** "why i delete courier charges and the tag is not remove?" You were right, and it was a real fault, not a display quirk.

**What was wrong.** The charge is two things: the amount, and **who paid it**. Putting **Who paid the courier** back to **Not recorded** felt like deleting the charge - and it is the obvious way to do it, because that picker is what decides where the charge goes - but it only deleted the answer to *who*, and left the amount behind. The order's

row kept a tag reading **Courier RM 8.00 · customer** for a charge nobody had assigned, and nothing you could do in that box would take it off. Worse, it named the customer as the payer when nobody had said so, so a row could claim they owed money they did not.

**What it does now. A charge is the amount AND the payer.** Put the payer back to **Not recorded** and both go: the tag comes off the row, the customer's total goes back to the bread alone, and the expense row goes out of your books. Clearing the amount still does the same, as before. And a charge with no payer recorded - one restored from an older backup, say - is tagged with nothing at all rather than being labelled as the customer's.

**Clearing a charge now reaches the customer's track card too.** It did not before: the card was only republished when a charge was added, so a customer could keep looking at a courier line you had just deleted. Now the card is republished whenever the charge changes, in either direction.

**Where to see it.** Orders -> an order with a courier charge -> **Note / tracking**, then set **Who paid the courier** back to **Not recorded** and Save. The tag goes from the row.

**One thing left as it is, so you know.** Deleting the **Delivery & fuel** expense from your Money screen does *not* take the charge off the order - that row is your own record of money going out, and the order is the record of what happened, so the two are deliberately separate. If you would rather that deleting the expense cleared the order's charge as well, tell me.

**19 Sep 2026 - engine v126 (no database step this time). The messages that ask your customer for money now show how the total is reached, and the app shows you the same arithmetic.**

**What you told me.** You pasted the confirmation for order #7E51BC - 4 × Focaccia 9"x12" 800g, self collect, RM8 courier, Total RM 72.00 - and said: "the message need to show the add up for rm72. And it should be the same for APP."

**What was wrong.** The RM72 itself was right. What was missing was any sign of where it came from. The line above it named the bread without a price, so the RM64 of foccacia and the RM8 courier charge were both invisible, and the total read as a figure picked out of the air. That is the worst place to leave it, because this is the message that asks them for the money.

**What it does now.** When the customer bears a courier charge, the messages no longer state the answer on its own - they show the sum, in the same order every time, with the total underneath:

- Items: Focaccia 9"x12" 800g x4 Items total: RM 64.00 Courier charge: RM 8.00 Total: RM 72.00

**The app says the same words, in the same order.** One piece of code works out those two figures and their total, and the confirmation, the payment reminder, the shipped message and your own screens all quote it. Two surfaces cannot drift apart when there is only one sum.

**On your side of the app.** The **Note / tracking** box's "The customer owes..." line and the **Edit** box's order total both now read the breakdown as **items total ... + courier charge ...** - the same two words the customer sees, so the figure in front of you and the figure they were sent are provably the same arithmetic.

**An order with no courier charge is untouched.** Its messages come out byte for byte as they did before, and the tests hold that.

**Where to see it.** Orders -> an order with a charge the customer pays -> **Send confirmation, Remind for payment** or **Send shipped message**. Your side: that same order's **Note / tracking** and **Edit** boxes.

**One thing left for you to decide - not changed.** The Money screen's "still to collect" counts the items only: RM64 on this order, not the RM72 the customer will hand over. That is deliberate, because the line measures what stays in your purse and the RM8 arrives and leaves again in the same breath. If you would rather it showed the full amount you are about to be handed, tell me and I will change it.

**19 Sep 2026 - engine v125 (no database step this time). The courier charge now appears in the two totals YOU read, and in the one customer message that was still missing it: the confirmation.**

**What you told me.** Straight after deploying v124: "if customer were to paid courier, the total is not shown to me and to customer." You were right, and it was three places, not one.

**What was wrong.** v124 did put a charge the customer bears into the payment reminder, the shipped message and the track card, and named it in each. But the **confirmation** - the very first message, the one carrying the TNG QR that asks them for the money - still quoted the items alone. And on your side of the app, the **Note / tracking / courier** box never showed what the customer owed, and the **Edit** box's order total left the charge out. So both totals you read were lower than the figure the customer had actually been asked for. A customer paying what the confirmation said would have come up short, and you would have had no way to see why.

**What it does now.** The **confirmation** names the charge on its own line just above the total, exactly as the payment reminder already did - so the message that asks for the money and every message after it agree. The **shipped message**, which named the charge but never said what the order came to, now ends on the total as well. In your app, the **Note / tracking / courier** box tells you what the customer owes as you type the amount, and it moves the moment you say the customer paid it - showing the arithmetic, items plus courier, so the figure explains itself. The **Edit** box's order total counts the charge the same way.

**A charge you bear still appears in none of the customer's totals** and in none of the figures they are shown - that part has not changed, and it is still deliberate. Your own books are untouched by a charge the customer pays, exactly as v124 had them.

**Where to see it.** Orders -> open an order -> **Note / tracking** (the total sits under the question of who paid). **Edit** -> the Order total line. The customer's side: all four messages, and their track card.

**19 Sep 2026 - engine v124 (one database step: run supabase/courier\_fee.sql in Supabase BEFORE you deploy this build). The Note / tracking box now records the courier charge itself, and who bore it - and the two answers behave differently on purpose: a charge the customer pays is added to their total and named where they can see it, and a charge you pay comes off your profit under Delivery & fuel.**

**What was wrong.** You looked at the box behind an order's **Note / tracking** button and said the **Paid by** picker there was for courier charges. It was not. It records how the CUSTOMER paid YOU - Cash or TNG - and it is what puts an order's money into the right book on your Money screen. It has been there since v101, from your own words about needing to reconcile cash against the TNG app. Your reading was the more useful one though, because the thing you were looking for did not exist anywhere in the app: there was no field for what the courier charged, and none for who paid it.

**What it does now.** That box carries the **Courier charge** - what it cost to send that order - and, under it, **Who paid the courier**.

**When the customer pays it,** the charge is added to what they owe. It is named on its own line above the total in their payment reminder and their shipped message, and on their track card, so a total that runs a little more than the bread explains itself rather than looking wrong. Your books are not touched at all - and that is deliberate, not an oversight: in your purse that money arrives and leaves again in the same breath, so counting it as takings would only make a good day look better than it was.

**When you pay it,** a third box asks **How you paid the courier**, and the charge becomes an ordinary **Delivery & fuel** expense dated the day you paid it. It then shows up on your Money screen, in that account's own journal, and as a line on your profit statement - so it comes off your profit like any other cost, which is exactly what you asked for: off profit only if you paid it. Recording the same order twice updates that one expense rather than adding a second, and clearing the amount takes it away again.

**Either way, the order's row is tagged** with the charge and whose money it was - amber when it was yours, quiet grey when it was theirs, sitting beside the Cash / TNG tag.

**The box you misread is renamed.** The picker that was called **Paid by** now reads **Paid by the customer**, and it sits below the courier charge rather than above it, so the two questions cannot be confused again: one is what THEY paid YOU, the other is what YOU paid the courier.

**Where to see it.** Orders -> open an order -> **Note / tracking**. The customer's side: the payment reminder, the shipped message, and their track card.

**Before you deploy.** Run **supabase/courier\_fee.sql** once in Supabase (Dashboard -> SQL -> New query ->

Run). The app publishes each order's tracking row in a single call, so if that column is missing the call is refused whole and the customer's track page stops updating for every order - not only the ones with a charge. Run the SQL first, then deploy.

Nothing else moved - your prices, customers, delivery dates, the shop and the way messages are written are all exactly as they were.

**18 Sep 2026 - engine v123 (no database setup needed).** The product dropdown's middle section now groups sold-out and not-sold-that-day together, and a product the shop does not sell on a day is named by the days it IS sold instead of being given a count for a day it was never on.

**What was wrong.** The dropdown's middle section was called **Sold out** and read its count for the day you were adding to. But a product the shop does not sell on that day at all - a Saturday-only loaf on a Wednesday - was sitting in **On the shop**, and worse, could be labelled "sold out" from a count for a day it was never on the menu. Neither told you anything true.

**What it does now.** The middle section is **Unavailable**, and it holds both kinds together: a product that has **sold out** for the day you are adding to, and a product the shop **does not sell on that day**. The second kind is named by the days it IS sold - a Saturday-only loaf on a Wednesday reads "(only Sat & Sun)". The headings are now **On the shop, Unavailable, Taken down**.

**Why they are one section, not two.** They are the same thing to you: an active product you can still choose. You bake to a plan of your own, and as you put it - every day the kitchen may make 12 focaccia, ordering closes the day before at 6pm, and the 12 are not necessarily sold off, so some go to a walk-in or into the fridge. **The picker is a guide while you sell, never a gate:** nothing is blocked, and the order-by deadline you set per product is deliberately not counted, because that deadline stops a stranger ordering - it says nothing about what you may sell by hand.

**Where to see it.** Orders -> **+ New order** -> Items, and Orders -> open an order -> **Edit**.

Nothing else moved - your prices, customers, delivery dates, the shop and the way messages are written are all exactly as they were.

**18 Sep 2026 - engine v122 (no database setup needed).** The product dropdown now shows the three kinds inside the list itself: each kind sits under its own heading, and on a newer phone each section is tinted in its colour.

**What changed.** Adding items to an order - in **+ New order** and in **Edit** - the product list now comes in three labelled sections instead of one long run of names: **On the shop** first, then **Sold out**, then **Taken down**. The headings appear on every phone, because they are part of the list itself.

**The colour.** On a phone with the very latest system - a new iPhone or a recent Android - each section is also tinted inside the open list: **green** for on the shop, **amber** for sold out, **grey** for taken down. The menu then reads as three blocks you can tell apart without reading a word.

**On an older phone.** The open list is drawn by your phone itself, and an older phone will not let a page colour it. You get the same three headings in the same order, and the closed box still wears the colour of whatever it holds - you lose the section tint and nothing else. Nothing is broken, and nothing has moved; only the colour is missing.

**This replaces the v121 note below.** v121 put the colour on the closed box alone, because the open list could not be reached. Newer phones can be reached, so the list carries the colour too.

**Where to see it.** Orders -> **+ New order** -> Items, and Orders -> open an order -> **Edit**. The list is otherwise unchanged: the same products, the same "(hidden)" marker, the same "12 left" count.

Nothing else moved - your prices, customers, delivery dates, the shop and the way messages are written are all exactly as they were.

**18 Sep 2026 - engine v121 (no database setup needed).** The product dropdown now shows you which kind of product each choice is, in colour, and lists them in the order you would look for them.

**What changed.** Adding items to an order - in **+ New order** and in **Edit** - the product list is now sorted into the three kinds you already know, in the order you would reach for them: what is **On the shop** first, then anything that has **sold out** for the day you are adding to, then whatever you have **taken down** (still marked "(hidden)"). A product still in Draft stays out of these lists entirely, as it always has.

**The colour.** The closed box wears the colour of whatever is picked - **green** for on the shop, **amber** for sold out, **grey** for taken down. On an order with several lines that means one glance down the list tells you which lines need a second look, without reading each one.

**Why only the closed box.** The open list is drawn by your phone itself, and iOS does not let a page colour the rows inside it. So the colour sits on the box you see while the list is shut, which is where you read it anyway.

**Where to see it.** Orders -> **+ New order** -> Items, and Orders -> open an order -> **Edit**. The list itself is unchanged: the same products, the same "(hidden)" marker, the same "12 left" count.

Nothing else moved - your prices, customers, delivery dates, the shop and the way messages are written are all exactly as they were.

**18 Sep 2026 - engine v120 (no database setup needed).** One customer is one row. The app now recognises a phone number by its digits, so the same person can no longer appear twice just because their number was written with a + in front of it - and the duplicate already in your app joins itself into one the first time you open it after updating.

**What was wrong.** A customer used to be identified by their number written in exactly the same way every time. So **+60123456789** and **60123456789** looked like two different people, and the Customers list showed the same customer twice. Your messages always went to the right person - only the list disagreed. Numbers are now read by their digits, which means **012-345 6789**, **+60 12-345 6789** and **60123456789** are one person, the same way "012345" already found "012-345" in the finder.

**It tidies itself once.** The first time you open the app after updating, it joins any customer who was split this way, carrying their dog's name and photo, what they like and avoid, and their note across to the one that remains. There is nothing to do - you will simply find one row where there were two. If you are the careful sort, take a cloud backup first from **More -> Backups**: that is the way back if anything looks wrong, and it costs one tap.

**New: "Join with another customer".** Open a customer's history and you will find the button under their number. Tap it, pick the duplicate from the list, and confirm. Reach for it when two rows are split for some other reason - a misspelt name with no number beside it, say - which the tidy above cannot guess at. It moves the other person's orders under the name and number of the customer you were looking at, so open the row you want to KEEP. This one cannot be undone inside the app; your cloud backup is the way back, and the confirm box says so before it acts.

**Two things worth knowing.** Update both phones - a phone still on v119 can save a number the old way from its Profile card, and only the updated build does the tidy. And on the Profile card itself, the number you type is now stored in the one form every label, message and link already uses, so a "+" or a space can never split a customer again.

Nothing else changed - your prices, your delivery dates, the shop and the way messages are written are all exactly as they were.

**18 Sep 2026 - engine v119 (no database setup needed).** Start typing a customer's name in an order form and the people you have already served appear underneath - tap one and their name and WhatsApp number fill themselves in.

**How it works.** Type two letters - "aun" is plenty - and a short list appears right under the name box. Each row shows their number, how many orders they have placed, and the item they buy most, like **012-345 6789 · 5 orders · usually Focaccia**. Tap the right person and the name and the number both go in; the delivery day, the items and self-collect-or-courier stay yours to set. One letter on its own is not a search - it would offer you half your address book - and a name you have never served offers nothing at all. Typing a number without its dashes finds them too, so "012345" works the way "012-345" does.

**Both order forms do it** - the **+ New order** card and the **Edit** pop-up, the same way.

**Two things worth knowing.** It draws on the orders already in the app, so on a brand-new setup it has nothing to offer until your first order is saved. And on the **Edit** form, saving a changed name has always carried that name to every other order belonging to the person it was - that is the one-saved-name rule from v68 and it has not changed, but picking a name is now much easier than typing one, so it is easier to do by accident. If you ever pick the wrong person while editing an order and save it, tell me and I will add a confirmation that asks first.

Nothing else changed - the Customers screen, its finder, your prices and the shop are all exactly as they were.

**18 Sep 2026 - engine v118 (one SQL step in Supabase; nothing to re-install). The order alert on your phone now arrives as a real notification instead of a wall of raw text.** The ping that reaches you when a customer orders was showing the whole message as unreadable computer text - curly brackets, quote marks, the word "title" - instead of a notification. The cause was the way the message was handed to the ntfy service: the channel name has to travel inside the message for ntfy to lay it out as a title and a body. It was missing, so ntfy did the only thing left and printed everything as plain text.

**What you see now.** A notification titled **New order - Jien Luv 2 Bake**, with the delivery day written out in full (Sat 19 Sep 2026, not 2026-09-19), the total as money (RM 10.00), a bread icon, and a higher priority, so it is more likely to make a sound and pop up rather than wait quietly. A courier order also carries the address; a self-collect order no longer shows a delivery address it does not have.

**A tidy-up came with it.** Your channel name used to be typed inside the rule that sends the pings, and three leftover test rules from setting this up carried copies of it too. It now lives in one small table, and the three leftovers have been removed - so the name that guards your alerts exists in exactly two places: your database, and the ntfy app on your phones.

Nothing about ordering changed, and no phone needs re-subscribing.

**17 Sep 2026 - engine v118 (no database setup needed). The Paid step now stays in its place on every order, wearing an X until the money is in - and turns into a green tick when it is.** This replaces what v117 did, on your instruction.

**Your reasoning, and you are right.** The customer's track page is the design: they must see the same steps in the same places every time, and your own app has to read the same way. v117 left the Paid step **off** an order that skipped it - five dots instead of six - which meant the two lines did not line up, and a missing step is itself a thing to puzzle over.

**What happens now.** A regular who pays when she collects still never passes through Paid. But that step keeps its place in the line, and it wears an **X** - amber, deliberate - instead of a tick, and it never turns green while the money is outstanding:

> New tick · Confirmed tick · **Paid X** · Baked tick · Packed tick · Collected / Shipped dot

Press **Paid · Cash** or **Paid · TNG** when she hands the money over - at whatever stage the order has reached - and **the X becomes the tick**: the ordinary green tick, in the same spot, with the **Cash** or **TNG** tag beside the row. The order itself stays where it was: taking the money never drags it backwards.

**Both lines change together.** Your row's map and the **customer's own track page** draw the same six steps, so the X is on their page too, and it turns green there the moment you record the money. A step that is *deliberately gone past but not paid* now looks different from a step that has not been reached (grey) and from a step that is done (green) - which is exactly the distinction that was missing.

**17 Sep 2026 - engine v117 (no database setup needed). A regular who pays when she collects no longer has a Paid step on her order - and the Paid · Cash / Paid · TNG buttons stay on until the money is actually recorded, wherever the order has got to.** From your note about close customers paying by TNG or cash at pickup.

**What was wrong.** Move an order from Confirmed straight to Baked - no payment - and the row's map still showed **Paid with a green tick**. It was claiming money you had not taken. The app assumed "past Paid means paid", which is fine for her old orders but false for a regular who settles up at the counter.

## What happens now.

- **A bypassed order has no Paid step.** Her route reads New -> Confirmed -> Baked -> Packed -> Collected / Shipped: **five dots instead of six**, and no tick for a payment that never happened. **The Paid · Cash / Paid · TNG buttons stay on** at every stage from Paid onwards - Paid, Baked, Packed, and Collected / Shipped - until the money is recorded. So the moment she hands over the cash at the counter, wherever the order has got to, you press the button right there: **the order stays where it is** (marking an order paid never drags it backwards) and the money is stamped with the day it landed, which is the day the Money screen counts it. **And the Paid step comes back once it is paid** - a green tick, in its proper place in the line, with the **Cash** or **TNG** tag beside the row. So the map ends up telling the truth about that order either way. The buttons disappear the moment the payment is recorded, and they never appear before the order reaches the Paid stage. **The customer's own track page follows the same rule** - the same five dots, no Paid tick for a payment that has not happened, and the step appears there too once it is recorded.

One rule behind it, worth knowing: moving an order into Paid **or any later stage** without having pressed a Paid button is taken as *this order owes money*. That is what your older orders are protected from - nothing already in the app changes, because only orders you move now are marked.

**17 Sep 2026 - engine v116 (no database setup needed). A Day one form: your opening balance - the cash in the tin, the money on your phone and what is on your shelf - entered once, in one place.** From you telling me *"we need to enter opening balance"*.

**What an opening balance is here.** Four answers about the morning your books begin: what is in the tin, what is on your phone, what is on your shelf, and who still owes you. Until now the app could take all four, but in three different places and one ingredient at a time, which is a poor way to start.

**The form.** More -> Money -> the new **Day one** line (under **Books**) -> **Set**. It asks for:

- **Cash in your tin** and **Money on your phone (TNG)** - two boxes, and the day your books begin, which takes any date but should be the day you are starting from. **What is on your shelf** - your ingredients, one box each, already in the unit you use for that ingredient (kg for the flour, g for the salt - no converting). Ingredients you have marked as *not something I buy*, like labour, are left out: they have no shelf to count. **Who still owes you** needs no box - those orders stay unpaid and show under **Still to collect**.

**Nothing is half-saved.** Every box is checked before anything is written: if one has something that is not a number, nothing saves at all until you fix it. And **a box you leave empty is left exactly as it is** - so the form is safe to reopen later just to correct one figure, and safe to run again without wiping what you already had.

**What it writes.** The tin and the phone become ordinary money-in rows dated that day (so from then on the Money screen's **Net** is what you should really hold, and the weekly check works from the first week), and the shelf becomes each ingredient's **On hand** - the figure your shopping lists subtract, so your first list buys only what you are genuinely short of.

**What it deliberately does not do.** Money you **owe** - a loan for the oven, a supplier you have not paid - has nowhere to go: the app keeps a loan as a way of paying, not as a debt, so record what it pays for as it happens. Your own money in shows under **Capital you put in** on Profit, never as income. And the tin must not be entered as a sale: a sale has a customer behind it.

Section 3 of the **Money and Profit** manual has been rewritten around the form - it is attached.

**17 Sep 2026 - engine v115 (no database setup needed). The month arrows on Profit work both ways again.** From you telling me *"the profit month can move earlier but cannot move later"*.

**What was wrong.** Open Profit - it starts on this month, where the **right** arrow is correctly switched off, because there are no numbers after today. Press **left** to step back to August, and **right** stayed switched off too, so you could not get back to September. You could walk backwards through the months but never forwards again: to return you had to leave the screen and open it again. The arrow's state was worked out once, when you opened the screen, and then reused on every step - so it kept answering for the month you started on rather than the month you were looking at.

**Fixed.** The arrows are worked out fresh on every step, so **left** and **right** now move both ways between the months you have, and **right** switches off again only when you are back on this month. Verified by stepping back and forward repeatedly. The same fault did not exist on the other calendars - Orders, Delivery dates and the date pickers all recompute their arrows each time they draw.

**17 Sep 2026 - engine v114 (no database setup needed). Every spending line in Profit & Loss now opens - including the ones reading 0.00 - and the lines are twice as tall, so they are easy to tap.** From you asking *"in profit the expenses is not clickable, is that a bug?"*

**You were right, and it was my mistake.** The lines that open their journal were the ones with money in them. A line showing **0.00** was deliberately left dead, because there is no journal to show - but I made it look **exactly** the same as a live one: same colour, same font, no hint. So if your month has most categories at 0.00, nearly every line you tapped did nothing, and "not clickable" was the only sensible conclusion. A line that looks alive and does nothing is worse than no line.

**What it does now.** Every spending line opens: *Packaging, Utilities, Delivery & fuel*, the categories you have added, and **Total expenses**. A line with money in it shows its journal as before. An empty one opens and **says so in your own words** - *"Nothing recorded under Rent in September 2026"*, In / Out / Net at zero, and a line telling you it will fill up on its own as you record spending under that category. The statement's own totals (**Sales, Cost of sales, Gross profit, Net profit**) stay figures rather than doors - you did not ask for those, and they are not spending.

**Two more things found while fixing it.**

- **The lines were only 17 pixels tall** - a small target for a finger, and easy to land in the gap between two lines and hit nothing. Tappable rows everywhere (the Profit statement, the Money screen's cash rows, the Books list) are now **36 pixels**, comfortably thumb-sized. **The Total expenses journal did not say which category a row belonged to.** It read *"2 Sep · boxes · Cash"* with no way to tell what "boxes" was for. It now names it - *"2 Sep · Packaging - boxes · Cash"* - while a single category's journal still reads short, since its title already says which category it is.

**17 Sep 2026 - engine v113 (no database setup needed). Every way you pay now has a book you can always open - including a pocket that has been quiet - behind a new Books line on the Money screen.** From you asking *"where can i find pocket journals"*.

**What was wrong.** The journal rows live on the money card, and that card only shows a pocket's row **when that pocket moved money in the stretch you are looking at.** The card opens on **Today**, so a pocket that paid for something last week had no row today - and no way into its book at all. Cash and TNG never had the problem: their four rows (Cash in, Cash out, TNG in, TNG out) are always there. The pockets were the only books that could go missing.

**The Books line.** Under the two money cards there is now a second line beside *Categories & ways to pay*:

> **Books** > Every way you pay · 5 books, each opening into its own rows [ Open ]

**Open** lists **every** way you pay - Cash, TNG, Loan, Personal Pocket Kean, Personal Pocket Suan, and anything you add later - each with what moved by it in the stretch on screen, each opening into its own book **under the line you tapped**, so the list stays in front of you and you can step from one book to the next. The first method with anything in it opens ready; a quiet pocket says so plainly - *"Nothing moved this way in this stretch"* - with its In / Out / Net at zero.

Two things this keeps honest: a method you have since **renamed or deleted** still gets a line, because its money is still in the books and a figure you cannot open is a figure you have to take on faith; and every figure in the list comes off the Money screen's own rows, so the list and those totals cannot drift apart.

*A small thing the screenshot caught while building this: the first version printed the word "null" under each closed line. Fixed, and there is now a test that fails if it ever comes back.*

**17 Sep 2026 - engine v112 (no database setup needed). You can pay a personal pocket back out of the till in one go, and the category list on the expense form is no longer cut off.** From your question - *"can my*

*own withdrawal payback to the cash register like Personal pocket Kean or Suan?" - and your note about the categories.*

**Pay back a pocket.** On **More -> Money**, beside **+ Put money in**, there is now **+ Pay back a pocket**. It is for the case where a pocket of yours - **Personal Pocket Kean, Personal Pocket Suan**, a loan, or one you add later - paid for something, so the money left you rather than the till; that pocket's line on the Money screen then reads **-RM 40.00**, which means *the till owes it 40*. Open the form and it comes up **already on the pocket that is owed**, with the amount filled in - Kean's RM 40 sitting in the box, and a line saying so. Say whether the money came out of **Cash** or **TNG**, check the day, and press **Pay back**.

It writes both halves at once, which is the whole point: the **till goes down** by that amount (it shows in Cash out and in the Net, and in the Cash journal with your note), and the **pocket's line comes back to zero**. One pocket can be paid back without touching another. Because it is your own money going back to you, the till's side is recorded as a withdrawal - so it never counts as a cost and your **profit does not move**. On the pocket's own list it reads *"Paid back by the till"*, not *"From my pocket"* - the two are different things and now say so. Paying it back in part is fine: pay RM 25 of the RM 40 and the pocket reads -15 after.

**The category list was being cut off - fixed.** Thank you for catching this. On the expense form, the row of categories ran off the side of the screen: it was laid out as one single line, so at phone width everything past about the sixth category - *Salary (you), EPF / SOCSO, Marketing, Equipment & tools, Other, My own withdrawal*, and the **+ New category** chip - was off the screen and impossible to reach. Measured on a 375-pixel phone, the row was 744 pixels wide inside a 343-pixel box. The pills now **wrap onto as many lines as they need**, the same fix applies to the ways-to-pay row and to the new pay-back form, and the whole list is visible and tappable. Nothing about the categories themselves changed.

**17 Sep 2026 - engine v111 (no database setup needed).** Every figure on the books is now a door: a journal behind each line of spending in Profit & Loss, and a book for every way you pay - **Cash, TNG, Loan, Personal Pocket Kean, Personal Pocket Suan, and anything you add later**. From two notes close together: *"the expenses items in Profit & Loss should reveal its journals"*, then *"each CASH, TNG, LOAN, Personal Pocket Kean, Personal Pocket Suan, and others that might be added in future need a journal"*.

**Profit & Loss - tap a spending line and see what made it up.** On **More -> Profit**, the running-cost lines (Packaging, Utilities, Delivery & fuel...) are now tappable, and so is **Total expenses**. Tapping one opens its journal for the month on screen: every expense behind that figure, oldest first, each reading "*<day> · <what it was for> · <how it was paid>*" and ending on the total the statement itself shows. So *"Packaging RM -58.00"* opens two lines - the boxes paid from a pocket and the bags paid in cash. A line with nothing in it that month is not tappable; there is nothing behind it. The header now says **Running costs · tap a line to see the spending behind it**.

**Money - one line and one book per way of paying.** The single *"Paid by loan / other"* row is gone. In its place, every way of paying that is not cash or TNG gets **its own line** with what moved by it, and its own journal behind it: **Paid by Loan, Paid by Personal Pocket Kean, Paid by Personal Pocket Suan, Paid by Bank OD** - whatever you have called them. Money of your own you put in that way reads as a plus on that line; what was paid out of it reads as a minus. Cash in, TNG in, Cash out, TNG out and the Net are exactly as they were, and none of these other ways is in the net - it never came out of your purse.

**Why they appear by themselves.** The lines are read off your own transactions, not off a fixed list, so a method you add tomorrow needs nothing from me: record one payment with it and it has a line and a book. And a method you have since **renamed** keeps a line for the name its older rows were written with - those rows are still in the books, and a figure you cannot open is a figure you have to take on faith. Each book ends on **In / Out / Net** for that method.

**One small correction while here:** the note at the foot of the Profit screen about your own hours now matches v110 - either pay yourself a **Salary (you)** expense (with EPF / SOCSO as their own category), or mark Labour as a not-bought ingredient and put the hours into the recipes. Count them one way, never both.

**17 Sep 2026 - engine v110 (no database setup needed).** An ingredient you never buy - **your own labour**,

**electricity, gas - can be marked as a cost, and it then never appears on a shopping list again.** From your note: *"certain ingredient we dont purchase, in ingredient we can set that as a non purchase item, like labour and electricity"*.

**What it does.** Open **More -> Ingredients** and tap **Edit** on the ingredient (or fill in the New ingredient card at the top). Above the cost there is now a switch: **Not something I buy - it only ever costs. It still counts in a recipe's cost, but it never appears on a shopping list or a purchase order.** Turn it on for Labour, Electricity, Gas, your own time.

**Where you see it.**

- The ingredient's **card** stops showing an On hand and a Keep-at-least line, and says instead: *Not bought - a cost in your recipes, never on a shopping list.* Nothing else about the card changes, and the cost you typed is untouched. Every **shopping list** (the preview, the list you save, and an "orders changed" follow-up) leaves that ingredient out - and says so at the foot of the list: *Not on this list: Labour, Electricity - marked as not something you buy. Their cost still counts in the products that use them.* So it reads as deliberate, not as something the app forgot. The **recipe cost** is unchanged: a recipe using Labour at RM 8 an hour for 15 minutes still adds RM 2.00 to that product's cost, and that runs on into the Profit screen's cost of sales.

**Switching it OFF restores the ingredient exactly as it was** - the app removes the mark rather than storing a "No", so nothing that exists today changes unless you turn this on.

**And a word on the books, since this touches Profit:** if you mark Labour as not-for-purchase because it is your own unpaid time, it is already inside each product's cost of sales through the recipe - so do **not** also record it as a Salary expense, or the same hours get counted twice. The **Salary (you)** category is for when you actually pay yourself money out of the till.

**16 Sep 2026 - engine v109 (no database setup needed). A journal for each way the money moves - the cash book, the TNG book - and a bug that was hiding in the TNG column.** Two things, both from you asking how to see the journals.

**A bug first, because it matters.** Every order you marked **Paid · TNG** since v106 was landing in "Paid, no method" instead of the TNG column. The app was checking for the old way of writing it ("tng") while the paid buttons write the list's own label ("TNG") - so the TNG in figure read RM 0.00 with the money in your phone, and the same money showed as unaccounted for. Fixed, and there is now a test that would have caught it. If your TNG column has looked wrong since v106, that was it.

**Tap a figure, see its journal.** On the Money screen, **Cash in / TNG in / Cash out / TNG out** (and the loan line) can be tapped. Each opens that method's book for the stretch: every order paid that way with its code and customer, everything you spent out of it with its note, anything of your own you put in - in date order, ending on In / Out / **Net**, and a line saying what that is meant to be (your purse, your phone, or money that never went near either).

**16 Sep 2026 - engine v108 (no database setup needed). Where the two lists live, and how to change one.** One fix and one addition, both about the categories and the ways to pay.

**Why.** v106 put them in a Settings card and you could not find it, and there was no way to change an entry once it existed - "dont put the setting separately, it should be at where it suppose to be."

**They are on the Money screen now**, as a line under the money lists reading "Categories & ways to pay · 11 categories · Cash, TNG, Loan" with an **Edit** beside it. Tap Edit and the lists open: every category with its kind (ingredients / running cost / your own money) and every way to pay, each line saying "edit".

**Tap a line and it turns into the form:** rename it, change what kind it is, or delete it - in place, with the rest of the list still in front of you. Renaming moves what you have already recorded with it, so a slip of the thumb does not split your history into "Packaging" and "Packing".

**And the + chips sit where you are working:** the Add an expense form and Put money in each end their pills with **+ New category / + New way to pay**, which opens the same small form in place - the amount and the note you had already typed stay put.

**No window inside a window.** These forms expand in place instead of opening a pop-up of their own, the same rule the date field follows: the app has one pop-up layer, so a second one would wipe out the form underneath it.

**16 Sep 2026 - engine v107 (no database setup needed). The item line on an order reads properly again.** One fix, on the two forms where you type an order's items.

**What was wrong.** When the selling price joined the line (v101) it took most of the row. Measured on a 375px phone: the price box was 183 pixels wide and the product dropdown was squeezed to **2** - so the name of what you had picked was pushed onto a line of its own and unreadable, and the whole line looked broken.

**What it does now.** Each item line is two clean lines: **the product across the top**, full width so its name reads, and its controls under it - how many, the selling price on the right, and the X to remove. The same shape in the + New order card and in the Edit pop-up, since both are the same row.

**16 Sep 2026 - engine v106 (no database setup needed). A note on every transaction, and two lists you shape yourself.** Three small things, all on the money screens.

**A note on each one.** Add an expense now takes a note ("Mydin run, 2 boxes") - the Put money in form already had one - and both lists on the Money screen show it beside the amount, so a row reads as what it was, not just a figure.

**The categories are yours.** More -> Settings gains a **Categories & ways to pay** card. Every category shows what it means to the accounts (ingredients / running cost / your own money), each has a X to delete it, and you can add your own - "Baking class", "SSM licence" - choosing which of the three it is. Deleting one never touches what you have already recorded: those rows keep their label and still count, printed at the end of the statement's cost list rather than dropped.

**And the ways to pay are yours too.** The list is Cash, TNG and **Loan** - the third choice you asked for - and you can add your own, a **Bank OD** or a cheque, so a shopping run nobody would pay for out of the till can still be recorded honestly. The same list appears on the Bought prompt (where the receipt is still in your hand) and on both money forms.

**One thing that follows from it.** Money that never came out of your purse - a loan, an overdraft - no longer sits in the net. The Money screen leaves it out of the cash figures and names it on its own line, "Paid by loan / other", with a sentence saying why. Net now means exactly what it says: what should be in your purse and on your phone.

**16 Sep 2026 - engine v105 (no database setup needed). The books: a profit and loss account, a month at a time.** One new screen, More -> Profit.

**Why.** Money told you what was in your purse. It could not tell you whether the baking made money - a RM250 flour-and-butter run makes one week look like a disaster while the bread it went into sells the week after.

**What it shows.** More -> Profit, month by month (arrows either side), reading down like a set of accounts: Sales, Cost of sales, Gross profit, the running costs one line each, Total expenses and Net profit - with the gross margin beside the order count. Then, kept apart from all of it, your own money: capital you put in, drawings you took out, and what is left in the business.

**The one rule that makes it accounting rather than a cash total.** Ingredient cost is what the baking cost to make, taken from your RECIPES as the bread sells - not the packs you bought. Buying and using sit in different places on purpose: a shopping run is money out on the Money screen and stock on your shelf, and it becomes a cost of sales here as the bread made from it goes out. Nothing is counted twice, and a big stock-up week stops looking like a loss.

**Your own money is neither income nor a cost.** What you put in is capital; what you take out is drawings (including the "My own withdrawal" category). Both move cash - Money counts them - and neither changes profit. If you pay yourself a proper salary, record it as **Salary (you)**, with **EPF / SOCSO** as its own category, and those DO count as running costs.

**The category list is now a chart of accounts:** Ingredients & shopping (stock), Packaging, Rent, Utilities, Delivery & fuel, Salary (you), EPF / SOCSO, Marketing, Equipment & tools, My own withdrawal, Other - the

same list the Add an expense form offers, so every row lands on the right line of the statement.

**Two things it will not pretend to know.** Your own unpaid hours are not a cost - if your time is worth paying, pay yourself a salary and it becomes one. And a product whose recipe prices to nothing is counted as costing nothing; the screen says how many lines that was, so it is visible rather than flattering.

**16 Sep 2026 - engine v104 (no database setup needed). Money you put in yourself, and taking it back out.** One change, on More -> Money.

**Why.** Some spending happens before any order money arrives - a packet of flour paid from your own purse, a float of change for the day - and there was nowhere to put it. The Money screen only counted what customers paid.

**Put money in.** More -> Money now has a **Put money in** button beside Add an expense: how much, the day you put it in, cash or TNG, and a short line saying what it was for. It counts into the Cash in / TNG in rows, because your own money really is in your purse and those rows are what you check the purse against - and one line underneath says how much of the money in was yours: "of the money in, RM 100.00 was your own". The entry is listed in its own **Money in** card, so you can see where it came from, and take it off again.

**Taking it out again.** Add an expense has a new category, **My own withdrawal** - money you take back for yourself rather than a cost of baking. It leaves through the same money-out list as everything else, so the net, the day lines and your backups count it with no new machinery. Your own money in and out is shared between your phones, like every other list.

**16 Sep 2026 - engine v103 (no database setup needed). Money out: what you spend, beside what you take in.** One change, on the shopping side of the app.

**Why.** v101 gave you the money coming in. Spending had nowhere to go: a purchase order worked out what a shopping run should cost, and tapping Bought put the packs on your shelf, but nothing recorded what you actually paid.

**A shopping run now asks.** Tap **Bought** on a saved list and, once the packs are on your stock, a small box opens: **What did you pay?** - pre-filled with the list's own total, with **Cash** or **TNG** beside it. Accept the guess with one tap, or type what the receipt really said. **Skip the money** leaves the stock added and records nothing, exactly as the app behaved before.

**Everything else goes in by hand.** More -> Money now carries an **Add an expense** form for what a purchase order never sees - packaging, delivery and fuel, utilities, equipment, or a market top-up - each with the day you paid it, so a receipt found in your bag later still counts on the right day.

**The Money screen adds it up both ways.** Cash in, TNG in, Cash out, TNG out and **Net** - what should be in your purse and on your phone for Today / This week / This month - with that stretch's spending listed underneath, each line showing what it was for and how you paid. What is still to collect stays out of the net: it is money owed, not money held. Expenses are shared between your phones, and counted in your backups, like every other list.

**One fix along the way.** A payment taken just after midnight was being counted as the day before, because the app read the date off the stamp in UTC. It counts on your own day now.

**16 Sep 2026 - engine v102 (no database setup needed). Swipe across the calendar to pick a run of delivery dates.** One small change, on More -> Delivery dates.

You asked whether the delivery-date calendar could work the way a product's availability calendar does. It now does: **drag across days and the whole run fills in under your finger**, then **Add selected** puts them on the calendar. It is the same gesture as dragging a sell period on a product. A single tap still picks one day (or puts it back), and a day that is already a delivery date is left alone by a drag - it comes off by tapping it, the way it went on.

That makes four ways in: tap dates, swipe a run, tap a weekday letter for every one of that day in the month, or **Generate the next dates** on your Settings pattern.

**What I deliberately did NOT copy: the From / To boxes.** On a product's availability card those two boxes are how a SEASON is said - a sell period genuinely has two ends. A delivery date is one day, not a stretch, so a From/To here would only ever have been a slower way to do what swiping now does.

**16 Sep 2026 - engine v101 (no database setup needed). A price you can change on an order, and the money recorded as cash or TNG.** Three things, all about money.

**A price you can change on an order.** Every item line on the + New order form and in the Edit pop-up now carries its selling price. It opens on what the product costs; type over it and THAT order is sold at your price - the confirmation, the payment and pickup reminders, the receipt, the customer's own track page and every money number in the app follow it. Your menu price is untouched, and a menu price you change next week never rewrites a sale already made. Blank keeps "whatever the product costs", which is how an unpriced product has always behaved. Both forms also show the items total as you go.

**Cash or TNG, written down when the money lands.** The Paid button is now **Paid · Cash** and **Paid · TNG** - one tap each. The order remembers which, and when, and shows it beside its status. The Note / tracking box (now **Note / tracking / payment**) carries a **Paid by** box as well, for an order you marked paid before you could tell, or one to correct later. None of this reaches the customer: their page just shows Paid.

**Two places to check it against.** The delivery day's own header on Orders now carries a short till - "Cash RM 95 · TNG RM 120 · 1 to collect". And More -> **Money** is a screen of its own, with **Today / This week / This month**: cash collected, TNG collected, paid but no method recorded, and what is still to collect with the number of orders. Collected money is counted by the day it LANDED, so a transfer that comes in today counts today even when the order delivers on Friday; what is still to collect is counted by delivery day, because that is the day you hand it over. An order paid before any of this existed shows under "Paid, no method" rather than being guessed at.

**16 Sep 2026 - engine v100 (no database setup needed). Making delivery dates for a weekday is one tap again.** One change, on More -> Delivery dates.

**Why.** You found the function missing: adding your delivery dates for, say, every Monday meant tapping each Monday by hand. Nothing had been removed - the app's only "do the dates for me" button lived on Home, inside the "No delivery dates yet" message. That message shows only while you have no upcoming dates, and pressing the button creates the very dates that hide it: it was a one-shot button that took itself off the page the moment it worked. So once you had dates, which is every day since your first week, it was never there again.

**Three ways in now.** On More -> Delivery dates: tap dates and **Add selected**, as before; or **tap a weekday letter** - tap **M** and every Monday in the month shown is picked for you, tap it again to put them back, and the letter underlines solid once that whole weekday is picked; or press **Generate the next dates**, which is always there, follows the delivery days in your Settings, and adds the next six dates that are not on the calendar yet. Generate moves the calendar to the month the new dates start in, so you can see what it just did.

**One wrong line fixed with it.** Home's "no dates yet" message used to say "the next Mon/Wed/Fri dates" whatever your delivery days really were. It names your own days now (and so does the line under Generate).

**15 Sep 2026 - engine v99 (no database setup needed). The customer can tap the "Next available" line and be given that day.** One change, on the order page.

**Why.** A product you keep on the shop, on a day it cannot be ordered, tells the customer when it can be had - "Next available: Sat 19 Sep". Until now they had to go and find that date on the calendar themselves.

**What it does now.** The line is a control: tapping it takes that delivery day, so the product in front of them can be ordered straight away, and the page moves up to the calendar, where the chosen day is written out in words. It keeps its gentle pulse, and wears a small arrow so it reads as something to tap.

**The one time it does not take the day.** If the customer already has something in their basket, the line stays a plain label and tapping it answers instead: "Your basket is for Fri, 18 Sep. To order for another day, choose it on the calendar above." Taking a day then would silently move their whole order to a new date - and drop anything that does not fit there - which is not something a tap on a product line should ever do. Changing the day is what

the calendar is for.

**15 Sep 2026 - engine v98 (no database setup needed). The last status reads Collected / Shipped, and a placed order gains a Note / tracking button.** Two small changes on the Orders screen.

**One label for the last stage.** v97 named it per order - Collected on one the customer fetches, Shipped on one you post. You asked for the pair itself instead, so the stage now reads **Collected / Shipped** everywhere it is named: the row's status list, the row's own journey map, the day's status filter, and the map on the customer's track page (in all three languages). Which message the row offers is still decided by the delivery method - a courier order shows **Send shipped message**, a self-collect one **Send pickup reminder**.

**A way in for the two fields you reach for most.** Beside Edit, every order now carries a **Note / tracking** button. It opens a small box with exactly two fields - the order's **Note** and the courier's **Courier tracking number** - and a Save. Send, and the note lands on the order, while the tracking number also goes onto the customer's track card and into the shipped message. Nothing else moved: **Edit** still opens the full form (the delivery day, the customer, the address, the items, the quantities) whenever you need it. The tracking box that sat on the row itself in v97 is gone - the button is the quicker, tidier way to the same field, and it works on any order rather than only a courier's.

**15 Sep 2026 - engine v97 (one database line to run once). The last status is named for how the order leaves, and a posted order can carry its courier's tracking number.** One change, across the order row, the WhatsApp messages and the customer's track page.

**The last status is Collected or Shipped.** The final step used to say "Delivered" for everything. It now reads **Collected** on an order the customer fetches from you and **Shipped** on one you post - the same step in the journey, named for what actually happened. That is true everywhere the status is named: the row's status list, the row's own journey map, the day's status filter, and the map the customer sees on their track page (in all three languages). Nothing about the step's behaviour changed, and every order already sitting at that step keeps its place - only the word is different.

**A tracking number, typed where you post the parcel.** A courier order shows a **Courier tracking number** box under its details from Packed onwards, and a **Send shipped message** button beside the status. Type the number the courier gave you, press the button, and WhatsApp opens with the message already drafted: the order number, what was sent, and the tracking number. The box is in the Edit pop-up too, for a number you need to fix or read back over the phone.

**The customer sees it.** The number is published with the order, so a posted order's track page shows it on its own line under the delivery details - they can read it to the courier without messaging you. Self-collect orders show no such line. The shipped message is only offered on courier orders: a self-collect order keeps its **Send pickup reminder**, which is the message that fits that hand-over.

**What you need to run once.** The tracking number needs one column added to the tracking table - supabase/track\_no.sql (a single ALTER, safe to re-run). Until that runs, everything else works; the number just reaches the message and not yet the customer's page.

**15 Sep 2026 - engine v96 (no database setup needed). "Set day's availability" now shows how the day adds up.** One change, on the pop-up you already use.

**Why.** v95 made the day's total count only the products you sell that day, but the pop-up that sets those numbers still showed nothing about where the total came from - and a product you do not sell that day sat in the list looking as if it were part of it.

**What it does now.** Under the product rows, the pop-up ends with **How the day adds up:** one line per product on sale that day with what it contributes, and a final line with the total - **the same number the order page uses for that day**. Type a + or a - and the lines and the total move as you type. Underneath, the pop-up names what it left out ("Not counted: Saturday loaf - not sold on this day, so no order can go on them here") and says what is already booked, so you can read how much the order page can still take. A product not sold that day keeps its row so you can still set its numbers; it simply does not enter the total.

**One number, one place.** That sum and the day's capacity are computed by the same function the shop's numbers come from, so what you read on screen and what your customers get cannot drift apart.

**15 Sep 2026 - engine v95 (no database setup needed). A day's total now counts only the products you actually sell that day.** One change, on your side of the app, with one knock-on for customers.

**What it was doing.** The "x/y" on a day - the chip beside the date on Orders, the same number on the day in the month calendar, and the dials on Home - was adding up the daily limit of every product on your menu, whether or not that product can be ordered on the day you were looking at. So a day with one product on it (a limit of 12, say) still read **1/42**: thirty of those units were limits belonging to products that could never take an order that day.

**What it does now.** Only the products on sale that day count. A Saturday-only loaf adds nothing to a Wednesday, and a product kept on the shop as **Unavailable** adds nothing to a day it is not sold on. The same day now reads **1/12** - what that day can really take. Products with no sell marks are unchanged (they sell every day, so they always count), and a day where nothing on sale has a daily limit falls back to the **default capacity** in Settings, exactly as before - it does not drop to 0, which would make the app call the day Sold out.

**The knock-on worth knowing.** This same number is what the app tells the order page about a day, so a day now turns FULL at the moment every unit you are actually selling that day is booked, rather than at some larger number padded out by products that were never on sale. Past every real slot being taken, the order page stops offering that day - which was always the intent, and now happens at the right moment rather than late.

**Nothing else moved.** Your over-capacity warning, the "N ordered · N left" line on a delivery date, the purchasing plan and each product's own "N left" stamp all read the same number they always did for products that are on sale.

**15 Sep 2026 - engine v94 (no database setup needed). One sentence on the customer's greyed card, reworded.** Wording only - nothing behaves differently.

On a product you keep on the shop, the line saying why it cannot be ordered today now reads **Only available on Mon.** where it read "Only sold on Mon." The old wording was the seller's voice on a customer's card, and "available" is what the rest of that same card already says ("Only available for delivery from 19 Sep"). The Chinese and Bahasa Malaysia say the same thing as before, in their own words.

**15 Sep 2026 - engine v93 (no database setup needed). The track link now lands on the track card, lit.** One change, on the customer's side of the shop.

**Why.** Your confirmation message ends with a Track your order link. Tapping it opened the order page at the very top, with the tracking card somewhere below - the customer had to scroll and hunt for the very thing they had just tapped, on a page they have never seen before.

**What it does now.** The page comes up with the Track your order card already in view AND gently glowing, so it cannot be missed. The glow does not fade on a timer: it goes when their pointer or finger reaches the card, so it is still burning while they are looking around for it. That is the same rule your own app uses when you tap an order in the New orders box and it flashes the row - the flash is there until you arrive.

**Nothing else changed.** Typing a code into the box on the page behaves exactly as before (the card is already in front of them, so nothing moves), and a customer who simply opens the shop sees no glow at all.

**15 Sep 2026 - engine v92 (no database setup needed). The New product card on Products folds away.** One change, on one screen.

**Why.** The card that adds a product sat open at the top of the Products screen, and open it is a long form - it pushed your three lists (On the shop, Draft, Hidden) right off the bottom, so every look at what you sell began with a scroll past a form you were not filling in.

**What it does now.** The card arrives folded to a single line reading **+ New product**. Tap it and the form opens; tap the title again, or tap anywhere else on the screen, to fold it away. It stays open after you add a product - the toast still says it was saved as a draft - so you can type the next one straight away, and it starts folded again the next time you come to the screen. Nothing about adding a product changed: the form inside is exactly as it was, and it is still the same card that stays on the page while an Edit pop-up is open over it.

**15 Sep 2026 - engine v91 (no database setup needed). The Orders calendar now answers a tap it cannot act on.** One change, on one screen.

**Tapping a day you do not deliver on.** The month calendar at the top of Orders rings your delivery days in green with how booked each one is, and draws every other day of the month plain. A plain day never did anything before - tapping it was silent, which left you guessing at the reason. Now the calendar answers: a warm line appears under the grid naming the day and saying it is not a delivery day, and that delivery days are added in More -> Delivery dates. A marked day says its usual name above the day at the same time, so a holiday that is not a delivery date now tells you both what it is and why no order can go on it. Opening a real day, or tapping another plain one, replaces the line. Nothing can be ordered on a plain day, and nothing is created by the tap - the calendar simply says so now instead of staying quiet. A plain day already gone stays silent: there is nothing left to add to it.

**Where it appears.** The month calendar at the top of Orders, the day picker inside the + New order card, and the Delivery day picker in an order's Edit pop-up - every calendar in the app that offers only your delivery days.

**Nothing else changed.** Delivery dates are still made in exactly one place, More -> Delivery dates (or **Generate next delivery dates** on Home). Tapping a calendar elsewhere in the app does not create one, and a tap that could open a day before still opens it.

**15 Sep 2026 - engine v90 (no database setup needed). A product can stay on the shop on a day it cannot be ordered, with the date a customer can next have it.** One new switch, off unless you turn it on.

**Keep it on the shop.** A product's Availability card now opens with a switch at the very top, above the sell-day calendar. It is for the two or three hot items customers come back looking for - if it is not listed they wonder whether you still make it - but leave it on for everything and the menu fills up, so it is a decision you make one product at a time. The card's own title carries the state too, so you can read it while the card is still closed.

**What the shop shows with it on.** On a delivery day the product is not sold on, the card stays instead of vanishing: greyed, stamped **Unavailable** (a day you never sell it is not the same as a day it ran out), with the reason it already writes ("Only sold on Mon") and one more line - the next date it can be ordered. On a day it has sold out the card is exactly what it was, **Sold out** stamp and all, and gains that same line. The line reads **Next available: Sat 19 Sep**, and names how many are left that day when a daily limit is set; it pulses gently so it reads through the dimming, and sits still - simply lit - if your phone asks for reduced motion. Nothing on a greyed card can be ordered: the + and - do nothing, it never enters a basket, and a line already in a basket still leaves with a note when the day changes. The switch never overrides **Draft** or **Hidden** - those still take a product off the shop entirely.

**With the switch off, nothing at all changes.** Off is the default and no product you already have carries the setting, so every product behaves exactly as it does today until you switch one on. One consequence worth knowing: on a day where every product is unavailable, a shop with kept items shows those greyed cards instead of the "nothing is on the menu" line.

**14 Sep 2026 - engine v89 (no database setup needed). The Delivery Dates screen trades its list for the calendar: a ticked day comes off when you tap it again, and the past dates fold away into one group.** Three small changes, all on one screen.

**Tapping a ticked day takes it back off.** A delivery date that was already on the calendar used to come off only from the list underneath it. Now tapping that green day takes it off on the spot - the same tap that put it there undoes it, so the calendar is the one place you both add and remove dates. If that day already has an order on it the app asks first ("2026-09-16 has 2 order(s) on it. Delete the date? The orders are kept in your delivery history."), because that is the one removal worth a second look; a day with nothing on it just goes, with a small "Delivery date removed" note. The date is all that is ever removed - the orders stay in your history either way.

**The list of dates still to come is gone.** The calendar already draws every delivery date as a green tick, and now it lets you take any one of them back, so the list below was repeating what was already in front of you. That is most of why this screen reads so much lighter. Nothing else moved: the green tick, the coloured holiday tags and the "N ordered · N left" line are all as they were.

**The past dates are one group, folded away.** Dates already gone keep their green tick but cannot be tapped, so nothing on a past day can be changed by a stray touch. They now sit together under a **Past dates (12)** heading that starts folded - tap the heading to open the list, tap it again to fold it back. Every past date is in there now: the old list showed only the ten most recent and left the rest out of reach. Each one still opens its orders or can be deleted exactly as before, with the same "Del" button and the same question when orders sit on it.

**14 Sep 2026 - engine v88 (no database setup needed). The holiday tint sits on the day numbers at the Orders screen, and a mouse resting on a marked day names it.** Two small things, both about the Orders screen and the bubble.

**The tint now sits on the line of dates.** On the Orders calendar each delivery day carries its booking count, and that count sat *under* the date number, which pushed the number 5.5px above the middle of its row. The occasion wash is centred on the row, so on this one screen the tint sat lower than the numbers it was meant to be washing - and the dates themselves did not line up with each other. The count now sits at the foot of the day, every date number sits on one line, and the tint runs through that line. Nothing else about the calendar moved: the green ring, the chosen day's fill, and the counts all read exactly as before.

**On a computer the name comes up on hover.** v87's bubble needed a tap. With a mouse, resting the pointer on a marked day now shows its name on its own, exactly as the customer's page does. This is deliberately only for a pointer that can hover: a touch screen keeps the tap, so a finger sweeping a run of days across the Availability or Delivery Dates calendars never drags a bubble along with it.

**14 Sep 2026 - engine v87 (no database setup needed). Tap a marked day and it tells you its name - on every calendar in the app.** v85 drew your marks on all five of the app's month calendars, but only as colour: a wash and a band, with nothing to say which holiday it was. The customer's own shop page has answered this since its calendar arrived - tap a marked day and a small dark bubble above it says the name - so the app now does the same thing, with the same look.

**A tap names the day and still does that day's usual job.** The bubble appears above the day ("Malaysia Day") and goes away as soon as you tap anywhere else. Nothing else about the tap changed: on the Orders calendar it still opens the day, on a product's Availability calendar it still marks or unmarks the sell day, on Delivery Dates it still starts the range you are about to mark. A day that sits inside two marks takes the name of the shorter one - the same rule the colours already follow, so the name and the colour always belong to the same mark. A day already gone is never named, because no mark is drawn on a past day either.

**Two kinds of day could not be read at all before, and now can.** On the Orders calendar a marked day you do not deliver was a quiet number with nothing to tap, and on the Delivery Dates calendar a marked day that is *already* one of your delivery dates could not be tapped either - its tick is removed from the list below the calendar, never by tapping its square, so its tap was free. Both now say their name when tapped, and nothing else about them moved: a Delivery Dates square still never ticks or unticks a date.

**The Order date boxes name their day above the date, not above a day of the month.** Those two fields fold their calendar the moment a day is picked, so a bubble drawn on the day would vanish with it. The name sits above the date the box is showing instead, which also means an order recorded on a public holiday says so for as long as that date is on the field.

Nothing was added, moved or removed, and the shop is not touched: this is the same bubble the shop's calendar has been showing, arriving on your own calendars.

**14 Sep 2026 - engine v86 (no database setup needed). On the Orders screen your marked days now sit on their days, at the same depth as everywhere else.** v85 put your marks on every calendar in the app. On the Orders screen the wash did not line up with the days underneath it. Two things were behind that, and both came from the same fact: the days on that one screen are taller than the days anywhere else, because a delivery day there carries its booking count ("3/12") under its number.

**A holiday band was stretched to fill its week row.** Where every day is the app's usual height, a band came out the same 30px sheet a single-day box is, so the two shapes matched - which is how the rest of the app has always drawn them. On the Orders screen the rows are taller, so the band there came out deeper than a

single-day box, and deeper in the weeks that held a delivery day than in the weeks that did not, so one holiday's own band changed depth from row to row inside the same month. **A band is now a sheet of one depth, centred on its row** - the same depth a single-day box has, on every calendar, whatever height the days around it happen to be.

**A plain day kept the app's usual height inside a taller row**, so it sat at the top of the row rather than on its middle. A single-day mark was therefore drawn 6px higher on a day you do not deliver than on a day you do: the same holiday, on a different line, decided by whether you deliver that day. **Every day of the Orders month is now drawn at one height**, delivering or not, so a mark sits on the same line as the day it covers, beside the booking counts.

One more thing fell out of that. Today's soft glow is drawn around a whole day on the app's shorter calendars, and around a taller day it would have become a box again - the very thing that glow replaced. **Today on the Orders screen now glows around its number**, the same round glow a delivery day's number carries, so today reads the same whether or not you deliver that day.

Nothing else changed. The shop's calendar and your Delivery Dates calendar look exactly as before (their days are all one height, so they never had this), and no mark was added, moved or removed: this is the drawing only.

**14 Sep 2026 - engine v85 (no database setup needed). Your marked days are now drawn on every calendar in the app, not just the one you marked them on.** Until now a holiday only appeared on the customer's calendar and on the Delivery Dates screen you marked it on. Every other calendar in the app drew a plain month: the month calendar at the top of the Orders screen, the little **Order date** calendars in the new-order form and in the Edit-order pop-up, the delivery-day picker inside that pop-up, and a product's **Availability** calendar where you mark the days it sells. So a holiday you were planning around was invisible on every screen you actually plan on.

**Now all of them draw your marks, in the same two shapes and the same colours as the Delivery Dates calendar** - a see-through band across a run of days, deeper the shorter the run, and a box for a single day. A mark on a day you do not deliver is drawn too, on a dimmed day, since that is the only way a holiday on a quiet day can be seen at all. Nothing about the marks themselves changed: they are still only ever something to see, and a mark still never adds or removes a delivery date or changes what a product sells. Two small pieces of the drawing were tidied while moving it: the green tint on a day a product sells is now a see-through wash rather than a flat fill, so a holiday band running behind it still shows through instead of the mark disappearing on exactly the days you sell (the tint itself looks the same as before); and a marked day that happens to be today now keeps the soft glow that says "this is today", the way the shop's calendar has always drawn it.

**14 Sep 2026 - engine v84 (no database setup needed). A day you marked is now the same gentle wash everywhere, whichever calendar you are looking at.** A marked day was drawn one of two ways, decided by a single question: is this a day you deliver for? On a day you deliver for the mark went pale, so the green "you can order" ring stayed the first thing the eye read. On a day you do not, the mark was drawn at full colour, as a solid block with the number turned white. September's Malaysia Day sat on a delivery day and read as a quiet tint; October's World Animal Day fell past the delivery days published so far and read as a loud solid block - so the same kind of holiday looked like two different things, and a mark quietly changed its own appearance as the delivery window moved towards it.

**Now a mark is only ever a see-through wash of its own colour, and its depth follows how long the mark runs: a one-day holiday the deepest, a long school break the palest.** The customer's calendar and your own Delivery Dates calendar read that depth from the same place, so a day wears exactly the same mark on both sides of the shop and in every month. Nothing is ever a solid block of colour any more, which also means the date number and the green delivery pill always read on top of a mark, whether or not you deliver that day. Only the look changed: marking, removing, renaming and loading standard occasions all work exactly as before, and a mark still never adds or removes a delivery date.

**14 Sep 2026 - engine v83 (no database setup needed). Marking a holiday now tells the shop straight away.** A day you mark on the Delivery Dates calendar is drawn on the customer's own calendar (engine v79 to

v81). But saving a mark only ever sent your delivery dates to the cloud - it never sent the customer's page anything - so the marks reached the shop only by accident, whenever you next happened to save a product or a setting. Mark the holiday, open the shop, and nothing had changed. Found while you were looking for a holiday on the shop and finding none.

**Now the same save does both.** Marking a day, removing a mark, loading a set from the standard list, or renaming or recolouring one all republish the customer's page within a couple of seconds, exactly the way editing a product does. Nothing else about a mark changed: it is still only ever something to see while planning, it still never adds or removes a delivery date, and a name you typed yourself still never leaves your phone.

**One tap to catch up the marks you already have.** The fix sends the marks from now on; the ones already on your calendar have never been sent at all, so open More -> Settings -> Storefront and tap **Publish now** once. After that the shop's calendar follows your marks by itself. Nothing to run in Supabase.

**14 Sep 2026 - engine v82 (no database setup needed). A product's selling days are now marked on a little calendar inside the product's own screen - weekend, certain days, a season, or any mix of them.** Until today a product could say only two things about *when*: a notice period, and one from-date and to-date for a season. You asked to set it freely instead, so the two date boxes have become a card you mark.

**Open a product and tap the Availability card.** It sits under Daily limit and stays folded until you tap its title, exactly like the product-text card you already know. Folded, its title tells you what is marked - "Every day", or "Sat & Sun", or "1-24 Dec 2026" - so the Products screen alone tells you a product's selling days without opening anything.

**Inside, it is a month calendar and four gestures.** Tap a weekday letter (M, T, W...) and every one of that weekday **in the month shown** is marked: tap S twice and the product sells on Saturdays and Sundays. Tap one day to mark just that day. Slide your finger across several days and the whole run is marked. Tapping or sliding over a day that is already marked takes it back. Everything you mark is added together - mark Saturday and Sunday, then a week in the middle, and the product sells on all of them.

**Nothing carries over into the next month**, which is the point of it: marks belong to the days you marked and no others, so a December-only set stays December-only and January starts with nothing marked. Page to another month with the arrows and mark that one too if you want to sell then. And a product whose card you have never opened sells on every delivery day exactly as before, so nothing on your live shop moves until you mark it.

**A period that crosses a month or a year is one mark with two ends you can stretch.** The card lists your marks under the calendar; tap one and its **Starts** and **Ends** dates appear above it, and you can push either end as far as you like - a Sat & Sun run from 1 Dec to 4 Jan is one line, not two marks. Leave an end empty and the mark simply has no bound on that side ("from here on", or "up to here"). Type the two dates the wrong way round and they are put in order for you. A small x on a listed mark removes it whole.

**On the order page, a day a product is not sold for is not a note - the product is simply not there.** No card, nothing to want and not have. If a customer already had it in their basket and then switches to a day it is not sold for, it leaves the basket and a short line says why. The one rule that keeps its note is the notice period: a product needing 14 days IS sold that day, it only has to be ordered earlier, so it stays on the menu reading "Orders close 14 days before delivery - pick a later date" - the two are different questions, and only the second one hides anything.

**Two things worth saying plainly.** The old From/To season boxes are gone; a product that had them opens with that period already sitting in the card as one mark, so nothing you set before is lost. And a product whose selling days have all gone by keeps them, on purpose: your own rule is "if I have not indicated a selling date, it is not selling", so a mark you leave on says exactly that, and the way to put a product back on sale every day is to take its marks off with the x. No database setup, and delivery dates, capacity, value packs, ingredients, suppliers and everything else are untouched.

**14 Sep 2026 - engine v81 (no database setup needed). A holiday you marked is now drawn on the shop's calendar exactly the way your own calendar draws it: a pale band across a run of days, a solid box for a**

**single day.** The marks were already reaching the shop; it was only their shape that differed, and a mark that looks different on the customer's page than on yours is a mark your customer has to learn twice.

**On the shop, a marked day now wears the mark your own calendar gives it.** The shop had been drawing each marked day as its own flat pale square, so a holiday running over a week read as a row of separate blocks where your own calendar shows one continuous shape. Now a holiday running over **several days** is a **pale see-through band** across those days - one unbroken rounded band per week row it crosses, in its own colour, and **deeper the shorter the run**, so a one-day holiday still stands out inside a long school break. A **single-day** holiday is a **solid box** in its full colour on its one day, a touch narrower than the band so a longer band still peeks out at its sides. The eight colours are your own calendar's eight, and the strength steps are the same three, so a marked day reads the same to your customer as it does to you.

**The one place it gives way: a day you deliver for.** Green is the colour the delivery ring uses, and a solid box would swallow that ring - so on a day the bakery delivers for, the day a customer can actually order, the box goes pale there and the **green ring stays the first thing the eye reads**. The mark is still there on that day, and the day is still named on a tap. This is the one guard on the shop's calendar, and it covers every mark colour rather than green alone, so no colour can ever hide the days that can be ordered for.

**Nothing else moves.** Tap a marked day - or, on a computer, rest the mouse pointer on it - and the same small bubble names it, e.g. "Malaysia Day"; tapping anywhere else puts the bubble away, and still nothing is listed under the grid. Today still breathes its soft glow. A run still stops at today rather than colouring days already gone, exactly as your own calendar leaves past days alone. And your own typed-in days still never reach the shop: only a day that is, by name and date, one of the built-in standard days can be published, exactly as before. No SQL, no schema change. The calendar the shop draws with is still a copy of your app's own, and a test now pins the two copies' mark rules together as well as their month grids, so a day can never again look like one thing here and something else there.

**14 Sep 2026 - engine v80 (no database setup needed). Today glows instead of wearing a box, a holiday you marked is a soft tint you can tap to name, and the Orders screen's sideways date strip becomes the same month calendar the shop shows.** Three small things, all pulling the same way: a calendar should read the same on both sides of the shop, and nothing drawn on it should look like something it is not.

**On the shop, today no longer wears a little red rectangle.** It always meant to be a ring around the date, but on a day you do not deliver the app had no circle to draw that ring on, so it traced the day's number as a box instead - which is why it looked like a red square on the quiet days and only looked right on the days you deliver for. Today is now a **gentle glow** that breathes around the number, with no box at all, and it sits happily on a day you deliver for too: the **green ring** says you can order for this day, the soft brown glow says it is today. Under **prefers-reduced-motion** (a phone set to calm its animations) the glow holds steady instead of pulsing, so it is still there to find. The same fix reaches the app: the today marker on the Delivery Dates calendar, and inside every date picker, was a hard outline and is now the same soft glow.

**A holiday you marked is now a soft wash, and its name comes out on a tap.** The shop's calendar used to mark a standard day with a small coloured dot and then list the month's names in a "Holidays" line under the grid - which crowded the very line that line was there to keep clear, and made a holiday compete with the days a customer can actually order for. A marked day now wears a **pale wash** of its own colour over the whole day. The wash is deliberately neither a ring nor a filled dot, because a ring is what "you can order for this day" looks like, and the two must never be read for each other. Tap a tinted day - or, on a computer, rest the mouse pointer on it - and a small dark **bubble** above it names the day, e.g. "Malaysia Day". Tapping anywhere else puts the bubble away. The "Holidays" line under the grid is gone. Your own typed-in days still never reach the shop: only a day that is, by name and date, one of the built-in standard days can be published, exactly as before.

In the app the same holiday is named **beside the day** instead of in a list: open a delivery day and, if it carries a mark, its name sits under the date in the mark's own colour. Nothing at all is drawn on a day you have not marked.

**The Orders screen's date strip is now a month calendar.** The row of date chips at the top of Orders grew a

little longer every week. It is now the same **month calendar** the customer sees - one month at a time, with left and right arrow buttons either side of the month's name. The days you deliver for are ringed in **green**, and under each one is **how booked it is**: the same "3/12" the Home dials show, or **FULL** when the day has no room left. A day whose orders have already closed shows its count in **red**; a day already gone is dimmed but still opens, because you backfill and look back at old days. Every other day of the month is plain and cannot be tapped. Tap a green day and the app's view moves to it - that day is filled in on the calendar and its orders appear below - and tapping a day in the list below moves the calendar with it, so the two always agree. The arrows reach only the months your delivery days fall in, so paging never strands you on an empty month.

**The + New order card arrives folded, and leads with the calendar.** It used to sit open on every delivery day, which is not what that screen is for day to day. It now starts **shut**, showing just its title, "+ New order" - tap the title to open it, and tap the title again (or tap anywhere else on the screen) to fold it away. Opened, it reads in the order you think in: **the month calendar first**, the same one the shop shows, already sitting on the day you are looking at, so tapping another green day moves the whole screen before you have typed anything; **then the customer's details** - name, WhatsApp number, self-collect or courier, a note; **then the items**, with "+ Add another item" and "+ Add order" at the end. It stays open after you add an order, so you can type the next one straight away, and coming back to the screen later starts it folded again.

**The Edit pop-up now reads the same way.** Its **Delivery day** control was a button you had to tap to reveal a calendar; the calendar is simply there now, already open on the order's current day. The pop-up then reads the day, then the customer's details, then the items - the same order as the card. Every day still to come is tappable, plus the order's own day even if it has passed, so an old order always shows where it is, and the soft notes re-read underneath the moment you pick a different day. Moving an order is unchanged in every way that matters: still one delivery day per order, and capacity, the confirmation and the payment reminder, the track page and the WhatsApp messages are all exactly as they were. The button-style day picker the app used in these two places has been deleted along with its tests, because nothing calls it any more.

**14 Sep 2026 - engine v79 (no database setup needed). The customer picks a delivery day on a calendar, and so do you in the app.** The shop used to show its delivery days as a row of date chips laid out sideways. Now, under **Pick a delivery day**, the customer sees a **month calendar** - one month at a time, with the arrow buttons either side of the month's name to look ahead or back. The days you are taking orders for are picked out in **green**; every other day of the month is plain and cannot be tapped. They tap a green day, and the day they chose is written out in words just underneath - **"Your delivery day: Wed, 16 Sep"**. That is the one line of text below the calendar, exactly as asked: no dates floating outside the grid, and still **one delivery day per order**. The first open day is already chosen for them when the page opens, so ordering can never be blocked by forgetting to tap. Nothing else about taking orders changed: a day you have cut off is still not shown at all, and a day that is full still greys out with a red **Sold out** badge and cannot be tapped.

**Only the standard days you have marked reach the customer.** If you have loaded days from the built-in list onto your own Delivery Dates calendar, those days now wear a small coloured dot on the customer's calendar, and a short **Holidays** line under the grid names the month's ones (e.g. "16 Sep · Malaysia Day"). They follow the colour you gave them. A day **you typed yourself** - a birthday, a promo, a school break - never appears: only a day that is, by name and date, one of the built-in standard days can be published, so your own notes stay private to your phone. Mark nothing in the app and the shop's calendar simply carries no dots and no caption.

**In the app, the same little calendar now stands wherever you name a date.** Three places changed. In the **Edit** pop-up, **Delivery day** is no longer a drop-down list: it names the order's current day, and tapping it opens a small month calendar beneath it - every day still to come is tappable, plus the order's own day even if it has passed, so an old order always shows where it is. The **Order date** box in that pop-up is a calendar too, with a **Today** shortcut for an order you are typing in right now. And the **+ New order** card has gained a **Delivery day** picker at its top: tap another green day and the whole screen moves to that day, so the product list you are choosing from is always the one that is actually sellable for the day you picked. The soft notes that help you decide - the change/cancel window, a warning, a closed day, too little left - all still sit under the calendar, and none of them stop you.

The calendar expands in place under the button rather than opening in its own window, because the app's pop-ups share one layer and a calendar opened inside one would wipe out the Edit pop-up it was opened from; expanding in place also reads better on a phone. No database setup needed - nothing about how orders are stored, counted, bought for or messaged changed.

**13 Sep 2026 - engine v78 (no database setup needed). The two confusing "Add" buttons on the Delivery Dates calendar now say what they do.** On the Delivery Dates screen, **Mark an occasion** used to carry a small button reading **" + Add occasion"**. The name did not tell you that it opens a ready-made list of Malaysia's holidays and fun days, all ticked and waiting - so you had to open it to find out. It now reads **" + Load standard occasions"**, and the window it opens is titled the same.

Inside that window there were **two buttons both called "Add"** - the big one at the bottom that files the days you ticked, and a smaller one in the **My own day** strip just above it, which files a day you name yourself (a birthday, a one-off promo). Reaching for "Add" and hitting the wrong one gave you **"Type a name first"**, for a day you never meant to name. The smaller one is now spelled out as **"Add my own day"**, so the only plain **Add** in the window is the one that adds the ticked days.

Nothing else moved. Ticking a row and pressing the big **Add** still files exactly what you ticked and nothing you did not; **Untick all / Tick all** and the heading tick boxes work as before; the **My own day** box still takes any name and any date. No database setup needed.

**13 Sep 2026 - engine v77 (no database setup needed). The greyed line in the translated-text card now says what happens if you leave it blank.** Engine v76 turned the product screen's Mandarin / Bahasa Malaysia text into one folding card, and each empty line showed its translation on offer. That greyed line read "e.g. <the translation> - blank keeps English". It now reads **"e.g. <the translation>.....if blank, it will be filled with English"** - the same words you asked for, saying plainly what the customer gets rather than asking you to work it out. The line in the card's own explanation says the same. Deleting what is in a line brings the greyed line and its arrow straight back, exactly as before. Nothing else moved: the card still folds away when you tap outside, the -> still takes the words, the circular-arrow button still translates that line again.

**13 Sep 2026 - engine v76 (no database setup needed). The translated text on a product is now one quiet card that folds away, filled in for you one line at a time.** The **Product text for your customers** part of the New/Edit product screen - the same words in **Mandarin** and **Bahasa Malaysia** - was the busiest corner of the whole app. It was always open, so a long screen got longer before you needed any of it; eight lines each carried its own **"Translate this one"** button; two **"Fill all"** buttons sat above them; and a little **"auto"** tag marked the machine-made lines, in a word you had to remember the meaning of.

Now the card starts **shut** and stays shut - it is the setup part of the screen, not the part you touch every day. **Tap its title to open it.** Opened, each line already has its translation worked out and waiting as the same pale grey suggestion you know from every other box, with the little -> at its right edge: **tap the -> and the suggested words drop in** - exactly the gesture you already use. The moment you take them, the arrow is replaced by a small **circular-arrow button** - tap it and that one line is translated again on the spot, which is what the old "Translate this one" button did. **Type your own words over any line and the circular-arrow button goes away** for that line: it is yours, and it is never overwritten. **Leave a line blank** and that language keeps the English, the same as before. And when you are done, **tap anywhere outside the card and it folds away again.**

The **"Fill all Mandarin" / "Fill all Bahasa Malaysia"** buttons are gone, the eight **"Translate this one"** buttons are gone, and there is no **"auto"** tag to read any more - a line showing the circular-arrow button is machine text, and a line with no button is yours. Nothing about what your customers see has changed, and nothing needs setting up: a product is still translated for the shop the first time you save it and again when you Publish, exactly as before, so a product you never open this card for still reads in all three languages on the order page.

**12 Sep 2026 - the TNG QR box no longer claims to be on the order page (no new engine; your phones did not change).** The box in **Settings -> Storefront** holding your Touch 'n Go QR image link said it was **"shown on the customer's track page"**. It never was - your payment QR reaches the customer in the

**WhatsApp confirmation** and the **payment reminder**, and nowhere else. The help text now says that, and the order page's own settings no longer carry the QR at all, so it cannot be drawn there by accident. Nothing about how you take payment changed, and nothing about how the app works changed.

**12 Sep 2026 - engine v75 (no database setup needed). The glow on an order you have just jumped to now stays until you reach it.** Tapping a row in the **New Orders** inbox (or a result in **Find an order**) takes you to that order's delivery day and marks the order with a green glow - but the glow used to fade after less than two seconds, which is no time at all when your eye is still travelling down a long day, and you were left hunting for a row that was no longer lit. **The glow now keeps pulsing until you get to the row.** It stops the moment your finger or the mouse arrives on that order, or when you tap it open - so it is waiting for you however long you take, and it never lingers as a puzzle later. (If a phone has **Reduce Motion** turned on in its accessibility settings, the same green outline holds steady instead of pulsing - it still stays until you reach the row.)

**12 Sep 2026 - engine v74 (no database setup needed). The little arrow in an empty box now keeps working.** Engine v73 added the small right-arrow that sits at the right edge of a box showing a greyed example: press the **-> key** on a keyboard, or tap that arrow, and the example drops in as real text. On a phone that tap worked the first time and could then go dead - once you had tapped into a box, tapping the arrow on the next one did nothing.

Two things were behind it, and both are fixed. The tap was measured against the **page**, and the phone's keyboard shifts the page under your finger as soon as any box is focused, so the arrow was no longer where it had been. And taking a suggestion **opened the keyboard itself**, so the page jumped after the very first tap. The tap is now measured against the **box itself**, taking a suggestion no longer opens the keyboard at all, and the strip you can tap is about half as wide again - so a thumb that lands a little wide of the arrow still counts. A tap on the greyed text stays an ordinary tap that just puts your cursor there. **The arrow looks exactly as it did**, and nothing else changed.

**12 Sep 2026 - a new order now opens on the order itself (no new engine; nothing to set up).** Tapping a row in the **New Orders** inbox took you to that order's delivery day - and stopped there. On a busy day, with a long list of orders, you then had to read down the list to find the very order you had just tapped, and it was easy to open the wrong one.

The app now goes the whole way: it opens the delivery day, **scrolls the order to the middle of the screen and flashes it** for a moment, exactly as the **Find an order** search box already did. If the list had been narrowed with the status filter so the order was hidden, that filter is cleared first, so the order is always there waiting for you. Nothing else on the Orders screen changed.

**12 Sep 2026 - engine v73 (no database setup needed).** Five changes, all about the shop's promise to the customer and the baker's control over it.

Your rule is that an order is **not cancellable and money is not refunded, but it can be moved to another day**, and a customer may ask until a set number of days before delivery. That number now lives **per product**. Open a product in **Products** and, under its daily limit, a new box **"Changes or cancellations (days before delivery)"** - type 2, say, and that product's card on the shop reads **"Change or cancel up to 2 days before delivery."** It only ever *tells* the customer; it never blocks you, and you still move any order by hand. Leave it blank (or 0) and no note shows for that product. When one order holds several products with different numbers, the customer sees the **strictest** one - the largest, i.e. the earliest cut-off - so a mixed basket always gives one clear figure. The same sentence appears on the green **"Order received!"** card just after they place the order.

Second, **you can now move an order to another delivery day in the app.** Open an order with **Edit** and, in the pop-up, a new **"Delivery day"** box lists every day still to come, soonest first. Pick the new day and **Save changes**: the whole order moves as one - items, customer and the price it was sold at all stay put - and the customer's **"Track your order"** page follows to the new day straight away. Under the box are quiet notes to help you decide: the order's own change/cancel window, a reminder when the new day falls **inside** it, a note if that day's orders have already closed, and a note if that day does not have enough left for an item. None of them stop you - they only tell you - and the capacity check now counts the order on the **new** day, so you can see at

once if that day would go over. This is the **only** way to change an order's day: never delete an order and re-type it, because deleting reads as a cancellation. A day left empty stays in your list - to remove it, use **Del** on the Deliveries screen as before.

Third, a **Policies box** for your own cancellation and refund wording. It sits in **More -> Settings -> Storefront**. Type it once in English, tap **Translate**, and the Mandarin and Bahasa Malaysia boxes are filled for you (edit either and it becomes yours). It appears on the shop **under Track your order**, in the customer's own language and with the line breaks you typed, and it stays hidden until you write something.

Fourth, and everywhere in the app: **an empty box showing a greyed suggestion can now be accepted with one tap - the arrow**. A small right-arrow sits at the right edge of such boxes; press the **right-arrow key** or tap that arrow and the suggestion drops in as real text you can edit - the same gesture as accepting a suggestion in an AI chat. It is offered only where the grey text is a genuine recommendation; the sign-in and cloud boxes never offer it, so a made-up key or password can never be accepted by accident.

No database setup and no new SQL - the new product box and the Policies wording travel with your existing shared data.

**9 Sep 2026 - the wish list's automatic email went live (no new engine; the phones did not change).**

Adding a wish on **More -> Software wish list** now emails the **full wish list** - every wish, ticked or not, newest first, with the app version and the date - to the developer address(es) you set under Settings -> **Website & developer**, all on its own, whenever the phone is signed in and sharing data. Before today that automatic email needed a behind-the-scenes service that was not switched on yet, so only the **"Email the full wish list"** row sent it by hand; that row is still there as a backup any time.

**10 Sep 2026 - the homepage opens faster (no new engine; the phones did not change).** Two things about the public homepage were quietly costing every visitor time, and both are fixed.

The homepage's six photographs were stored *inside* the page itself, so the page was **249 KB before a single picture had even been asked for** - and every visitor downloaded all of it, on every visit, before they could see anything. The photos are now six separate files that the browser fetches and **keeps for next time**, which brings the page itself down to **22.7 KB**. The pictures are byte-for-byte the same ones, so the homepage looks exactly as it did, and each one now states its size up front so nothing jumps around as the page loads.

Second, the "What customers say" section used to fetch **every published review photo up front** - including the reviews a visitor never waited around to see. A photo is now fetched **as its review comes around**, one step ahead of time so it is ready before it slides in, and the reviews a visitor never reaches are never downloaded at all. On a phone that means less data and a quicker page, and it changes nothing about how the section looks or behaves.

A review photo is also shrunk a little harder before it is stored - to 1000 pixels across instead of 1600. The card on the homepage shows a picture no more than 340 pixels high, so the smaller copy looks just as sharp while being about **a third of the size**, which is kinder to the customer's mobile data and keeps your stored photos light. This applies to photos left from now on; the reviews already on your page keep the picture they were stored with.

Your sign-off line - **Jien Luv 2 Bake | Small Batch. Big Love.** - has moved from the very bottom of the homepage footer, where it sat shoulder to shoulder with the developer credit, up to the **top of the footer**, above your contact details and set off by a thin line. It reads a touch larger now, and in whichever language the visitor has chosen, like the rest of the footer. The "Website by ..." credit stays where it was and now stands on its own.

**10 Sep 2026 - the homepage's language buttons can no longer go dead (no new engine; the phones did not change).** On some phones the homepage's **EN / Mandarin / BM** buttons could stop responding - the page would sit in English and tapping Mandarin did nothing - often after the page had been left alone for a while, or refreshed. The culprit was a leftover background worker from when the backoffice was served from the top of the website instead of from /admin/. (A background worker is the machinery the app installs so it can open without a connection.) That old worker had cached the whole website, and whenever something failed to download it answered with whatever it had - including handing the browser **a web page where it had asked for a script**. A

page that receives a web page instead of its script never runs it, so the code that wires up the language buttons never ran; the English words you saw are written into the page itself, which is why the page still looked perfectly normal.

The old worker is now cleared away wherever it is still installed: a small cleanup file sits at the same address the old worker used, so the next visit replaces it, empties everything it had cached, and removes itself. The backoffice's own offline cache is deliberately left untouched, because the app needs it to open without a connection. The app's worker has also been taught, for good, never to answer a script or a picture with a web page - so the same fault cannot come back on any part of the site.

## **v72 - The shop's sold-out notes read in the customer's language (10 Sep 2026)**

---

The shop page switches language in place, but two corners of it had stayed English no matter which language a visitor chose: the short note on a product that **cannot be ordered for the chosen day** (that it only opens from a certain date, only runs up to one, or that orders close so many days ahead - with the date written in the page's language), and the notes that appear above the menu when the page **has to change a basket** - an item that just sold out, or a quantity trimmed to what is actually left.

Both now read in **English, Mandarin or Bahasa Malaysia** along with everything else. The reason a product is closed is now handed to the page as a plain fact - which kind of rule, and the day or number of days behind it - and the page writes the sentence itself, so no English can slip into a Chinese or Malay page. An English visit is word for word what it was.

## **v71 - The shop's Mandarin / BM buttons answer instantly (10 Sep 2026)**

---

On the shop page, tapping **EN / Mandarin / BM** used to reload the whole page. That re-downloaded the page and fetched the menu, the "left today" numbers and your saved storefront settings all over again, so each tap sat there for a moment before anything changed - and the slower the phone's connection, the longer the wait. A reload could also wipe a basket that was already half filled.

Now the shop switches **in place**, exactly like the homepage already did: the words change the moment you tap, and nothing is downloaded again. Whatever the customer has already done is left untouched - the items in their basket, the delivery day they picked, the name, WhatsApp number and address they typed - and if they are looking up an order, that card re-reads in the new language too. Nothing to set up.

## **v70 - An order keeps the price it was sold at (10 Sep 2026)**

---

Until now an order only remembered **which product** was bought, not what it cost or what it was called. Everything - the amount on the customer's confirmation, the total on their own "Track your order" page, their lifetime spend on the customer list, the Home estimate - was worked out from **today's** menu. So the moment you re-priced or renamed a product, every past order silently moved with it: last month's order suddenly read RM22 instead of the RM15 the customer actually paid, and a renamed loaf rewrote what people had already bought.

Now an order is a record of a sale. When an order comes in - from the shop or typed in by hand - it keeps the **name and the price it was sold at**, and those are what every message, the tracking page and the customer's spend use. Rename or re-price a product today and yesterday's orders stay exactly as they were. A line you deliberately swap to a different product when you **Edit** an order does pick up that product's current price - that is a new sale - but a line you leave alone keeps its old price.

Existing orders (taken before this version) are stamped once with today's values, which is exactly what they are showing right now, so they stop drifting from here on. Nothing to set up; the phones pick it up with the next sync.

## v69 - Either place updates the customer (10 Sep 2026)

---

Yesterday's fix put a customer's name and number back in one place - but it made the **customer card** the side that always won, so if the card and an order disagreed, the card's name was the one shown. That is no longer the rule: now it is simply **whichever you edited last**. Correct a customer's name with **Edit** on one of their orders and that is the name the list, the history pop-up and the "Hi {name}!" greeting use; open their card, type the name and save, and the card's is the one used. Either place updates the customer - neither is the boss. They normally read the same anyway, because each save writes through to the other side; this only decides the rare case where an old copy and a newer one are both sitting there. Nothing to set up; the phones pick it up with the next sync.

## v68 - One name and one number per customer (10 Sep 2026)

---

A customer's name and WhatsApp number used to be held in two places at once - on their orders, and on their customer card - and the two did not always agree. If you opened a customer and typed their name on their card, the customer list could carry on showing the old one, because it read the name off the order. And renaming someone who has **no WhatsApp number** on file could lose them: the name is what ties a numberless customer to their history, so the new name slid off their orders and appeared nowhere at all.

Now the name and number you save on a customer are the ones that count, and fixing them once fixes them everywhere: saving their card writes the name and number onto **every order they have**, so the labels, the WhatsApp messages and the customer list all follow. It works the other way round too - correct a wrong number with **Edit** on an order and that customer's card updates as well, so the two can never drift apart again. Correcting a number carries that customer's bring-a-friend credits across with it, and renaming someone who has no number no longer loses them.

Leaving a box empty means "leave this as it is" - it never wipes a number your confirmations depend on. And if you renamed anyone before today and it never stuck, the app puts their orders right the first time this version opens. Engine v68, guide v68.

## v67 - Naming a "No name" customer (10 Sep 2026)

---

A customer who reached you without a name - a storefront order that carried none - used to stay **"No name"** in your customer book even after you opened them and typed a name. Now it sticks: open them, tap **Add details**, type the name (and the dog's name, what they like, and so on) and save, and their row shows that name straight away - in the list, the finder and the 'Hi {name}!' WhatsApp greeting. A tidy-up; nothing else about your customers changes. Engine v67, guide v67.

## v66 - Auto-translate product text, plus Draft / Publish / Hidden states (9 Sep 2026)

---

Product **description**, **selling unit word** (the "RM15.00 / loaf" bit), **serving tip** and now the **name itself** read in English, Mandarin or Bahasa Malaysia automatically - a free translator fills the Mandarin/BM boxes for you the first time you save or publish a product. A small "auto" tag marks a machine translation; if one "sounds weird", open the product's edit screen, type over that box (it then becomes yours and is never overwritten), or tap a "Fill all Mandarin" / "Fill all Bahasa Malaysia" button or the circular-arrow button next to a single line to have it translated again now. Good translations are simply left alone, and you never hand-fill a whole product.

Products also gain a **third state**. A new product now starts as a **Draft** - fully built with its recipe, but **not on the shop and not orderable** until you **Publish** it. The Products screen shows **three separate lists**: **On the shop**, **Draft - not on the shop yet** (each with a Publish button) and **Hidden - taken down** (with an Unhide button). Hiding a product still keeps its history, recipe and purchase orders.

The **"Copy follow-up"** bring-a-friend message gains a small **EN / Mandarin / BM** choice, so the whole check-in you paste into WhatsApp - including the product's serving tip - can be in your customer's language. No SQL this version.

## v65 - Reviews preview + "waiting to publish" cue (9 Sep 2026)

---

More -> **Reviews** now shows each review **exactly as customers see it** on the homepage - the photo, the stars, the message, then the reviewer's name and the language and date, in the same warm card - and they play **one at a time**, like the homepage's "What customers say" carousel. The reviews waiting for your **Publish** come first; once you have worked through them, the ones already on the homepage follow. It never slides on by itself while you are deciding - a swipe, the arrow buttons, or a dot under the card moves it on, and Publish / Take down / Delete moves you straight to the next review. A line at the very top of the screen says how many are waiting ("3 reviews waiting for you to publish"). The **Reviews** row on the **More** menu shows how many are waiting too, and the app's **Home** screen gains a **"N new review(s) to publish"** card that opens the Reviews screen in one tap - so a new review is never left waiting unseen. No SQL this version.

## v64 - Review carousel, EN/Mandarin/BM site, developer contact + wish email (9 Sep 2026)

---

The homepage **"What customers say"** reviews now play as a **carousel** - one review fills the section at a time, then slides on every few seconds. Visitors can swipe or use the left / right arrows and the dots below to move through them, and it pauses while someone is reading. One review, or none, still shows simply.

The whole **homepage and storefront now read in English, Mandarin or Bahasa Malaysia** via an **EN Mandarin BM** switch near the top of each page (English is the default, and a visitor's choice is remembered next visit). Every product can carry its own **translated shop names** - Products -> Edit -> **Shop names** - so shoppers in Mandarin/BM see a proper translated name on the order page; a blank box keeps the English name, which always stays the real name in the app, labels and history.

The thank-you after sending a review is warmer and honest: it reassures the sender that their review and photo reached the baker and will appear once approved, and a critical review (3 stars or fewer) gets an extra line of thanks for honest feedback. A reviewer now attaches their photo **two ways**: **Take photo** (camera) or **Choose photo** (gallery).

Developer contact is now in the app: under More -> Settings -> **Website & developer**, type the developer's name and one or more email addresses. A small **Website by ...** credit links from the bottom of the homepage, the storefront footer and More -> About (each address tappable). Add the developer's optional **WhatsApp number** in the same card and that link opens WhatsApp (Business) with a ready "Hi!" - message the developer straight from the page - with the email links staying underneath. And adding a **software wish** on the More screen emails the developer the **full wish list** - every wish, ticked or not, newest first, with the app version, the project address and the date - to every address you listed; an **Email the full wish list** row on the card can send it from your own mail app too. No SQL this version.

## v63 - Customers tab, wish list, holidays fix (8 Sep 2026)

---

**Customers** is a real tab of its own, swapped with **Purchase Order** (now under **More -> Purchase Order**). The tab stays your automatic customer list - who ordered, how much, their favourite - and now every person can carry a **profile** (the **Edit** button in their history pop-up): their dog's name and a small photo (auto-shrunk to a thumb), what they like, what to avoid, and a note. Profiles save into a synced customers collection, so both phones know the same people, and they build up quietly for a future AI chat about your customers. A **finder** box at the top of the tab filters the list as you type - name, number, dog's name, likes, notes - and says how many match, just like "Find an order" on Orders. A saved profile shows as a paw or photo beside the name and a small line under the row.

On **Home**, the "Upcoming holidays" card is fixed: it used to vanish whenever nothing was marked on the Delivery calendar. It is now always there - up to three marks at a glance, a scroll for any more, and a friendly empty state that still opens the calendar so you can add days.

On **More**, a small green **Engine v63** pill (in the header card) shows which build this phone runs, a **Software wish list** sits at the bottom and behaves exactly like your to-do list (add, tick, reword, remove - a tick stays ticked and never touches the weekly routine), and **Full change history** opens this history as a PDF kept on your website. No SQL this version.

## v62 - "Not sharing" strip + read-only View (8 Sep 2026)

---

When a phone is not on the shared cloud - shared data off, never set up, or signed out - a thin amber strip at the very top of every screen reads "**Not sharing right now**" with a one-line reason and a **Fix it** tap straight to Settings -> Shared data. It never locks the app, and it disappears on its own the moment sharing is back on. (Set up once per phone: no SQL - the shared-data setup from v53 does everything.)

Also, every backup copy in Settings -> **Backup & safety** now has a **View** button: a purely read-only look inside that copy - its orders grouped by delivery date (customer, product, quantity, status), the product price list and ingredient stock at that time. Looking at a June price confirms the detail without moving today: no rewind, nothing written, ever.

## v61 - Cloud backups (8 Sep 2026)

---

Every time the app opens while signed in, it quietly saves a full **Daily** copy (first open each day), a **Weekly** copy (Mondays) and a **Monthly** copy (the 1st) to your own Supabase cloud, keeping the newest 7 daily / 4 weekly / 3 monthly. A "Back up to cloud now" button, and one "Before restore" copy saved before every restore, stay until you delete them. Each copy can be **Restored** (steps the phone - and the shared cloud - back to that copy; a "Before restore" copy is always saved first, so it is never one-way), **Downloaded** as jienluv2bake-backup-YYYY-MM-DD-KIND.json (which re-imports like an Export), or **Deleted**. Lives under More -> Settings -> "Backup & safety". The copies never contain the per-phone app-login email/password or the app password. Setup: run supabase/backups.sql once.

## v60 - Homepage customer reviews (8 Sep 2026)

---

Customers can leave a **What customers say** review on the homepage - name, 1-5 stars, a message in English / Mandarin / Bahasa Malaysia, and an optional photo. Every review lands *unpublished* in More -> Reviews, where **Publish** shows it on the homepage, **Take down** hides it, and **Delete** removes it - nothing becomes public until you tap Publish. Reviews appear on the homepage only. Setup: run supabase/reviews.sql once.

## v59 - Occasion import tidy-up (8 Sep 2026)

---

Added **World Animal Day** (4 Oct) to the fun/pet-day import, plus whole-group ticking - tick a heading box, or use **Untick all** / **Tick all**.

## v58 - Occasion import (8 Sep 2026)

---

A one-tap + **Add occasion** import of Malaysia's days plus fun days (pet, baking, kindness), and a **My own day** quick-add.

## v57 - Private notes per product ingredient (8 Sep 2026)

---

Each ingredient line in a product can carry a short private description *for that product only* (e.g. which flour that product uses). Typed and seen on the Products screens only - never on the shop, a label, or the product cards.

## v56 - "Keep at least" reserve (8 Sep 2026)

---

Ingredients can set a **Keep at least** level. The order list tops an ingredient back up to that level in whole packs when your stock drops more than 10% under it, and low items ride along even when today's bakes don't use them. The Ingredients screen shows a red low-stock strip. Guide synced to v56.

## v55 - Time & length units (7 Sep 2026)

---

**min/hr** and **cm/m** are pre-loaded in Units (under More), like g/kg - so bake times and pan sizes can be measured and converted (1 hr = 60 min, 1 m = 100 cm). Baking stock is guarded so a time or length unit can never distort a gram count.

## v54 - Ingredients on hand (7 Sep 2026)

---

Ingredients now have **On hand** stock. The PO buys only what you don't have (rows you already have drop out), "Bought" on a saved list adds its packs once, and marking an order **Baked** subtracts its ingredients (undo restores them).

---

*This history covers the two-phone cloud era (v54+). Earlier versions (pre-v54) were never recorded version by version, so they are not listed here rather than invented. Each new version is added here as it ships.*